BP3-QD dip 60 thrust, 0 to 1499.91 yr, events 1 to 16 (cut = the earlier end of the two runs; 365-day years)
onset = first time max slip rate over the 12 fault stations > 1e-3 m/s; dur = last time > 1e-3 m/s minus onset;
peak = max slip rate over the fault stations in the event; slip = each code's slip one year after its own onset, both codes
read at the same time after onset (m; dp = km x 10)

ev	code	onset (yr)	dur (s)	peak V	at	000	025	050	075	100	125	150	175	200	250	300	350
1	Tandem p=1 (Mac)	176.7347	48.0	0.6093	dp150	0.000	0.130	0.675	1.062	1.359	1.594	1.756	2.274	2.697	3.335	3.940	4.540
1	seas-mfem p=1	176.7347	47.9	0.6068	dp150	0.000	0.130	0.675	1.062	1.359	1.594	1.756	2.274	2.697	3.335	3.940	4.540
1	ours - T p1	+0.0000	-0.2	-0.0025		-0.000	+0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000	-0.000
2	Tandem p=1 (Mac)	288.5459	76.7	7.1124	dp000	4.396	4.485	4.579	4.672	4.749	4.787	4.953	5.479	5.889	6.498	7.081	7.749
2	seas-mfem p=1	288.5459	76.8	3.7544	dp000	4.396	4.485	4.579	4.672	4.749	4.787	4.953	5.479	5.889	6.498	7.081	7.749
2	ours - T p1	+0.0000	+0.1	-3.3580		+0.000	+0.000	-0.000	-0.000	+0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
3	Tandem p=1 (Mac)	343.4119	71.6	5.1305	dp000	6.792	6.882	6.977	7.070	7.150	7.198	7.258	7.723	8.087	8.619	9.133	9.722
3	seas-mfem p=1	343.4119	71.8	2.6757	dp000	6.792	6.882	6.977	7.070	7.150	7.198	7.258	7.723	8.087	8.619	9.133	9.722
3	ours - T p1	+0.0000	+0.2	-2.4548		-0.000	+0.000	-0.000	-0.000	-0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000
4	Tandem p=1 (Mac)	429.9763	71.5	5.4286	dp000	9.406	9.496	9.591	9.684	9.764	9.810	9.884	10.361	10.733	11.278	11.804	12.406
4	seas-mfem p=1	429.9763	71.8	2.6073	dp000	9.406	9.496	9.591	9.684	9.764	9.810	9.884	10.361	10.733	11.278	11.804	12.406
4	ours - T p1	+0.0000	+0.3	-2.8213		+0.000	-0.000	+0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000	+0.000	+0.000
5	Tandem p=1 (Mac)	517.7282	61.8	5.5614	dp000	12.124	12.214	12.309	12.401	12.481	12.526	12.608	13.089	13.464	14.015	14.546	15.154
5	seas-mfem p=1	517.7282	61.8	3.9305	dp000	12.124	12.214	12.309	12.401	12.481	12.526	12.608	13.089	13.464	14.015	14.546	15.154
5	ours - T p1	+0.0000	+0.1	-1.6309		+0.000	-0.000	+0.000	-0.000	+0.000	+0.000	+0.000	-0.000	+0.000	+0.000	-0.000	-0.000
6	Tandem p=1 (Mac)	605.8054	68.7	5.5647	dp000	14.883	14.973	15.067	15.160	15.239	15.284	15.369	15.852	16.228	16.781	17.314	17.924
6	seas-mfem p=1	605.8053	69.1	5.1309	dp000	14.883	14.973	15.067	15.160	15.239	15.284	15.369	15.852	16.228	16.781	17.314	17.924
6	ours - T p1	-0.0001	+0.3	-0.4339		-0.000	+0.000	-0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000	+0.000	+0.000
7	Tandem p=1 (Mac)	693.3267	66.7	5.5498	dp000	17.642	17.732	17.827	17.919	17.999	18.044	18.128	18.612	18.988	19.541	20.074	20.684
7	seas-mfem p=1	693.3267	67.1	3.6711	dp000	17.642	17.732	17.827	17.919	17.999	18.044	18.128	18.612	18.988	19.541	20.074	20.684
7	ours - T p1	-0.0000	+0.3	-1.8787		-0.000	-0.000	+0.000	-0.000	+0.000	-0.000	+0.000	+0.000	+0.000	+0.000	-0.000	+0.000
8	Tandem p=1 (Mac)	781.1846	69.5	5.5923	dp000	20.409	20.499	20.594	20.687	20.766	20.811	20.896	21.380	21.756	22.310	22.843	23.453
8	seas-mfem p=1	781.1847	69.8	4.0337	dp000	20.409	20.499	20.594	20.687	20.766	20.811	20.896	21.380	21.756	22.310	22.843	23.453
8	ours - T p1	+0.0000	+0.3	-1.5586		+0.000	+0.000	+0.000	+0.000	+0.000	-0.000	+0.000	+0.000	+0.000	-0.000	+0.000	-0.000
9	Tandem p=1 (Mac)	868.7331	68.1	5.6042	dp000	23.172	23.262	23.357	23.450	23.529	23.574	23.658	24.142	24.518	25.072	25.605	26.215
9	seas-mfem p=1	868.7331	68.1	2.7123	dp000	23.172	23.262	23.357	23.450	23.529	23.574	23.659	24.142	24.518	25.072	25.605	26.215
9	ours - T p1	-0.0001	-0.0	-2.8919		-0.000	-0.000	-0.000	-0.000	+0.000	-0.000	+0.000	-0.000	-0.000	-0.000	+0.000	-0.000
10	Tandem p=1 (Mac)	956.5280	69.4	5.5774	dp000	25.939	26.029	26.124	26.216	26.296	26.341	26.426	26.909	27.286	27.839	28.372	28.983
10	seas-mfem p=1	956.5280	69.6	3.1044	dp000	25.939	26.029	26.124	26.216	26.296	26.341	26.426	26.909	27.286	27.839	28.372	28.983
10	ours - T p1	-0.0000	+0.2	-2.4730		-0.000	-0.000	-0.000	+0.000	-0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000
11	Tandem p=1 (Mac)	1044.1173	67.9	5.6052	dp000	28.703	28.793	28.887	28.980	29.060	29.104	29.189	29.673	30.049	30.602	31.135	31.746
11	seas-mfem p=1	1044.1172	68.3	5.1885	dp000	28.703	28.793	28.887	28.980	29.060	29.104	29.189	29.673	30.049	30.602	31.135	31.746
11	ours - T p1	-0.0001	+0.4	-0.4167		-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	+0.000	-0.000	-0.000	-0.000	-0.000	-0.000
12	Tandem p=1 (Mac)	1131.8751	69.1	5.5842	dp000	31.469	31.559	31.654	31.746	31.826	31.871	31.956	32.439	32.816	33.369	33.902	34.513
12	seas-mfem p=1	1131.8750	69.3	3.4423	dp000	31.469	31.559	31.654	31.746	31.826	31.871	31.956	32.439	32.816	33.369	33.902	34.513
12	ours - T p1	-0.0001	+0.2	-2.1419		-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
13	Tandem p=1 (Mac)	1219.4936	68.4	5.5540	dp000	34.233	34.323	34.418	34.511	34.590	34.635	34.720	35.203	35.580	36.133	36.666	37.277
13	seas-mfem p=1	1219.4936	68.7	3.0188	dp000	34.233	34.323	34.418	34.511	34.590	34.635	34.720	35.203	35.580	36.133	36.666	37.277
13	ours - T p1	+0.0000	+0.2	-2.5353		-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	+0.000	+0.000	+0.000	+0.000	-0.000	-0.000
14	Tandem p=1 (Mac)	1307.2267	69.0	5.5871	dp000	36.999	37.089	37.184	37.276	37.356	37.401	37.486	37.969	38.346	38.899	39.432	40.043
14	seas-mfem p=1	1307.2266	69.2	4.1259	dp000	36.999	37.089	37.184	37.276	37.356	37.401	37.486	37.969	38.346	38.899	39.432	40.043
14	ours - T p1	-0.0001	+0.2	-1.4612		+0.000	+0.000	+0.000	-0.000	-0.000	-0.000	+0.000	-0.000	-0.000	-0.000	+0.000	-0.000

BP3-QD dip 60 thrust, 0 to 1499.91 yr, events 1 to 16 (cut = the earlier end of the two runs; 365-day years)
onset = first time max slip rate over the 12 fault stations > 1e-3 m/s; dur = last time > 1e-3 m/s minus onset;
peak = max slip rate over the fault stations in the event; slip = each code's slip one year after its own onset, both codes
read at the same time after onset (m; dp = km x 10)

(continued)

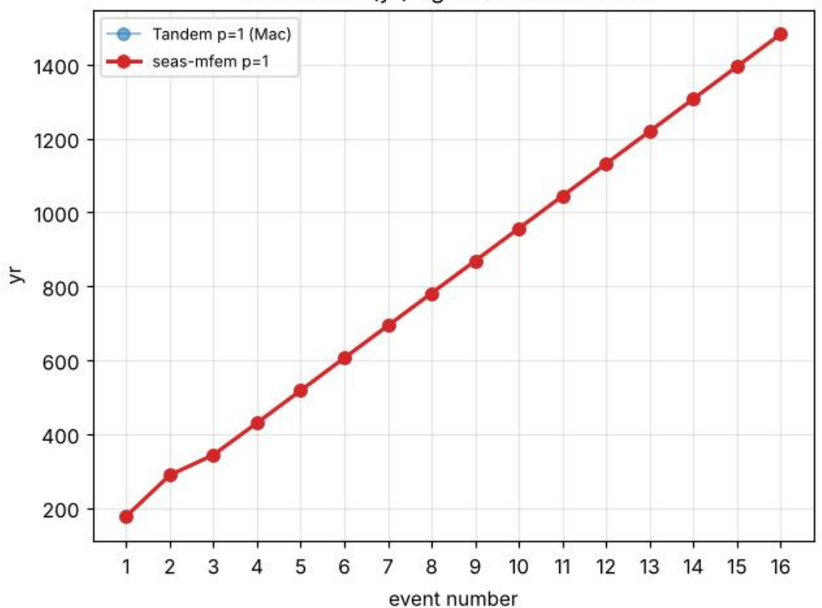
15 Tandem p=1 (Mac)	1394.8652	68.5	5.5916 dp000	39.764	39.854	39.948	40.041	40.120	40.165	40.250	40.734	41.110	41.663	42.196	42.807
15 seas-mfem p=1	1394.8652	68.7	4.5122 dp000	39.764	39.854	39.948	40.041	40.120	40.165	40.250	40.734	41.110	41.663	42.196	42.807
15 ours - T p1	-0.0000	+0.3	-1.0794	-0.000	-0.000	-0.000	+0.000	+0.000	-0.000	+0.000	+0.000	+0.000	+0.000	+0.000	-0.000
16 Tandem p=1 (Mac)	1482.5815	69.2	5.5755 dp000	42.529	42.619	42.714	42.807	42.886	42.931	43.016	43.499	43.876	44.429	44.962	45.573
16 seas-mfem p=1	1482.5814	69.1	3.2065 dp000	42.529	42.619	42.714	42.807	42.886	42.931	43.016	43.499	43.876	44.429	44.962	45.573
16 ours - T p1	-0.0001	-0.1	-2.3690	-0.000	+0.000	+0.000	-0.000	-0.000	-0.000	+0.000	+0.000	-0.000	+0.000	+0.000	-0.000

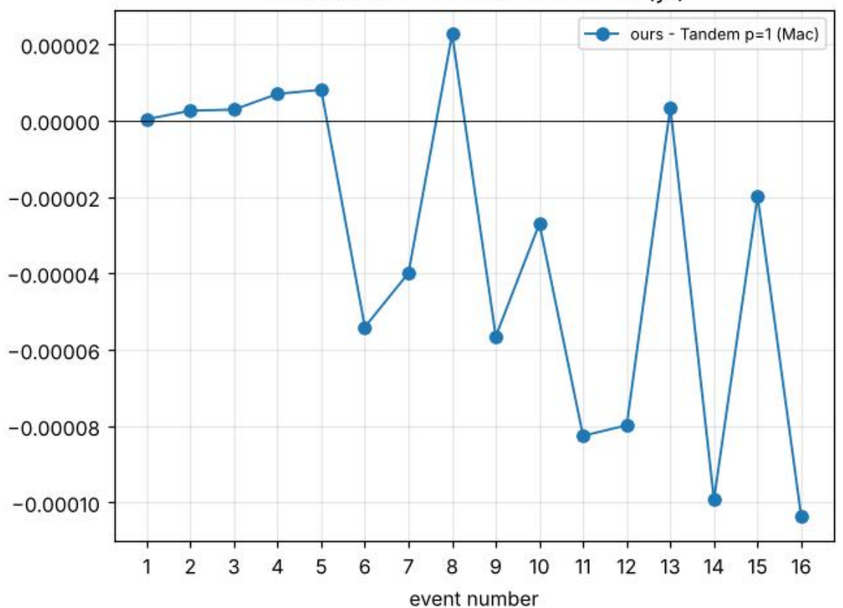
mean recurrence (yr) and onset drift

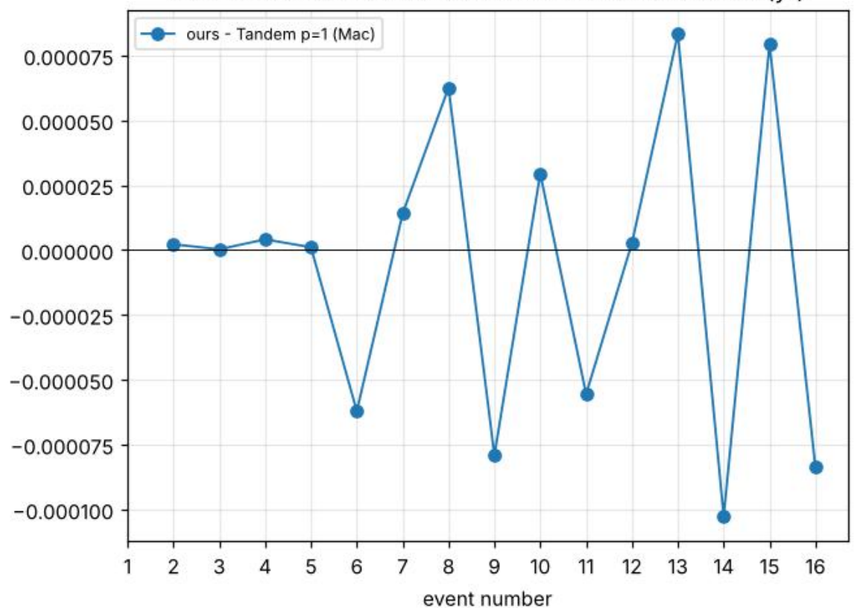
code	from ev 2	from ev 3	ours-code onset ev1	ev16	slope (yr/event)
Tandem p=1 (Mac)	85.288	87.628	+0.0000	-0.0001	-0.00001
seas-mfem p=1	85.288	87.628	+0.0000	+0.0000	+0.00000

(mean recurrence from ev 2 = mean of onset(k+1) - onset(k) for k >= 2, i.e. intervals ev2->ev3 ... ev16; from ev 3 likewise)

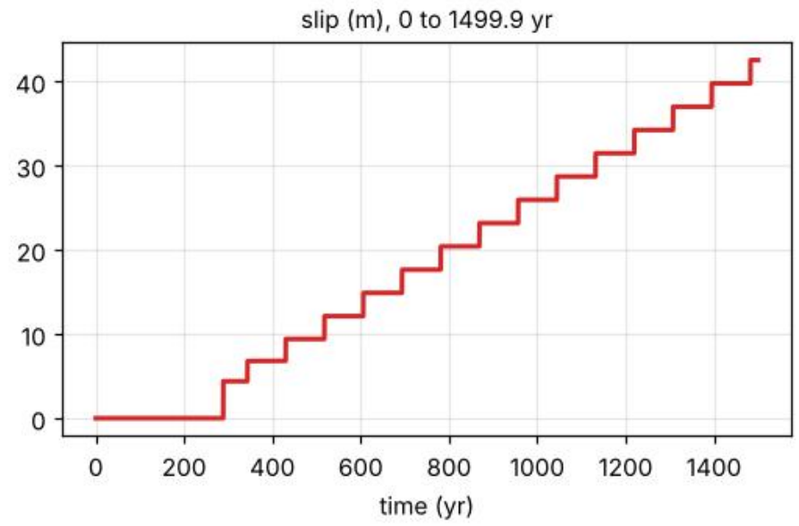
BP3-QD dip 60 thrust, events 1 to 16: mean recurrence from ev 2 / ev 3 (yr): Tandem p=1 (Mac) 85.29/87.63, seas-mfem p=1 85.29/87.63

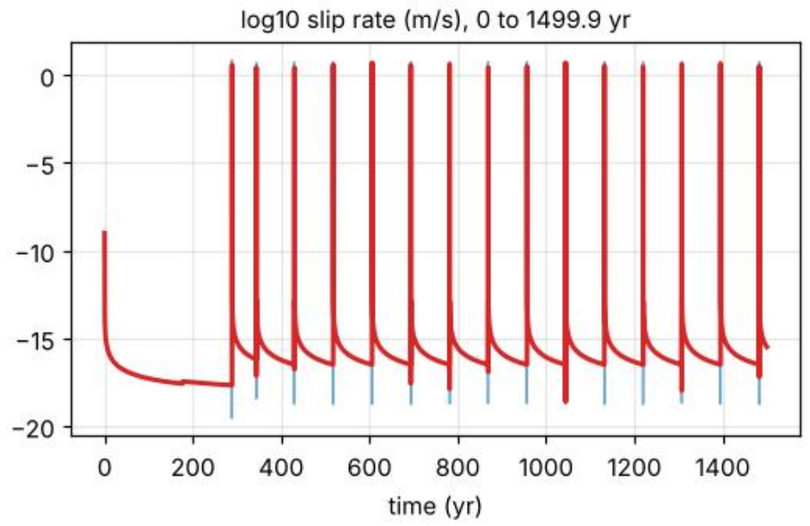


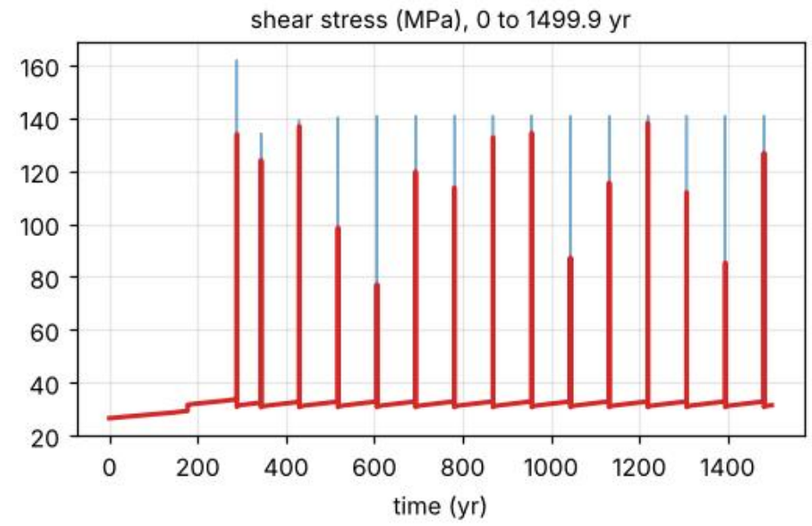


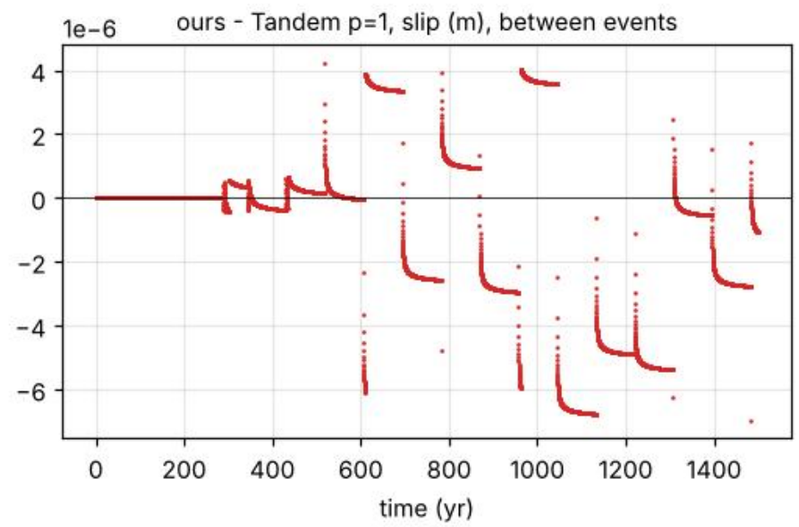


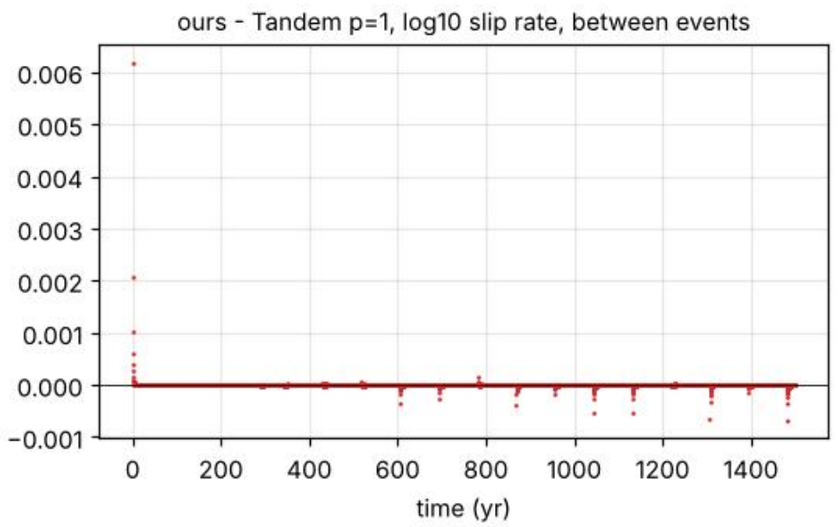
BP3-QD dip 60 thrust, fault station dp000 (0 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

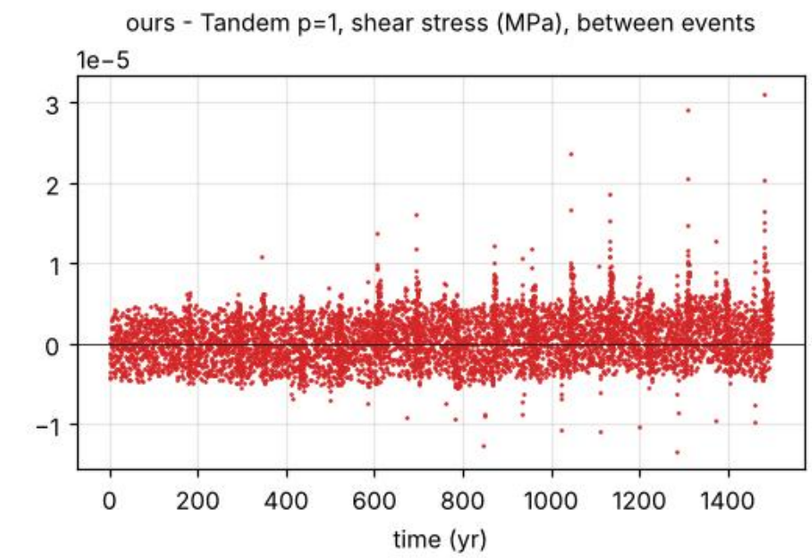


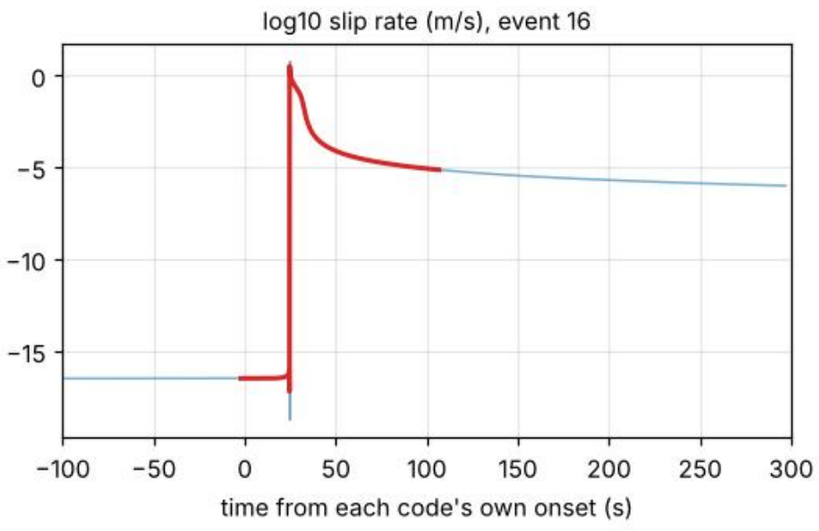


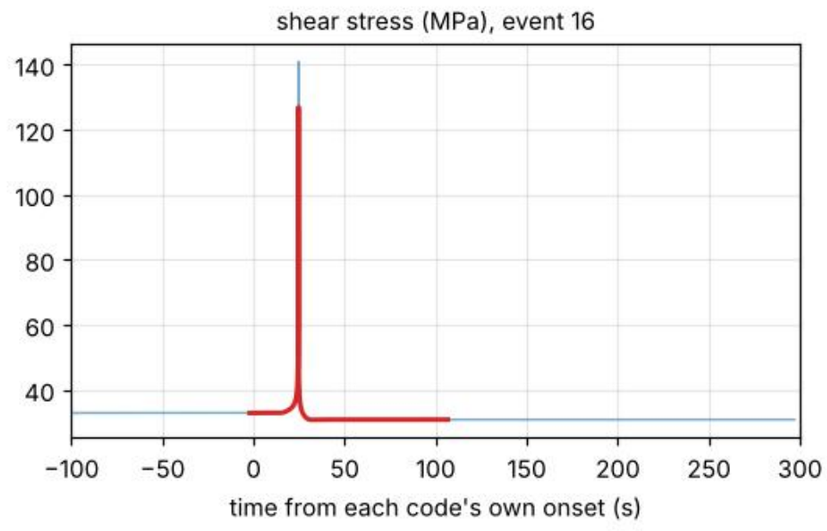


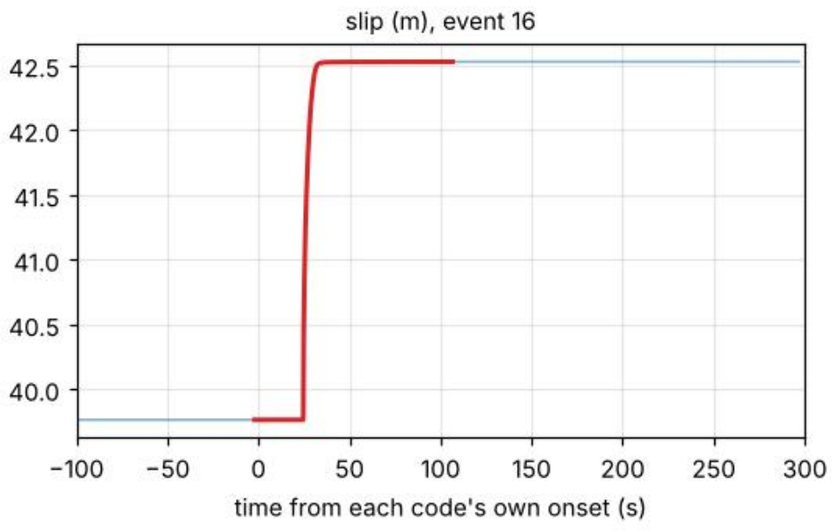


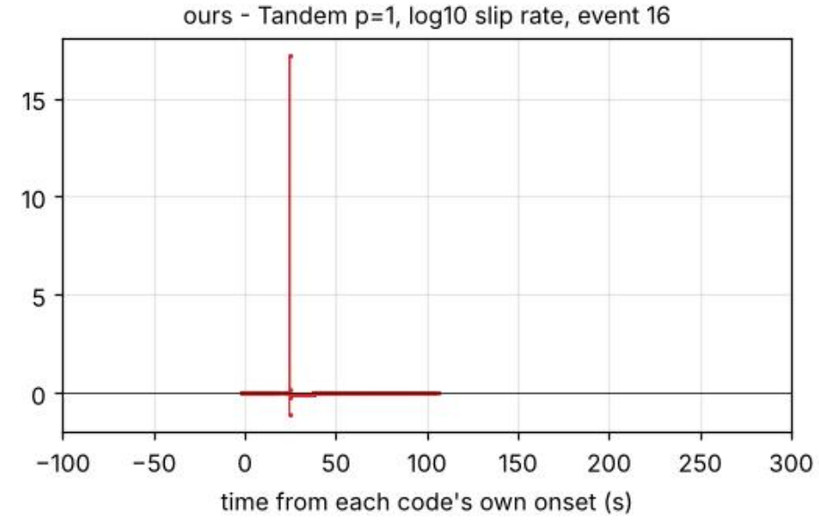


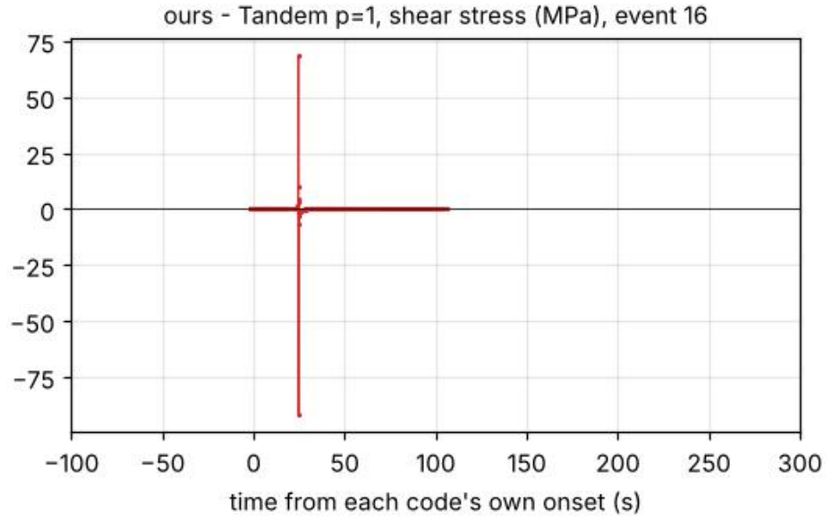


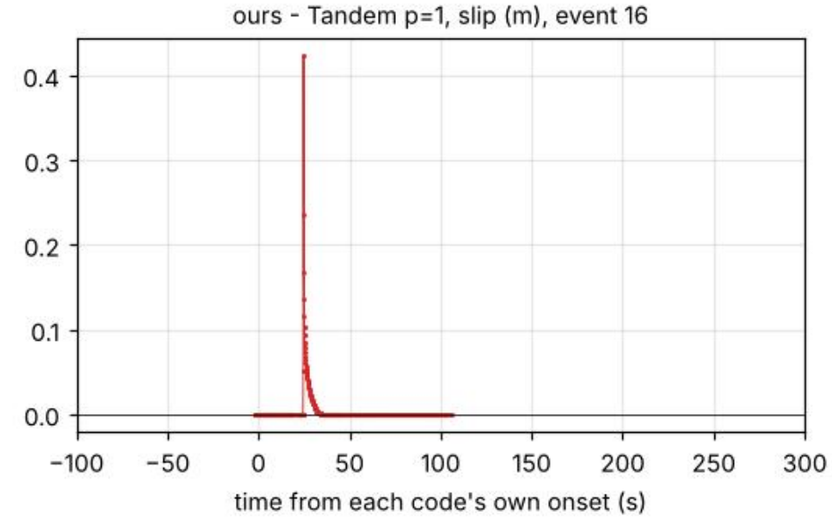






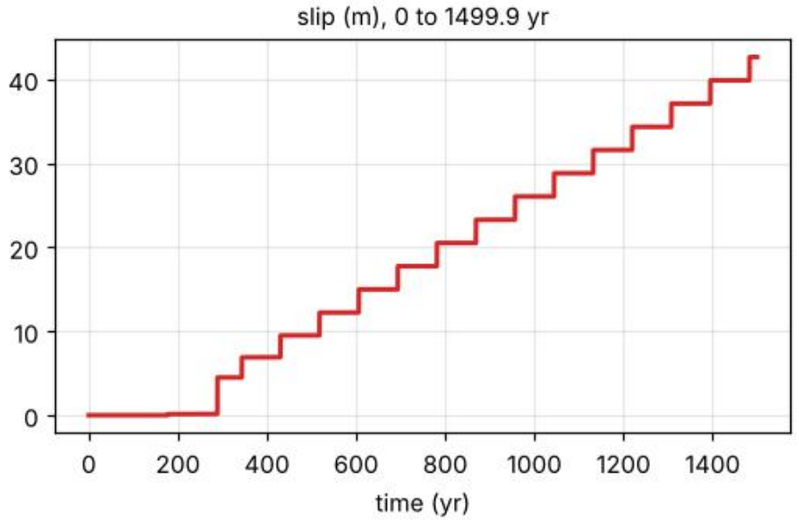


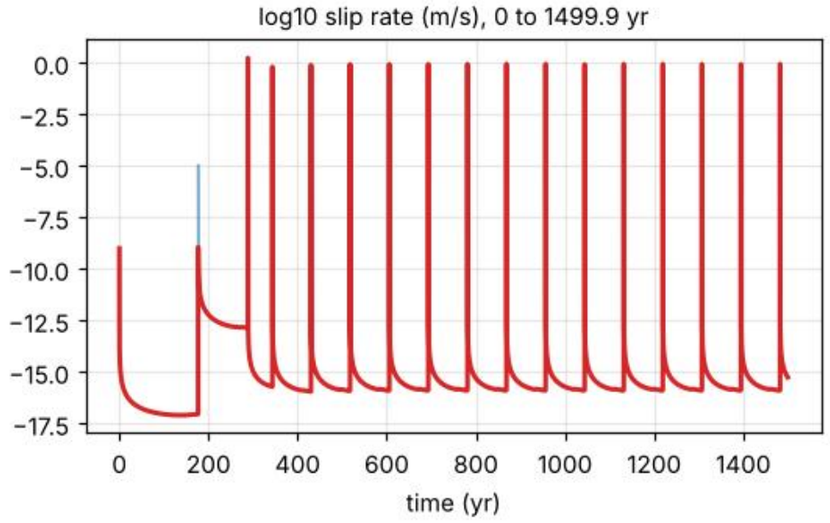


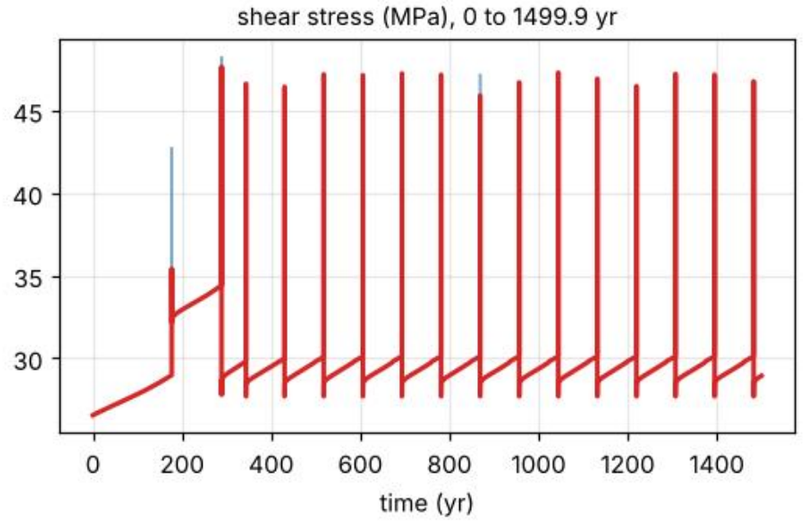


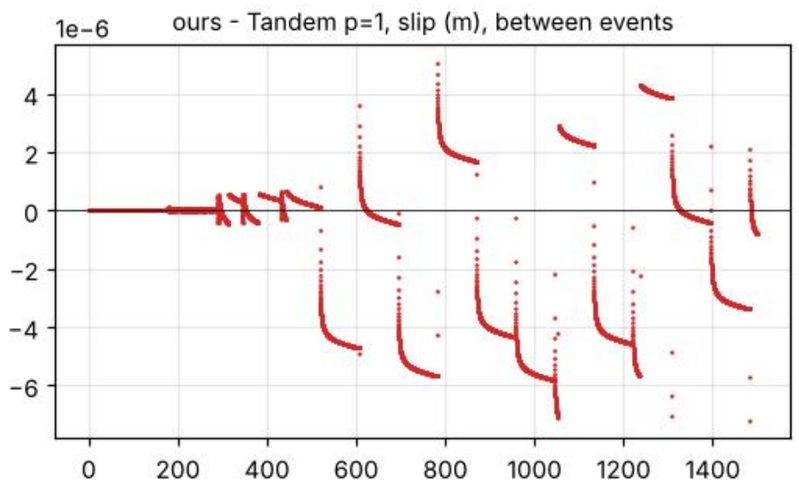


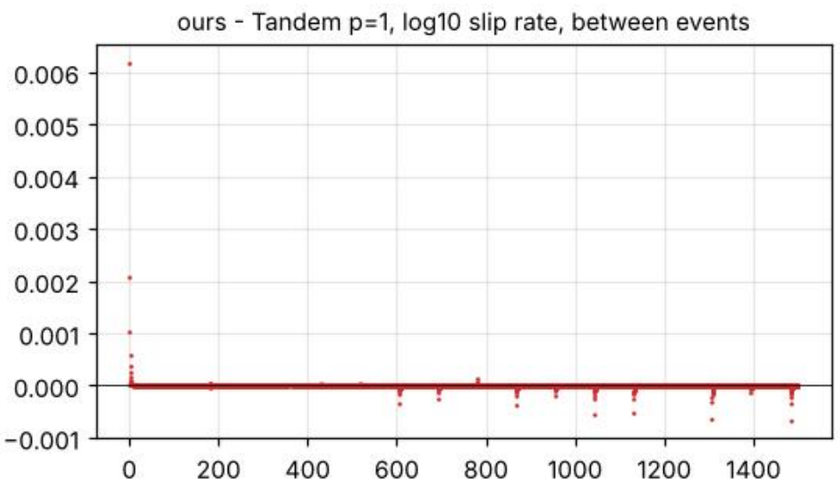
BP3-QD dip 60 thrust, fault station dp025 (2.5 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

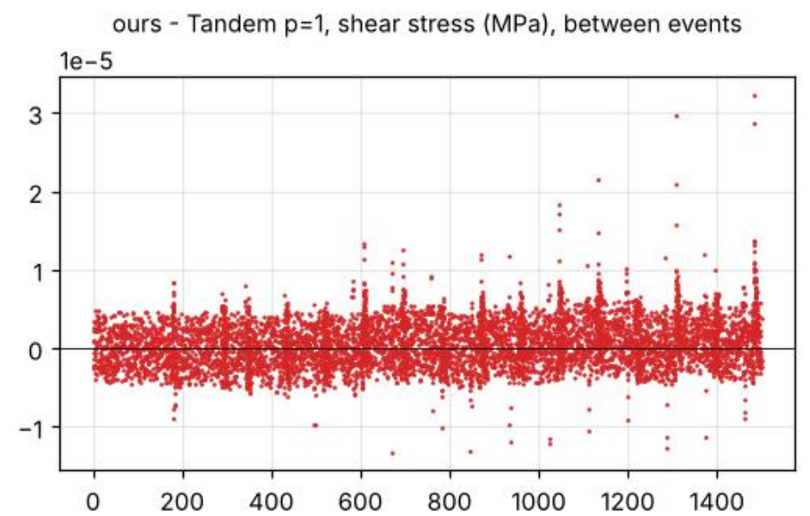


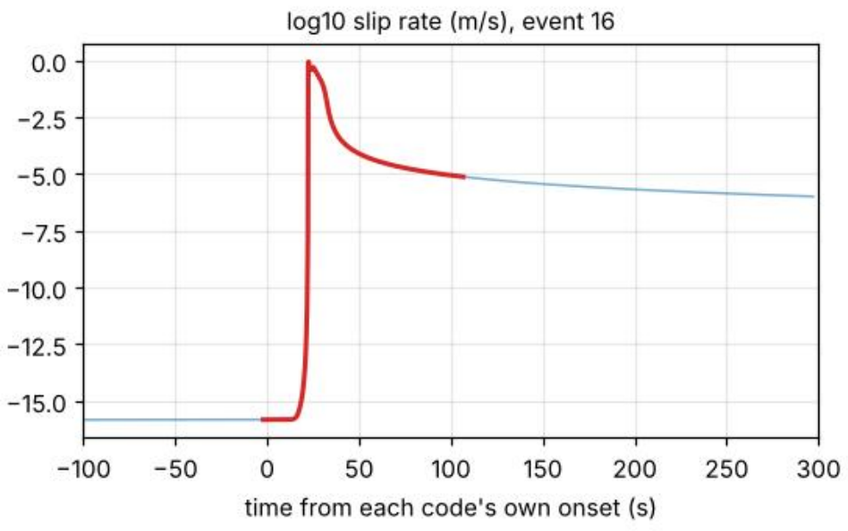


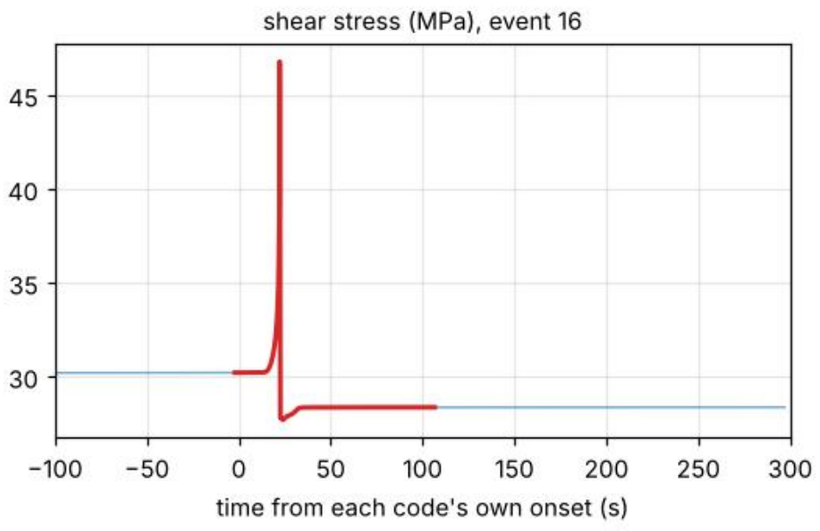


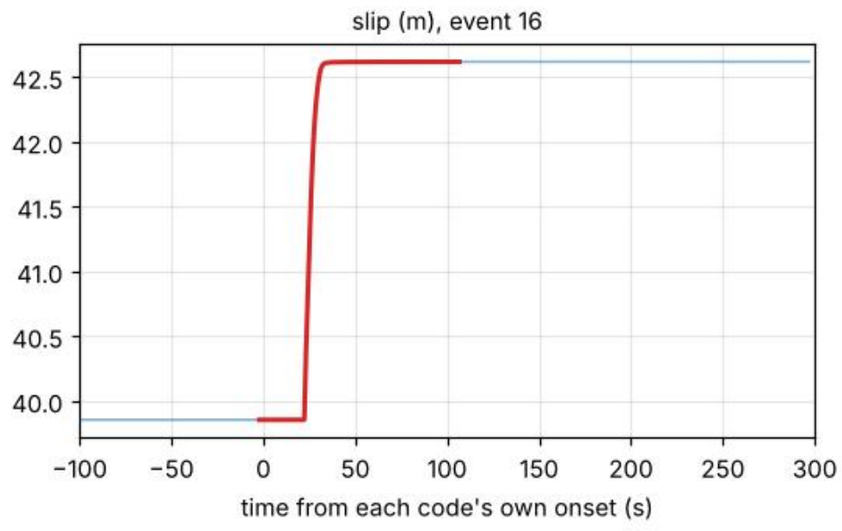


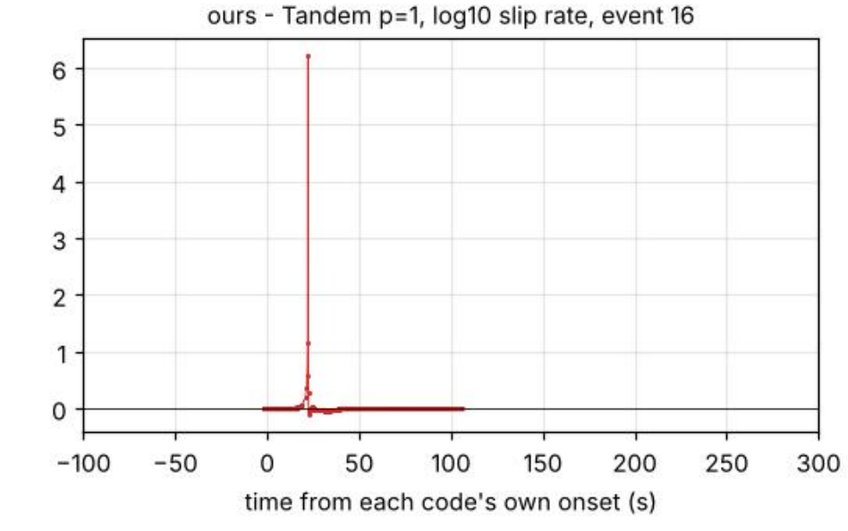


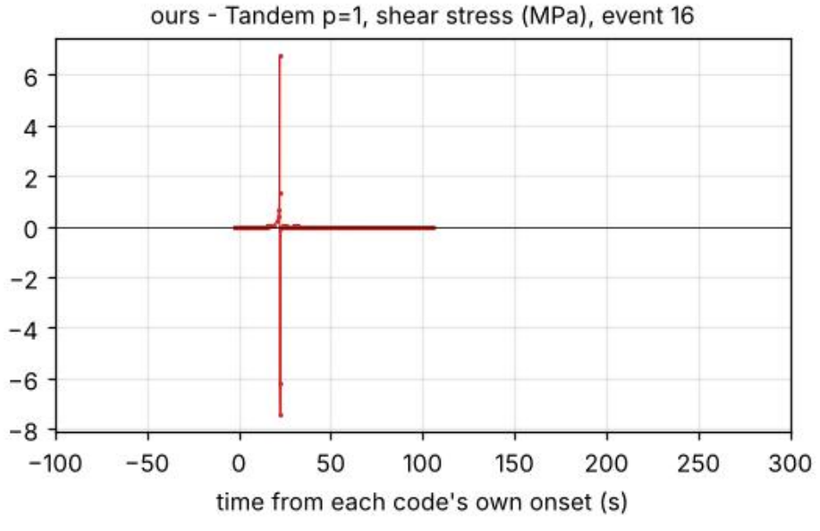


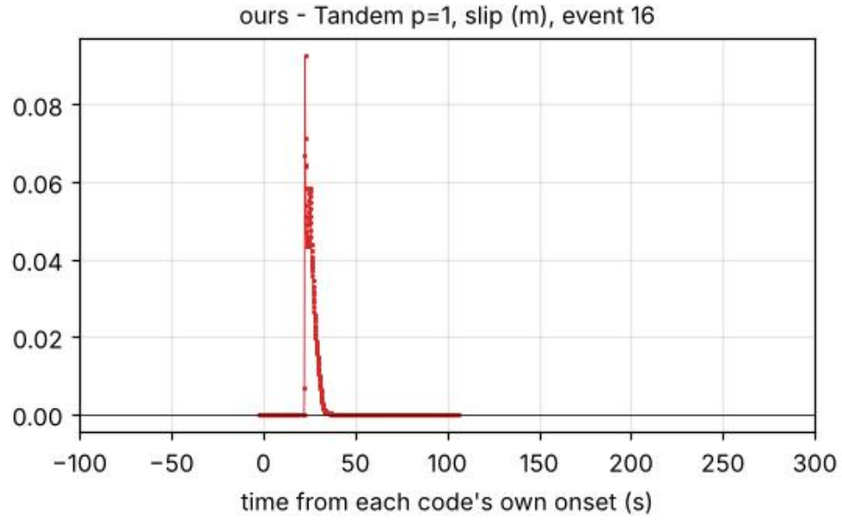






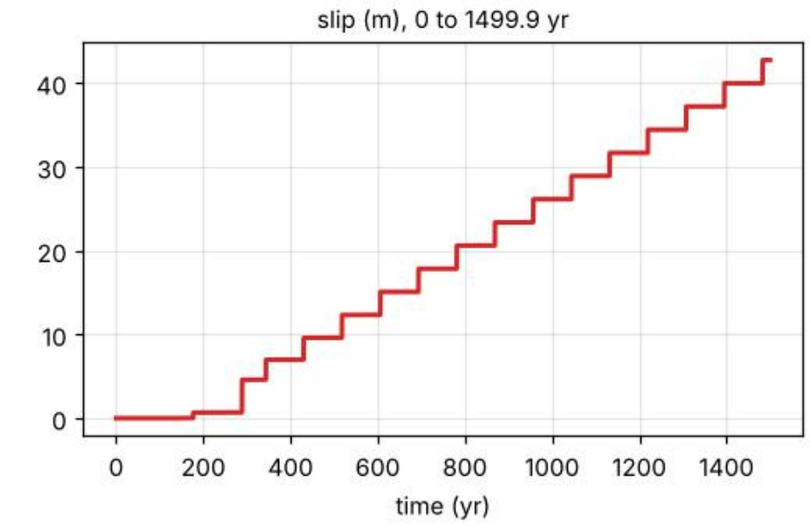


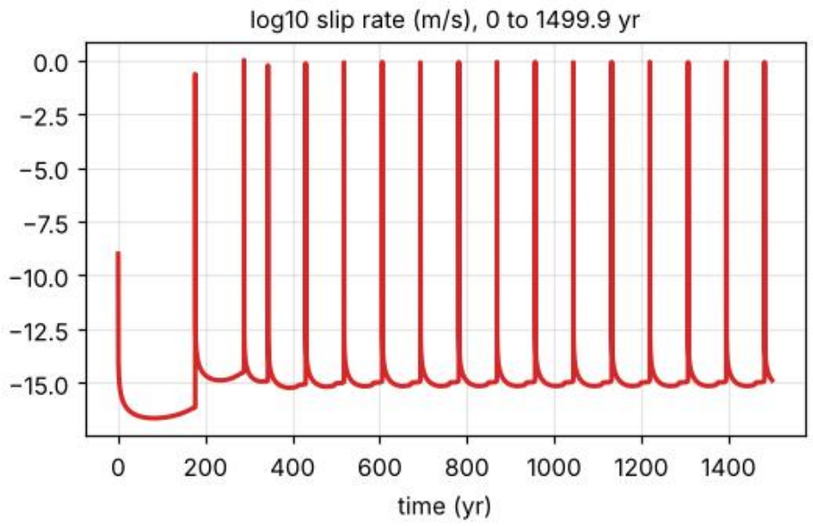


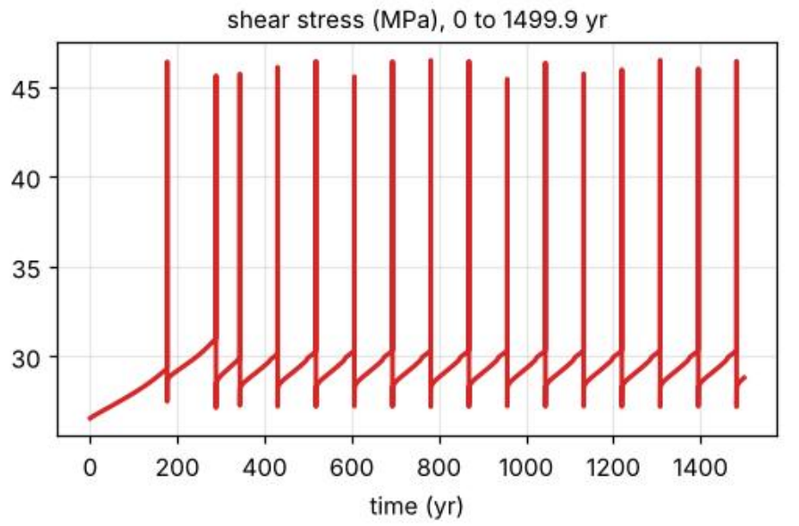


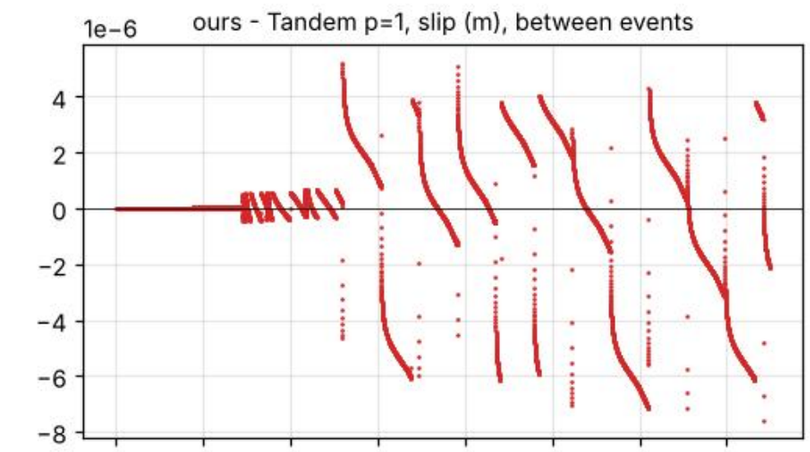


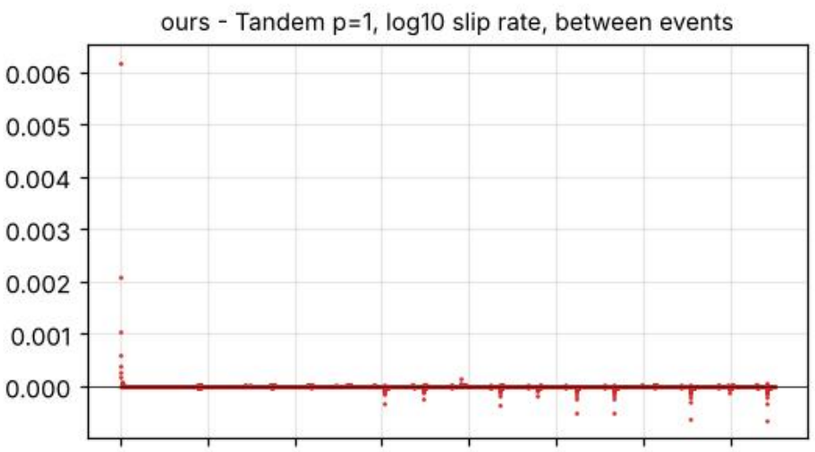
BP3-QD dip 60 thrust, fault station dp050 (5 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

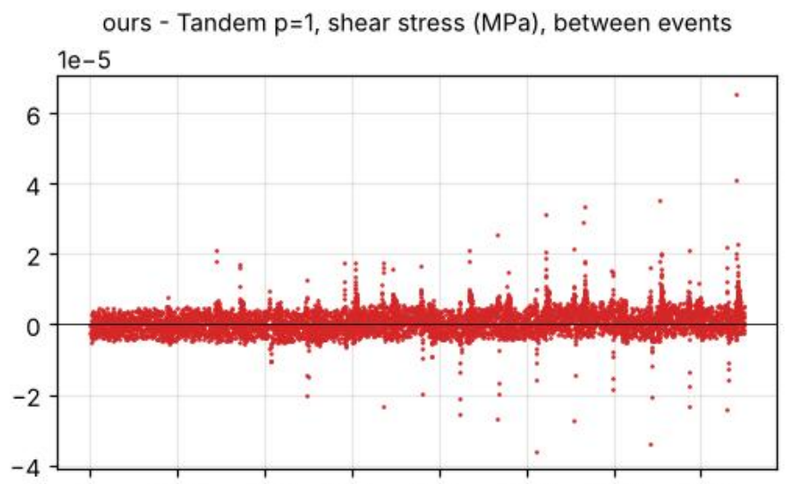


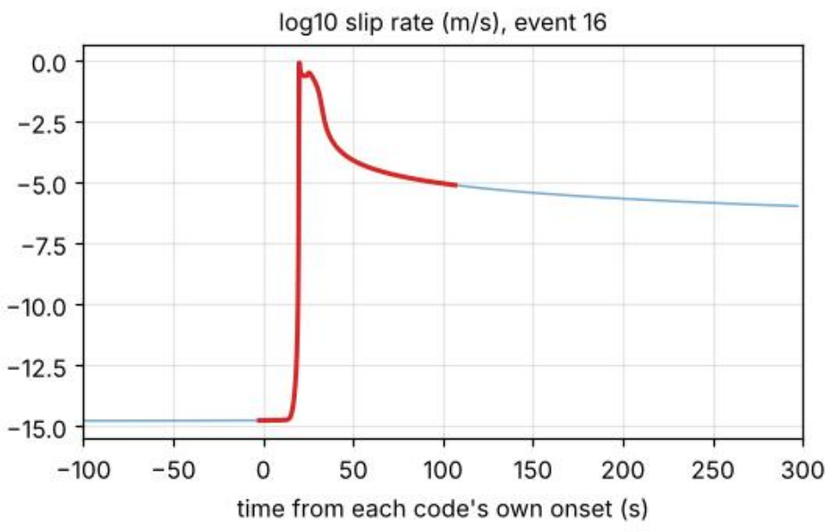


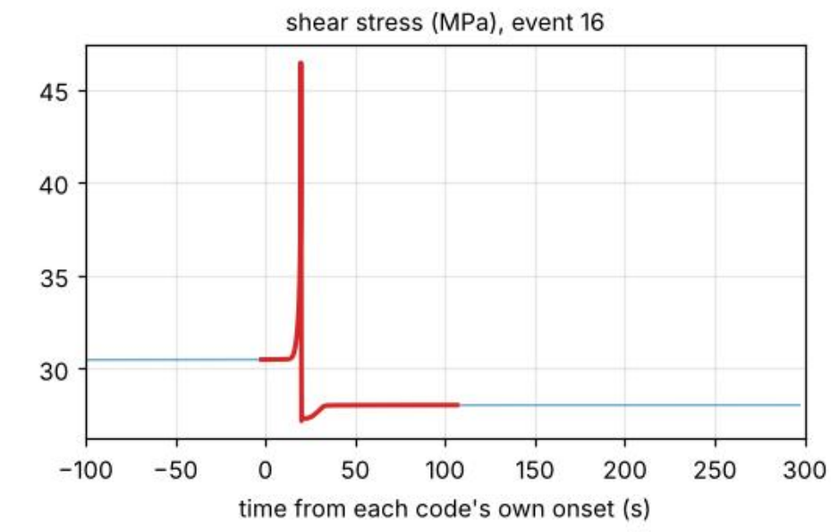


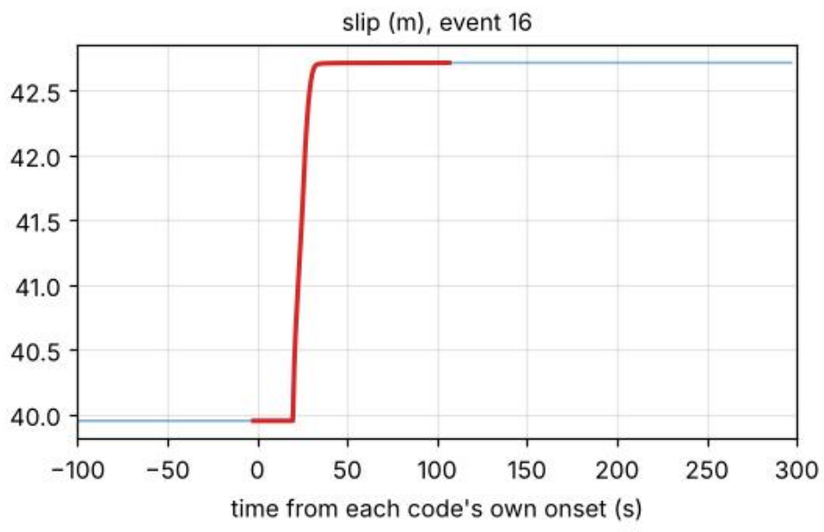


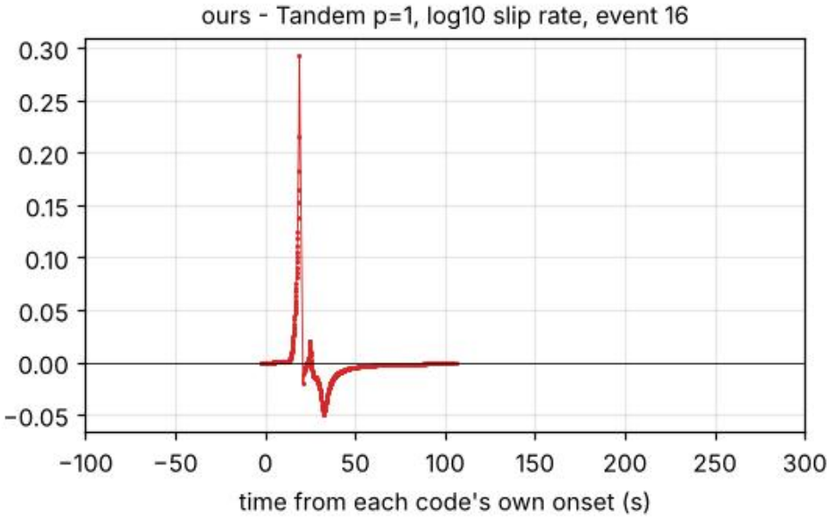


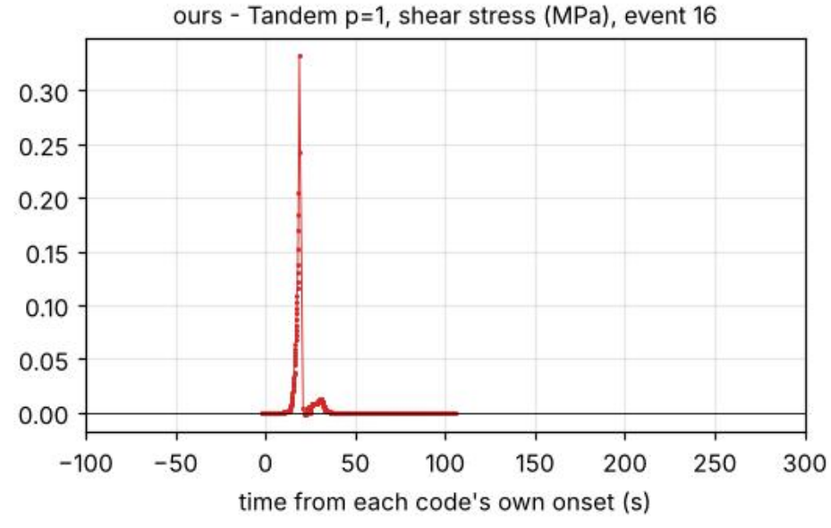


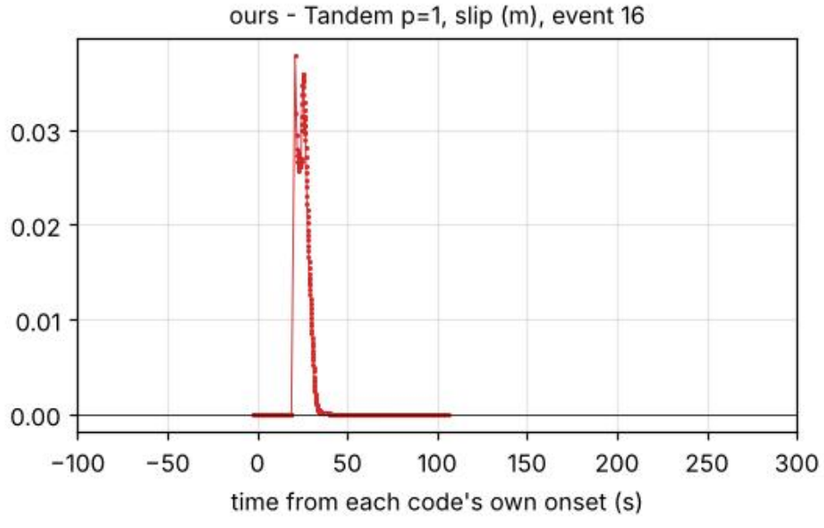






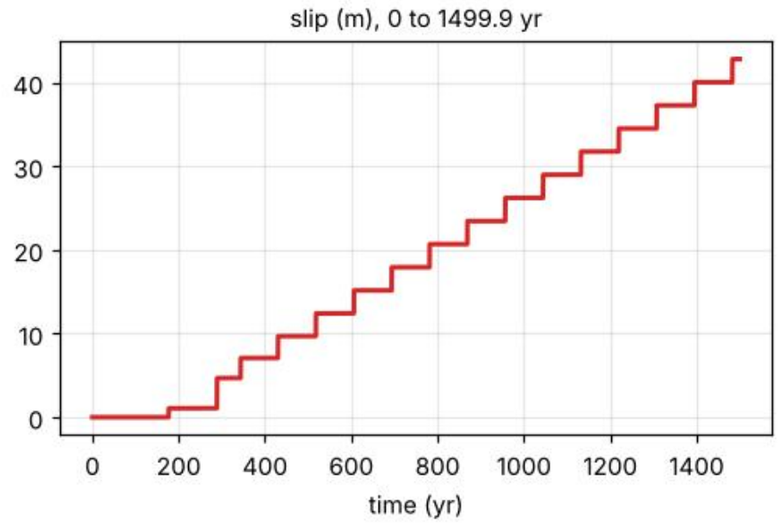


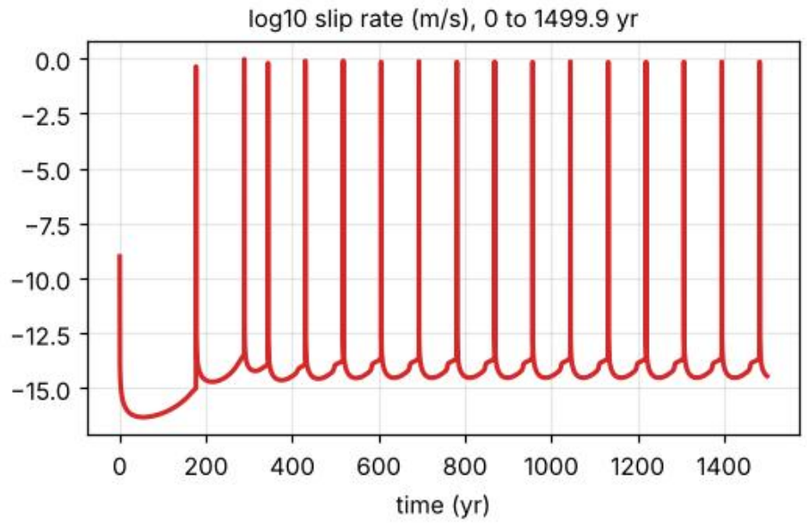


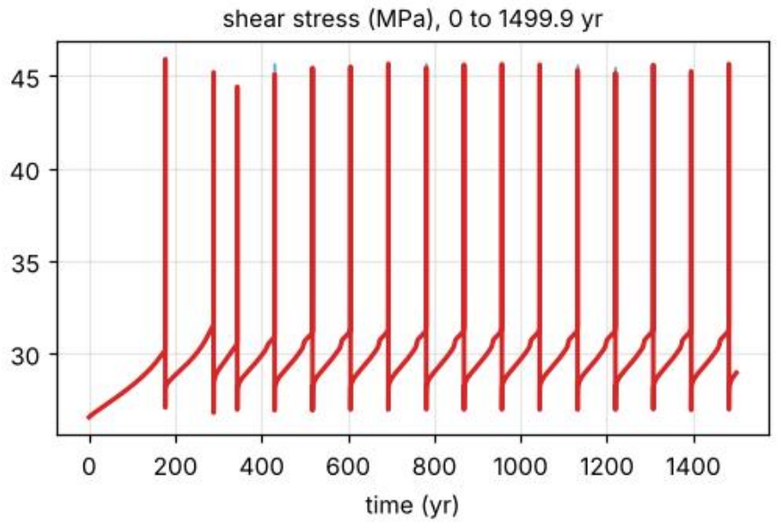


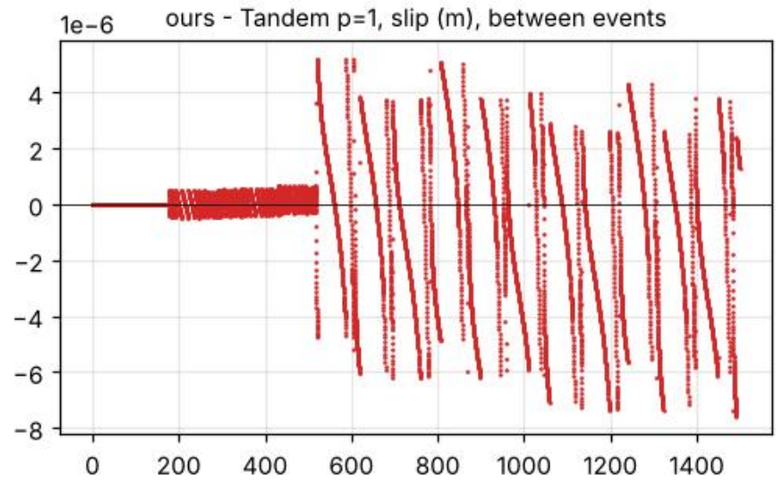


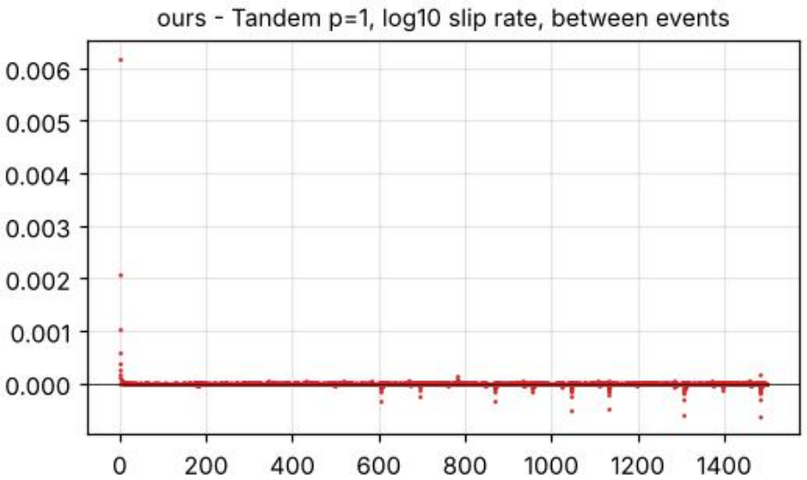
BP3-QD dip 60 thrust, fault station dp075 (7.5 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

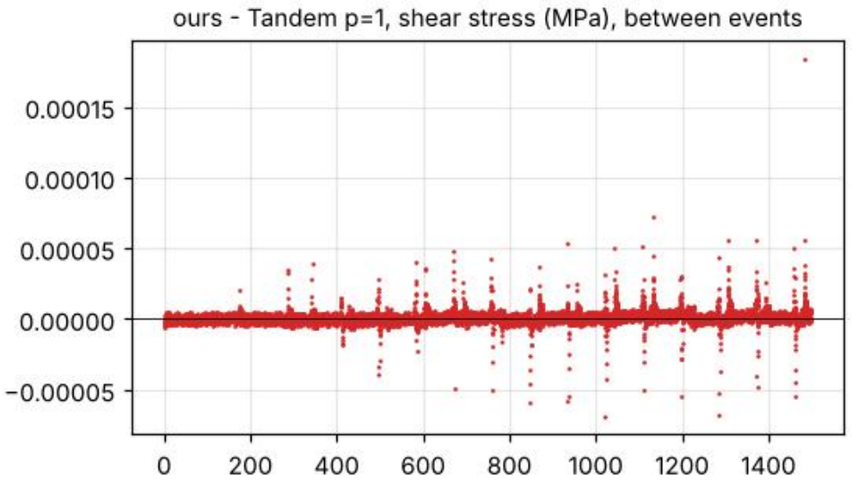


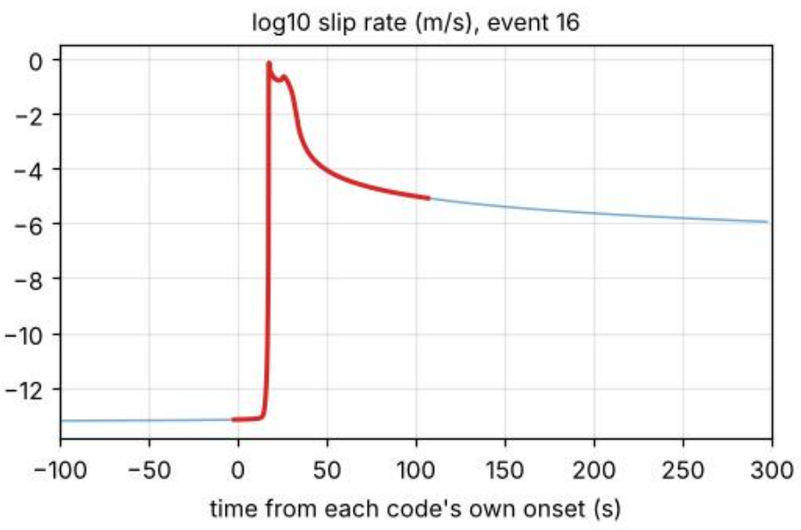


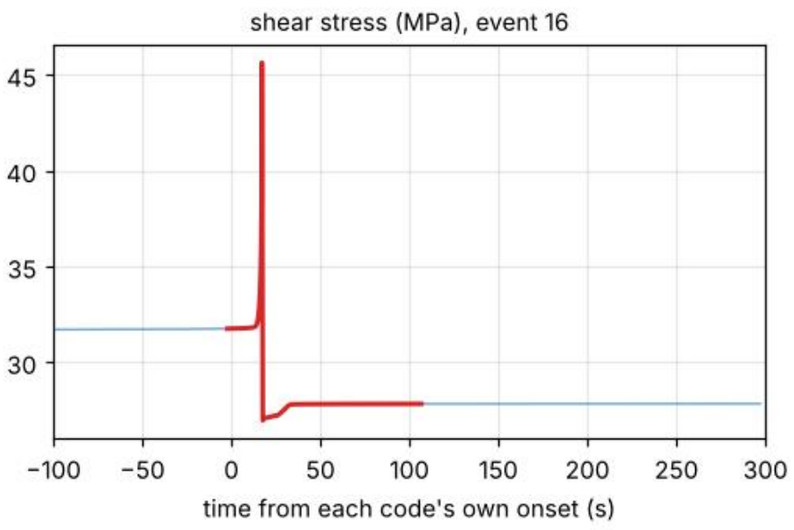


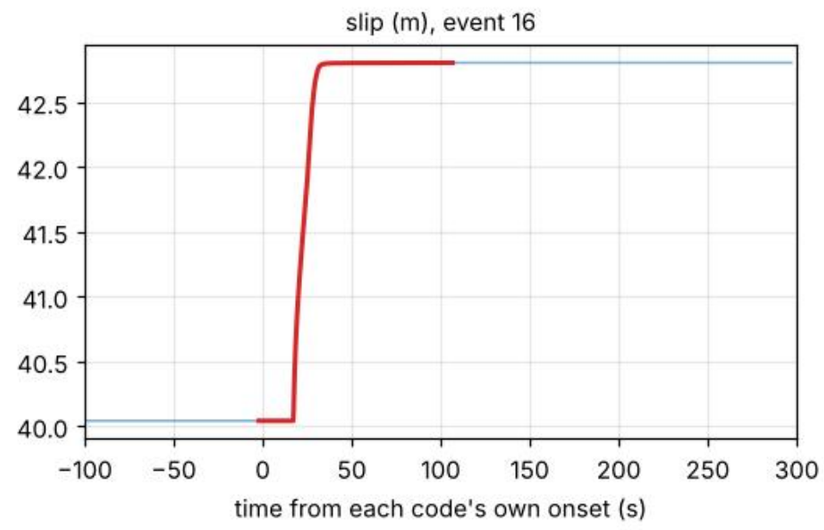


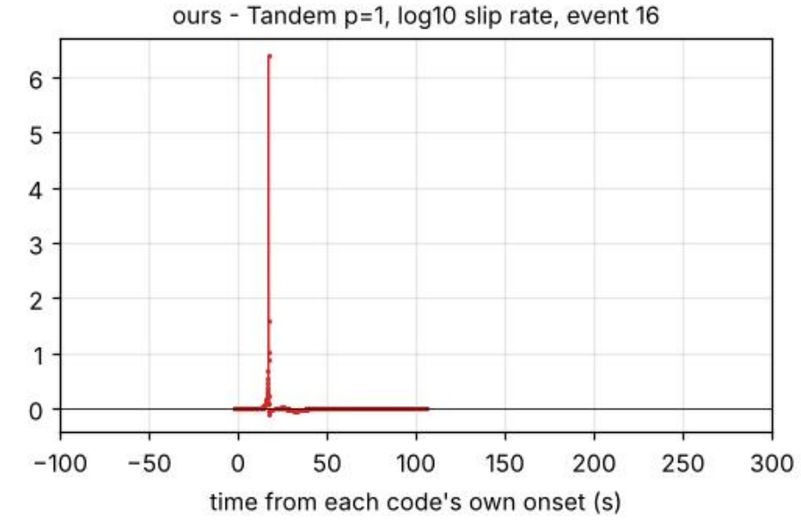


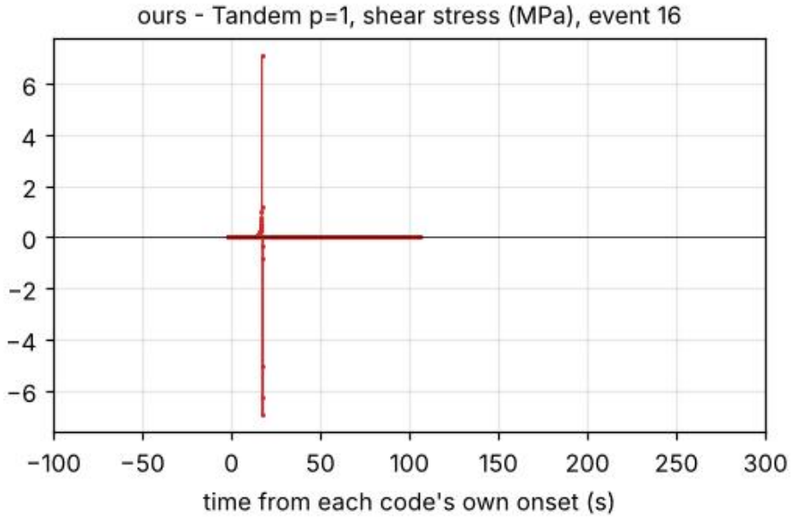


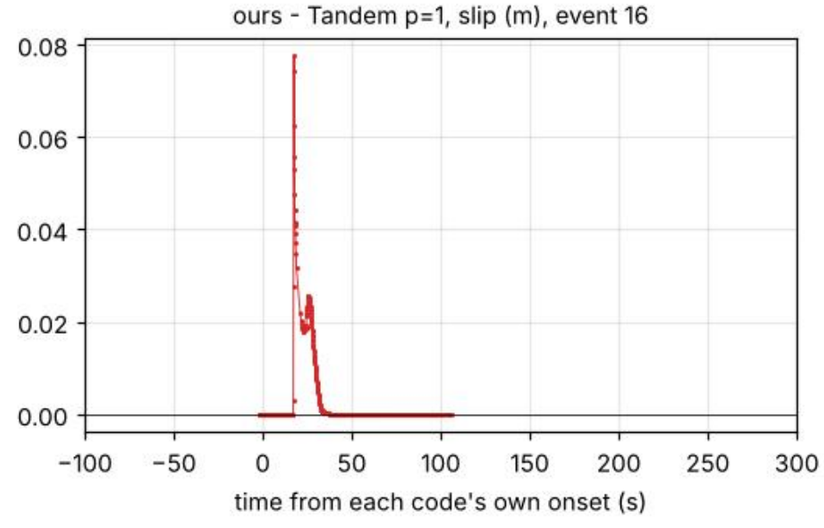






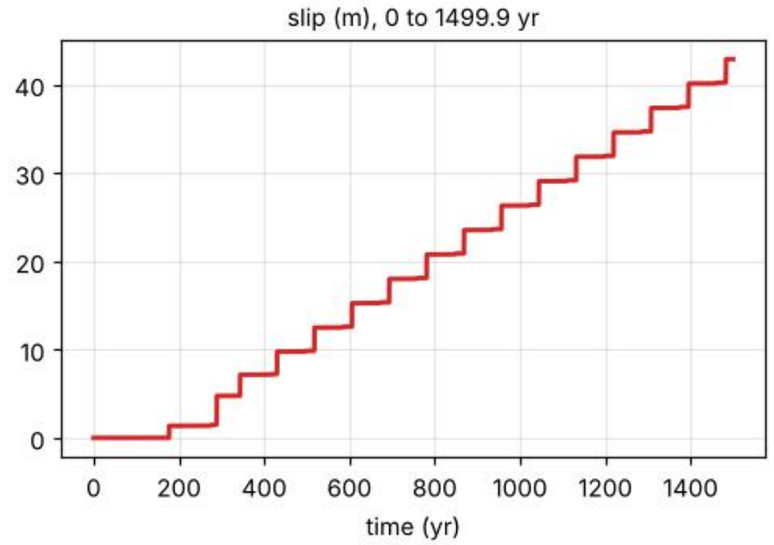


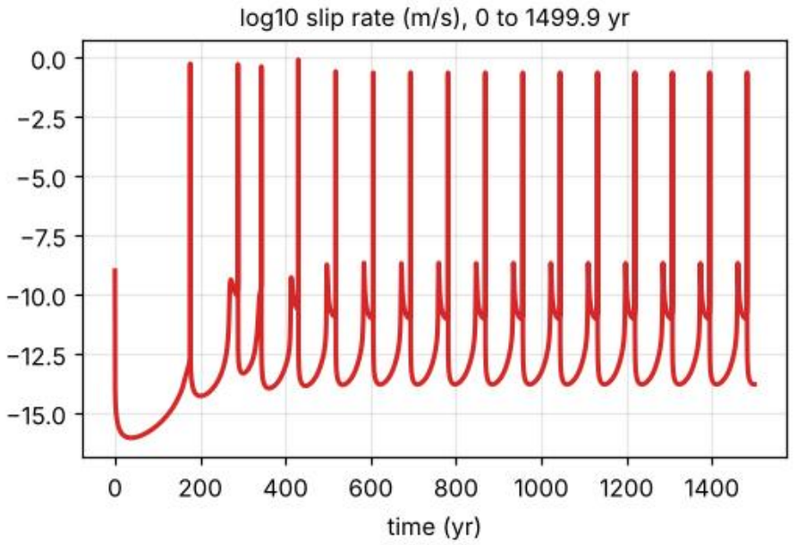


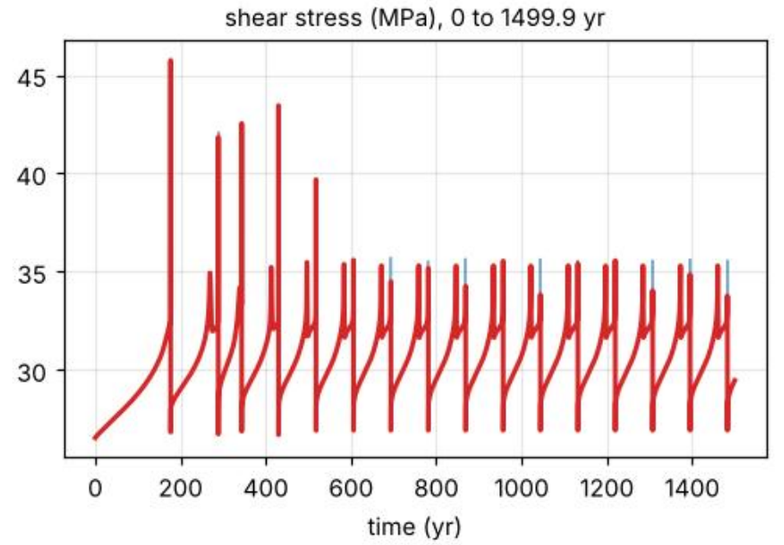


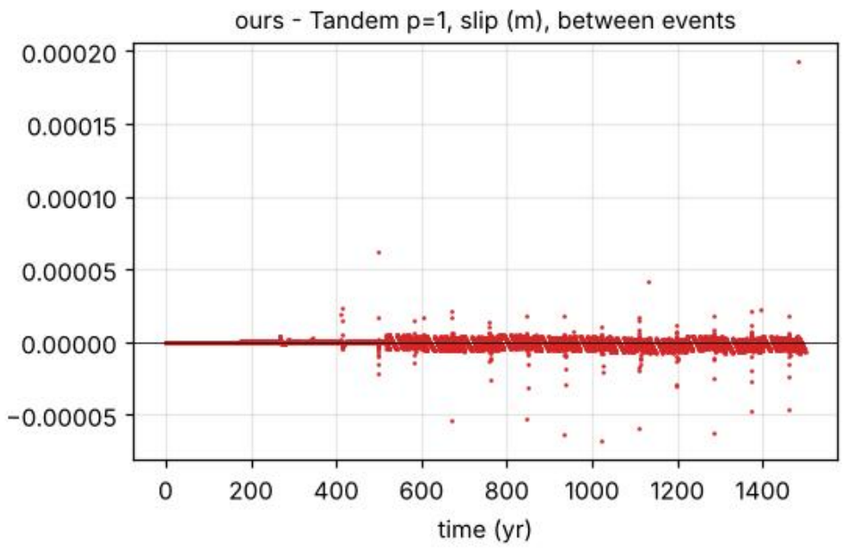


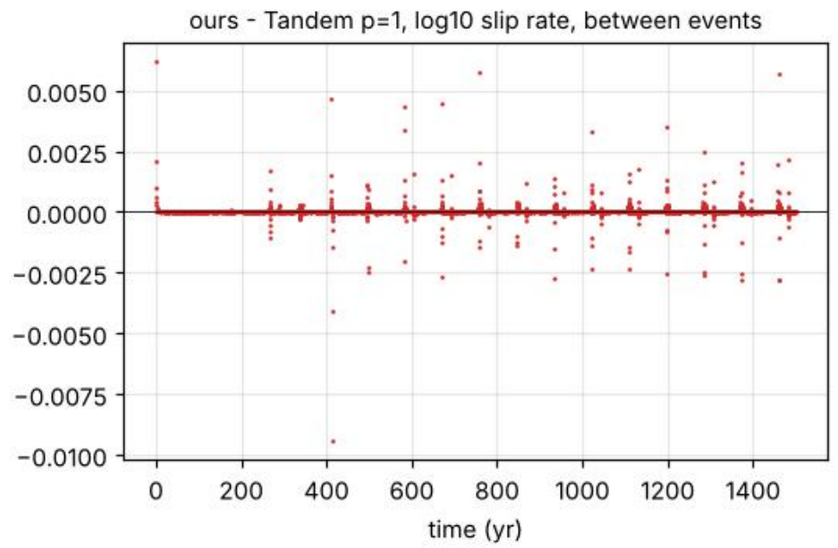
BP3-QD dip 60 thrust, fault station dp100 (10 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

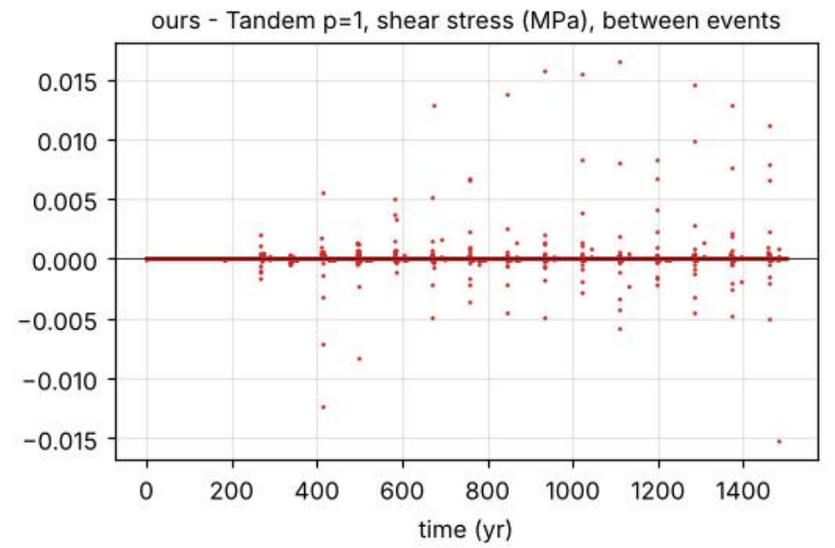


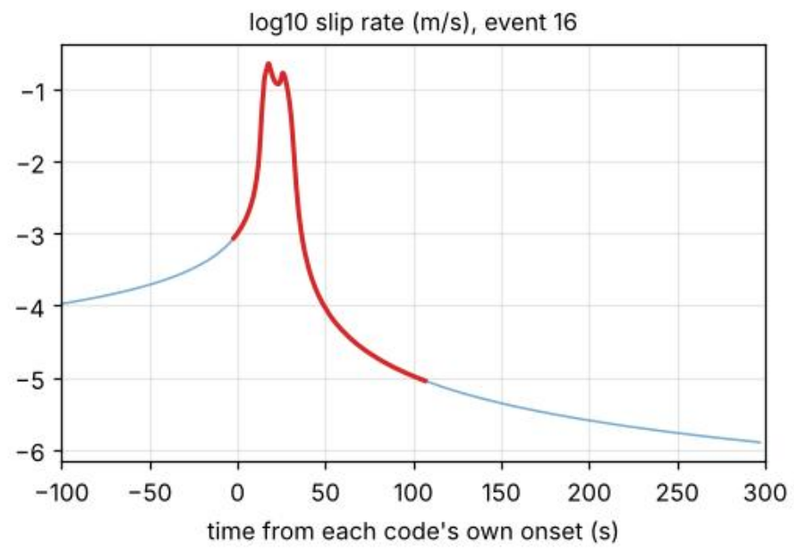


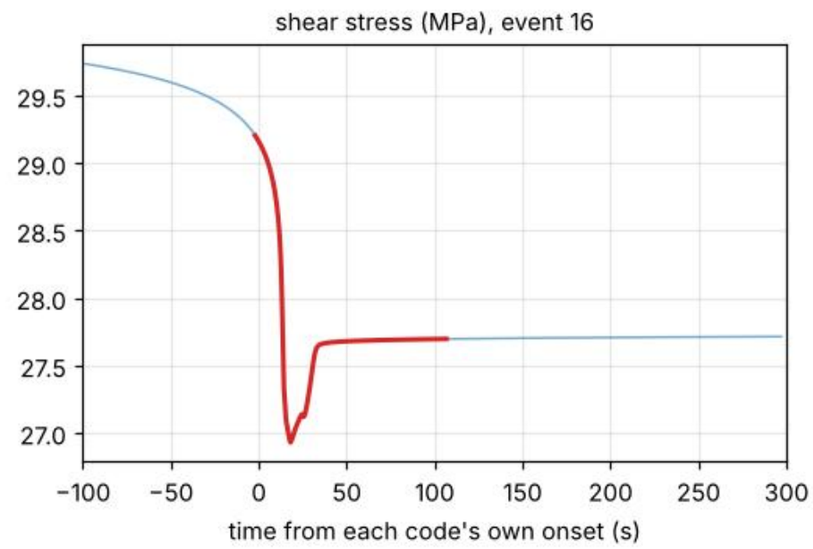


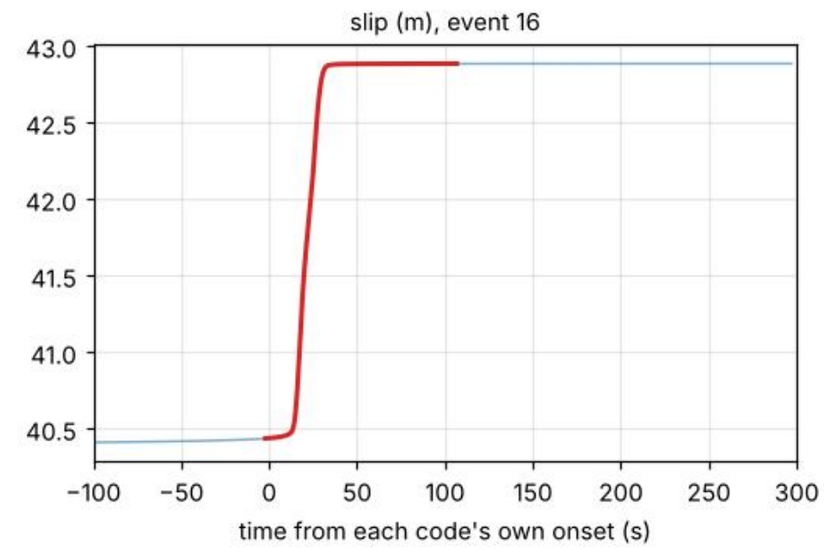


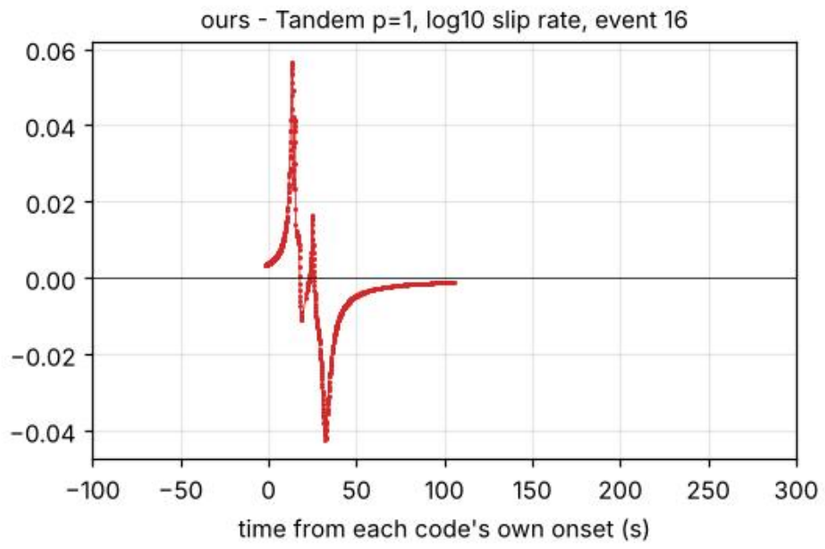


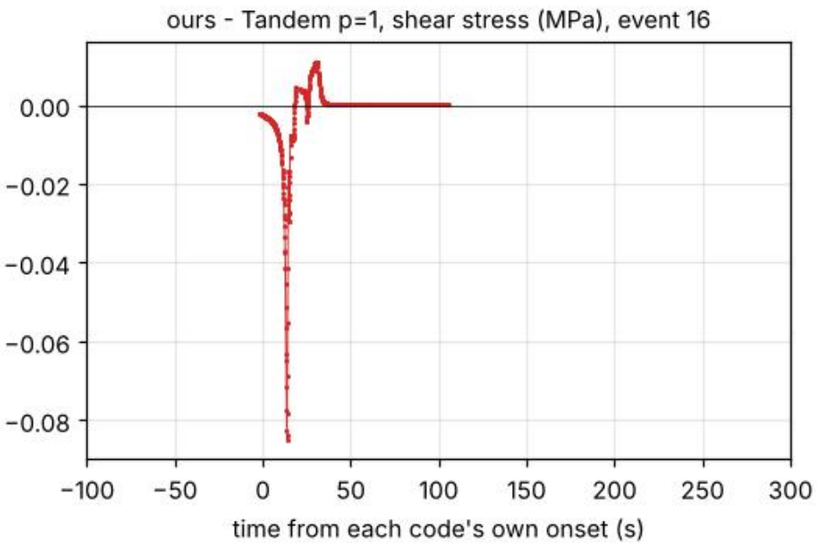


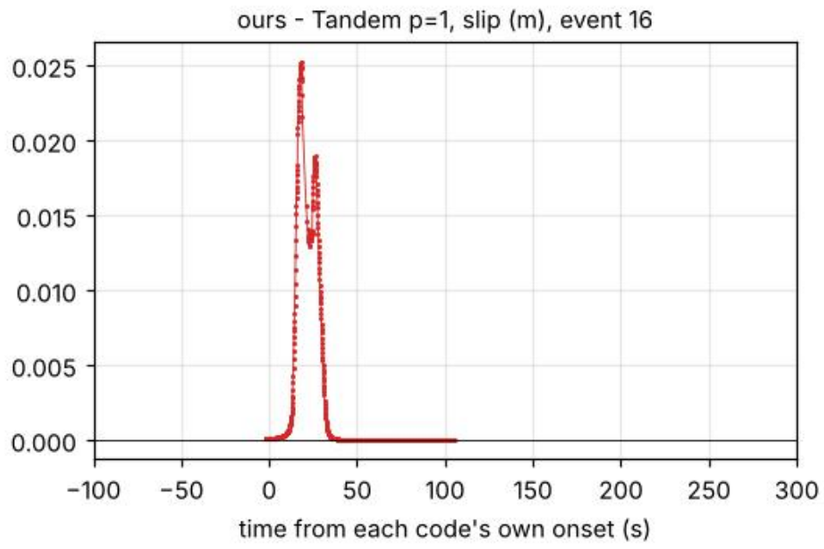






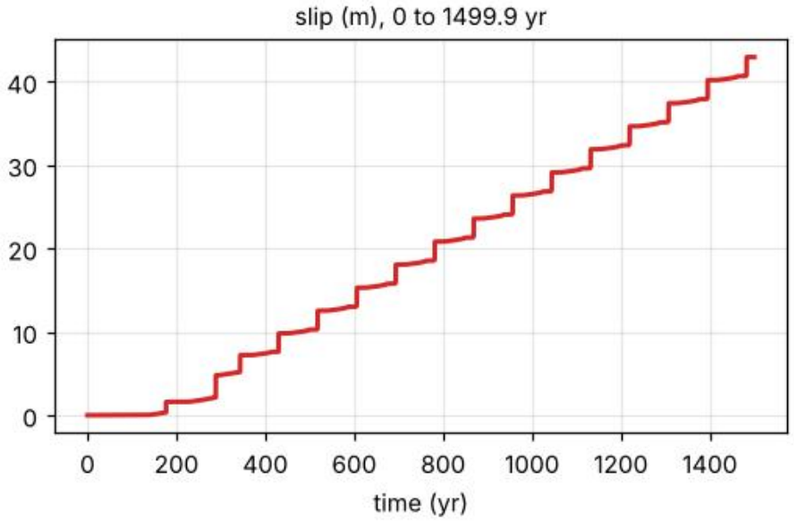


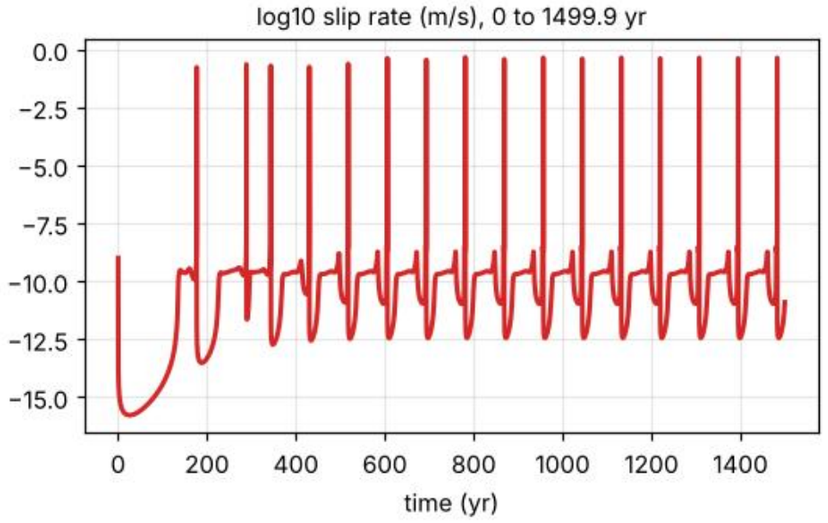


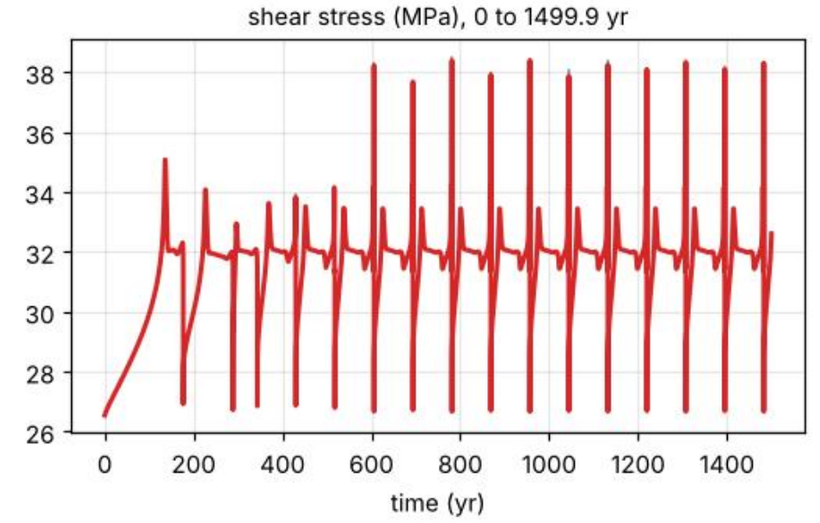


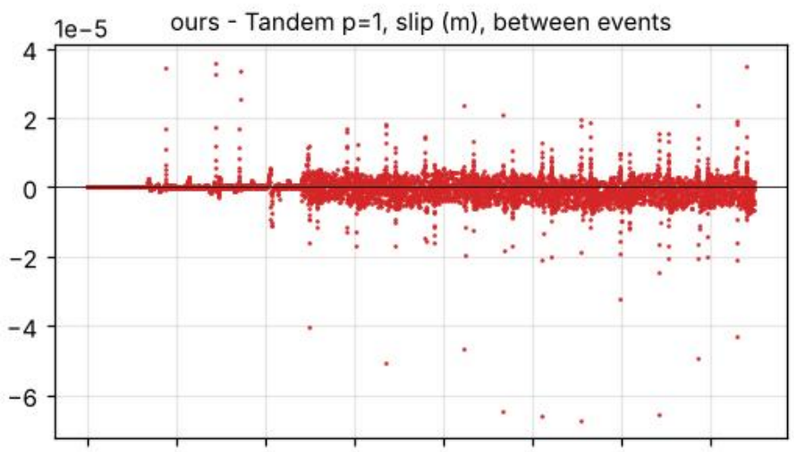
Tandem p=1 (Mac) seas-mfem p=1

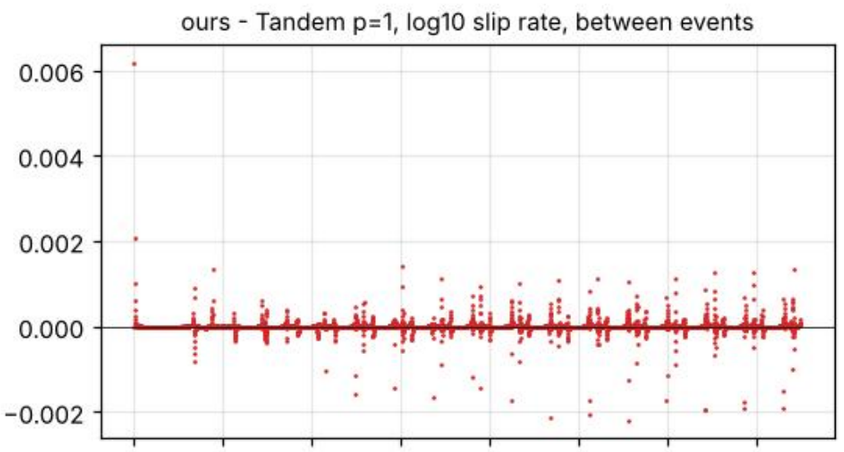
BP3-QD dip 60 thrust, fault station dp125 (12.5 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

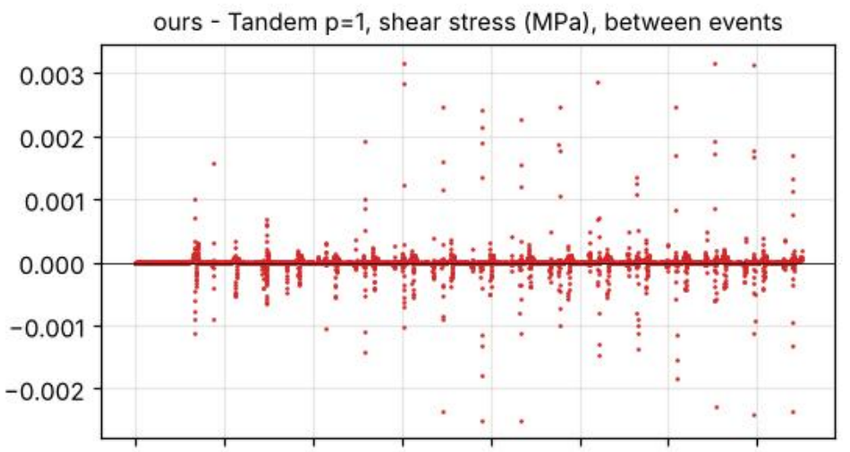


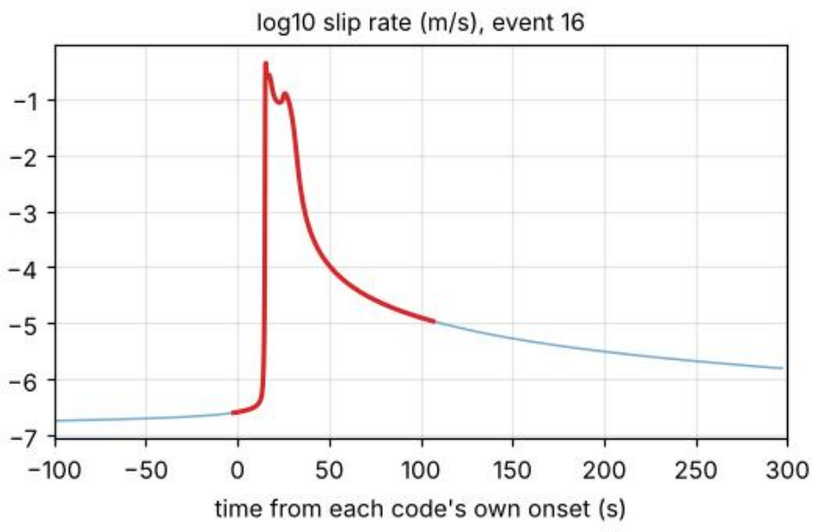


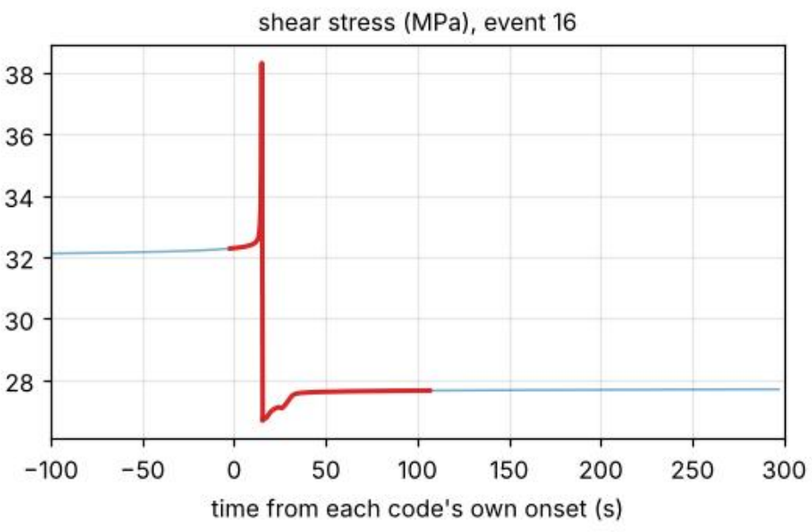


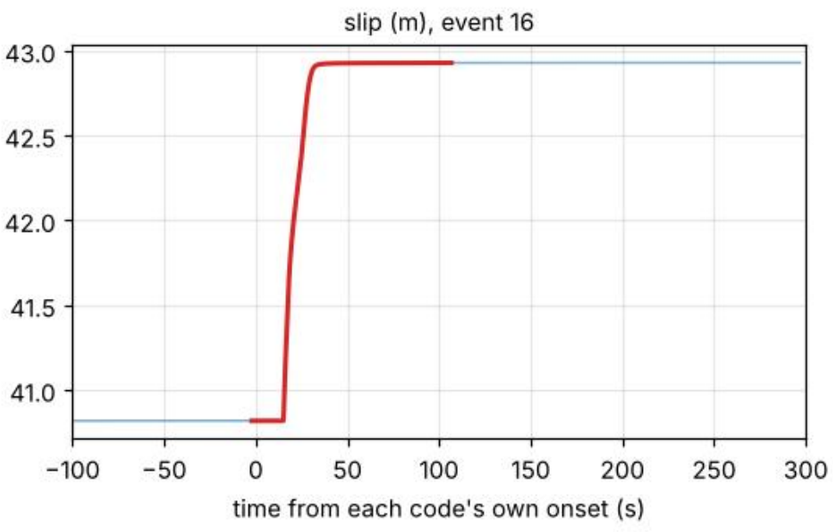


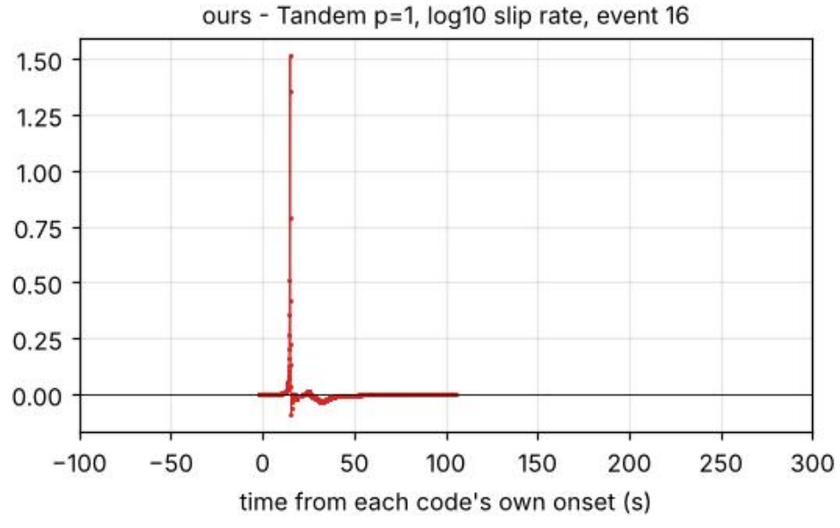


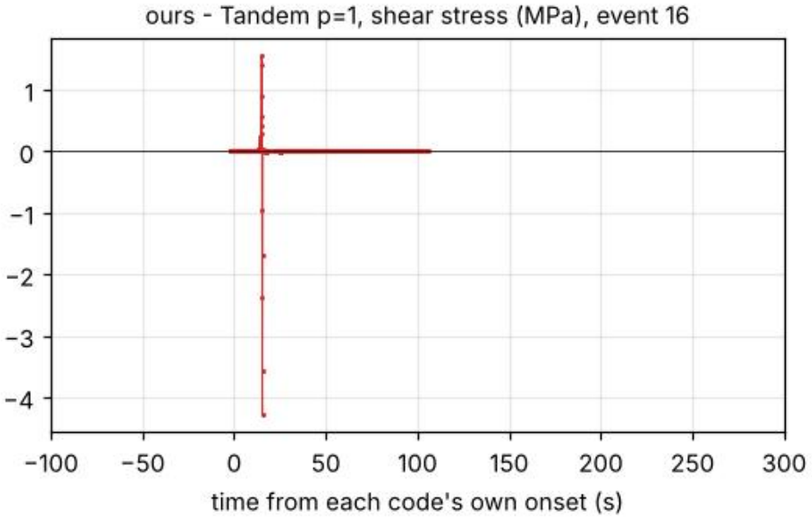


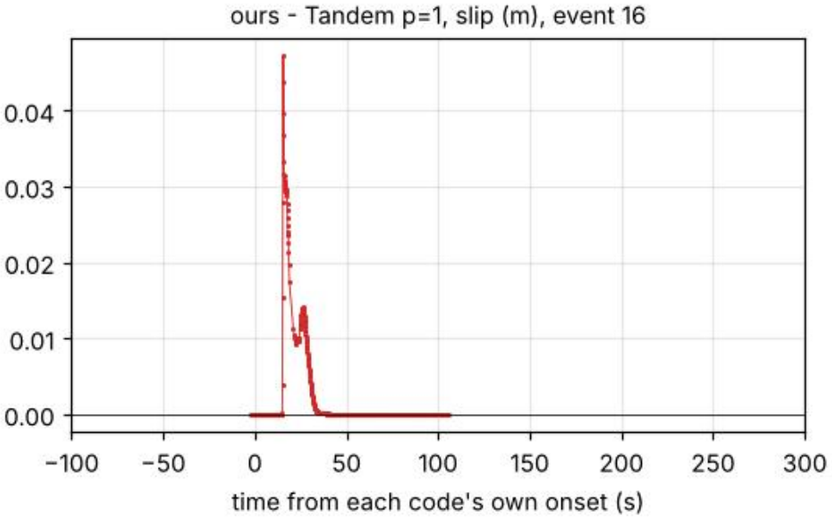






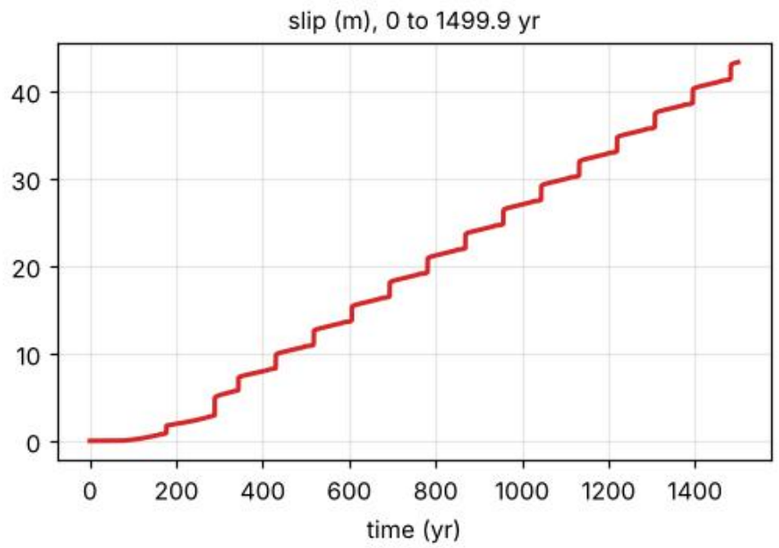


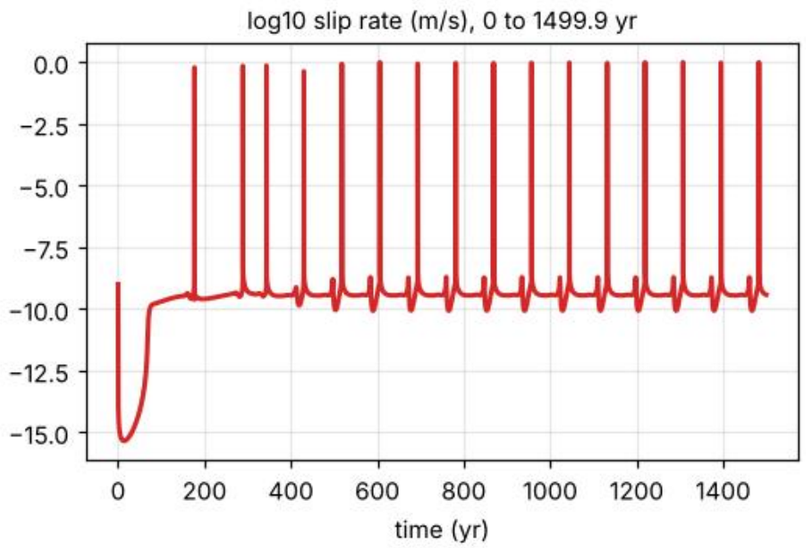


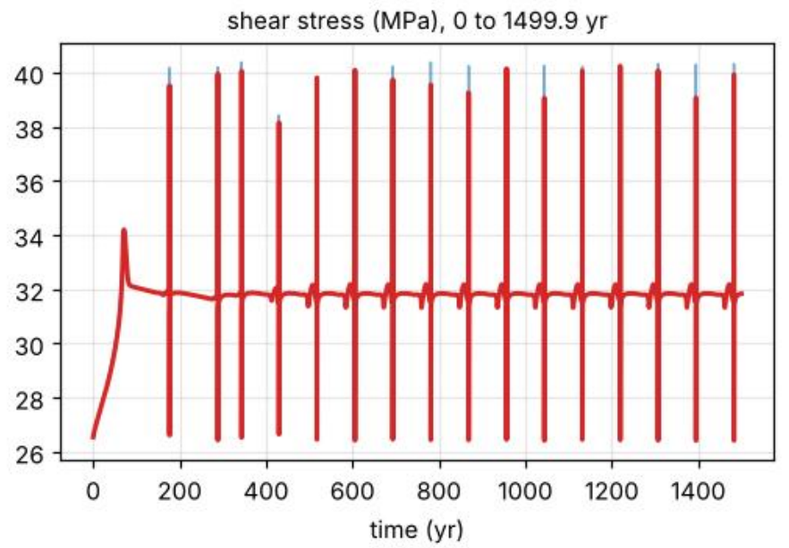


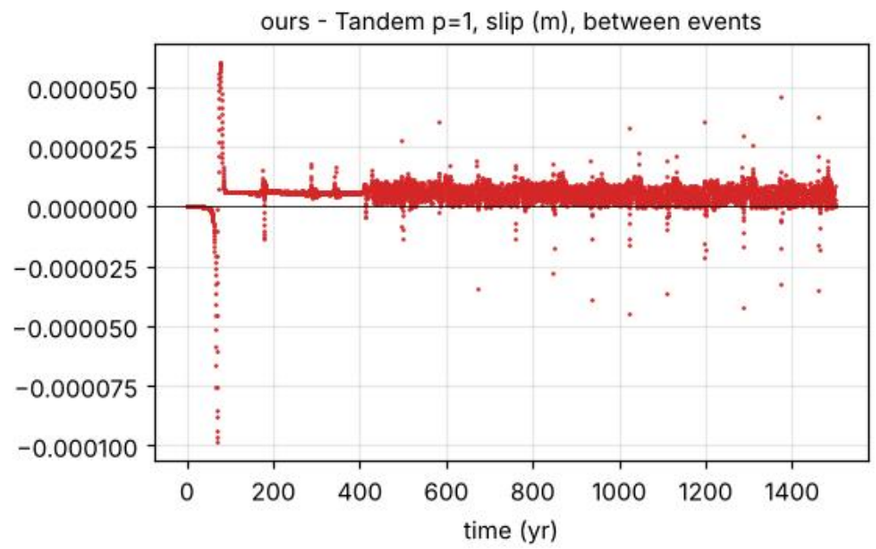


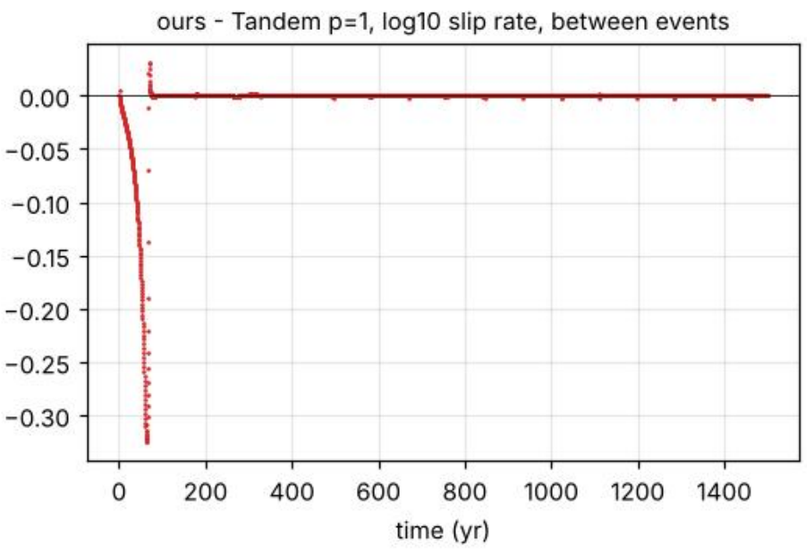
BP3-QD dip 60 thrust, fault station dp150 (15 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

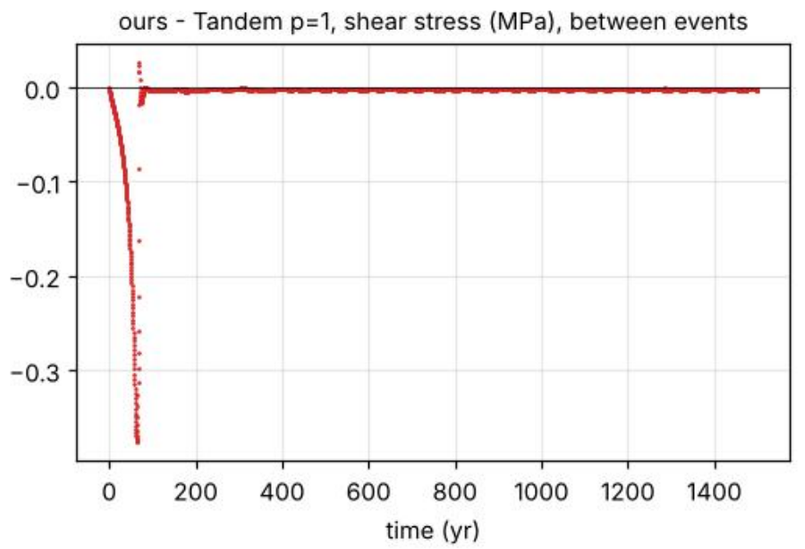


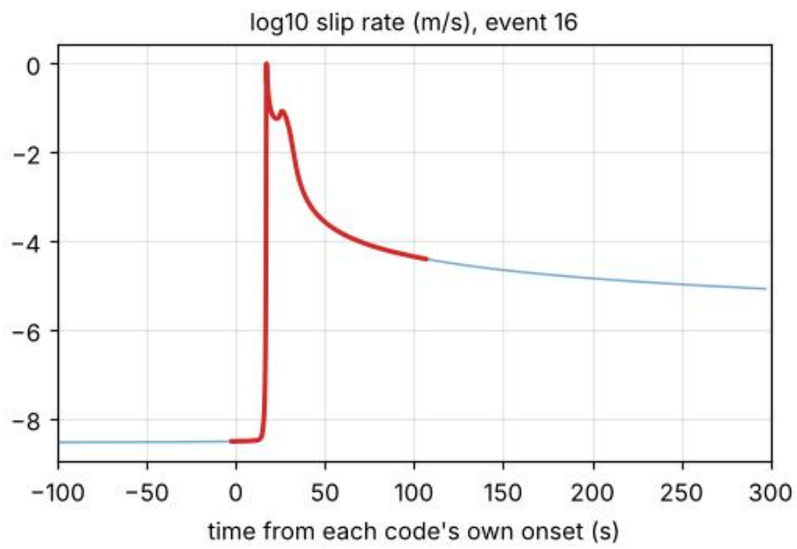


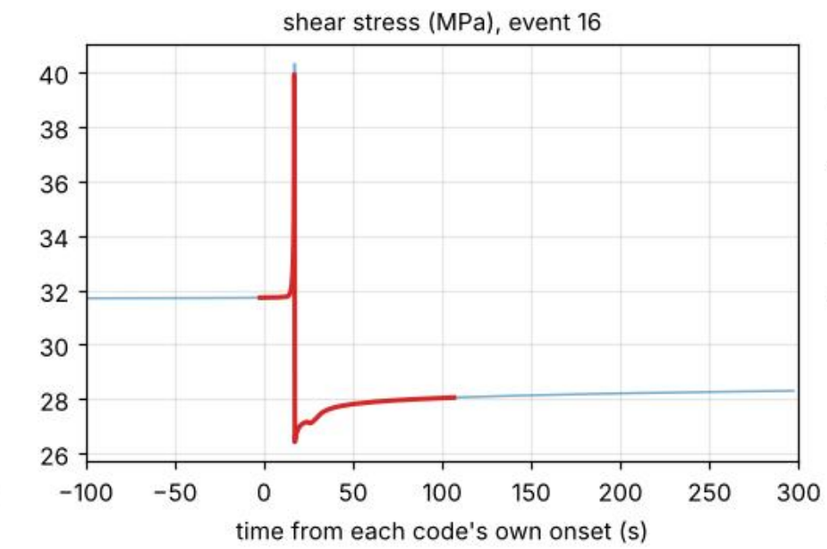


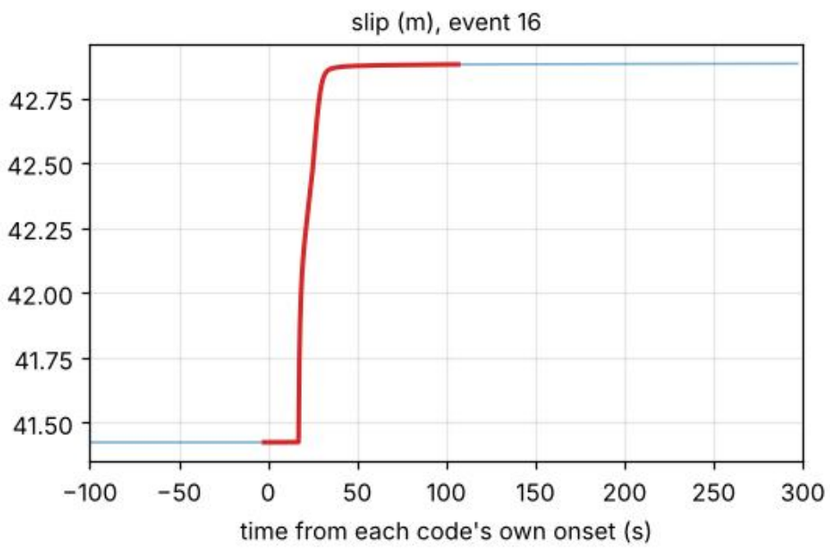


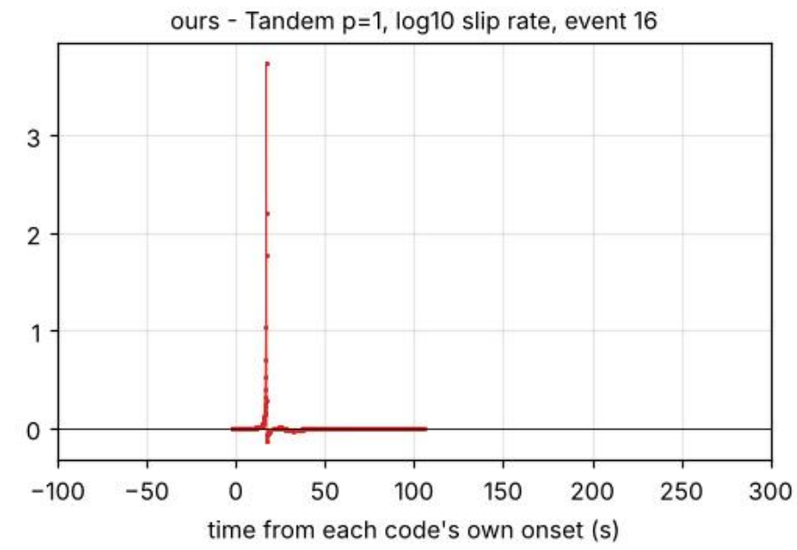


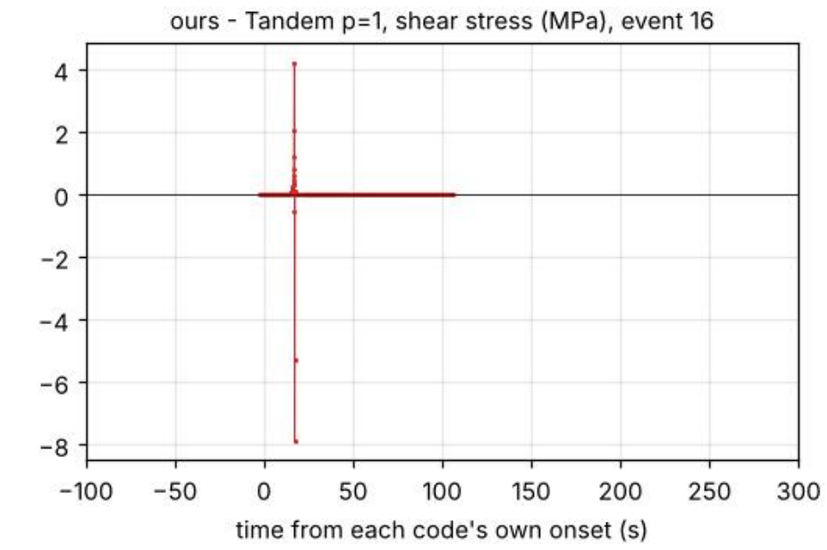


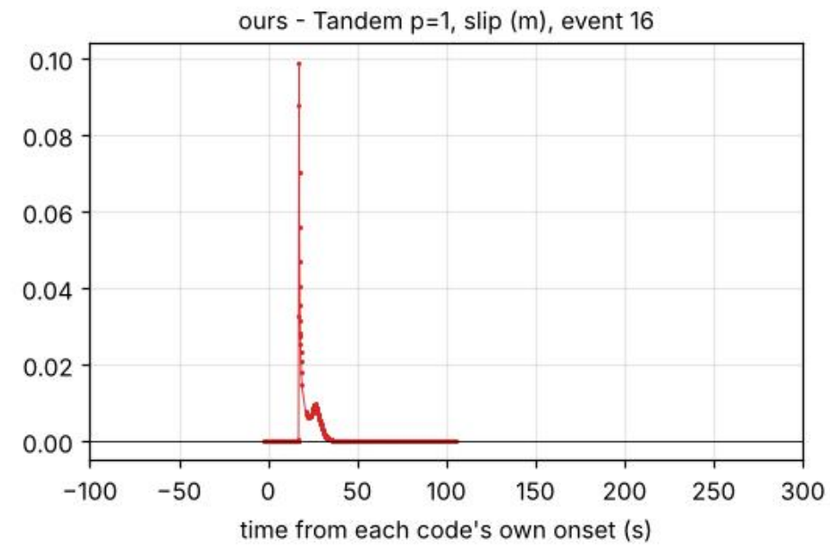






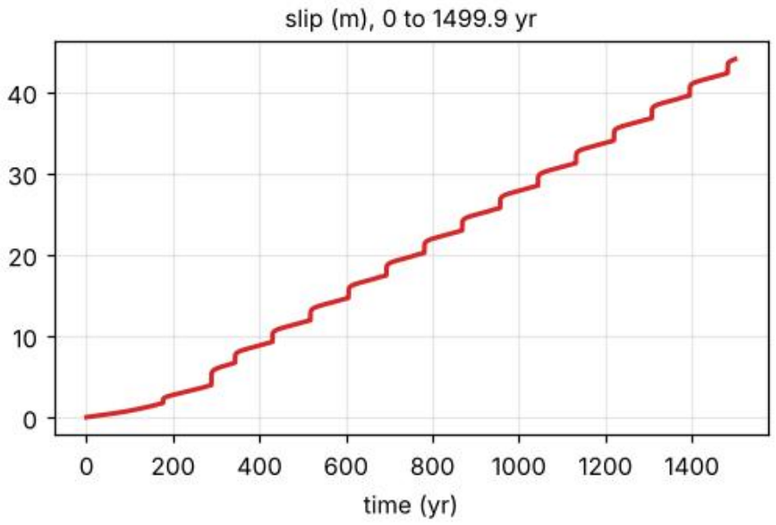


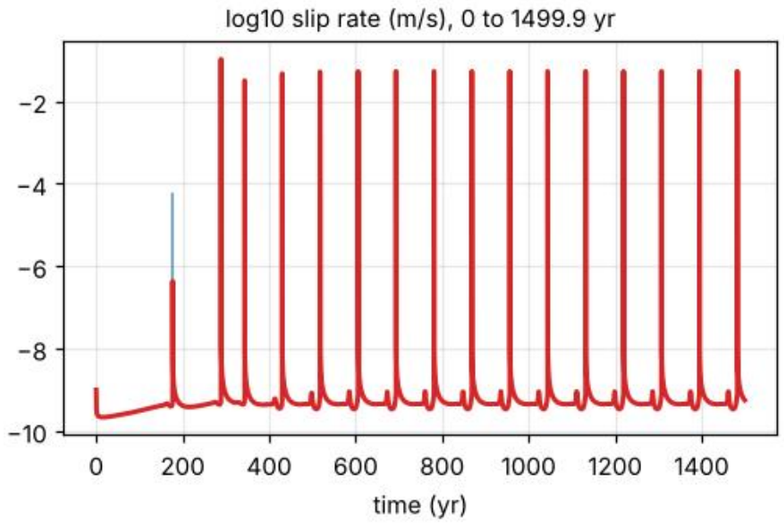


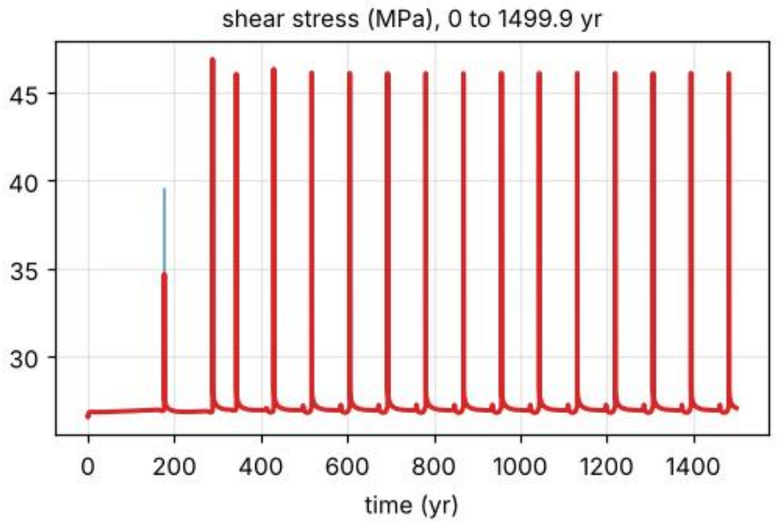


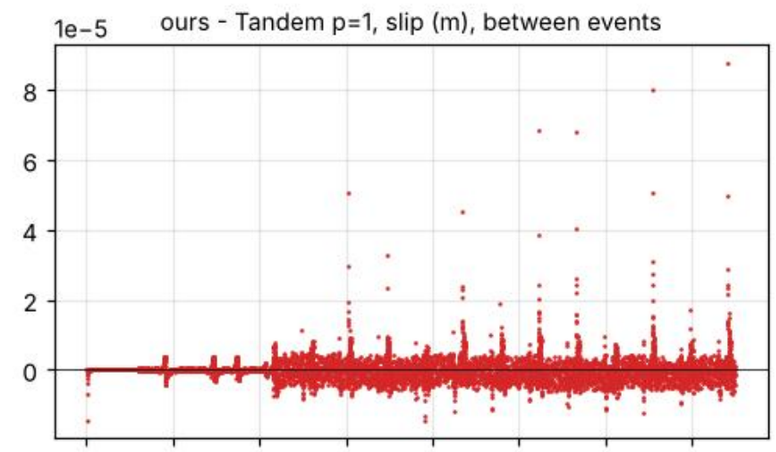
Tandem p=1 (Mac) seas-mfem p=1

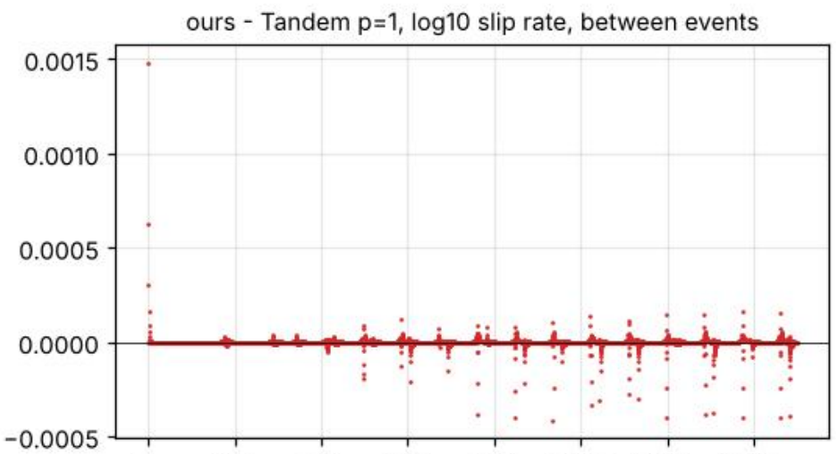
BP3-QD dip 60 thrust, fault station dp175 (17.5 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

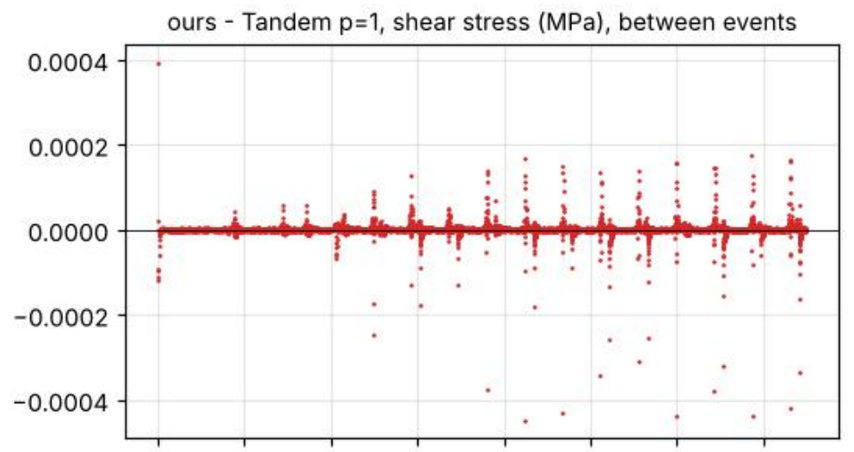


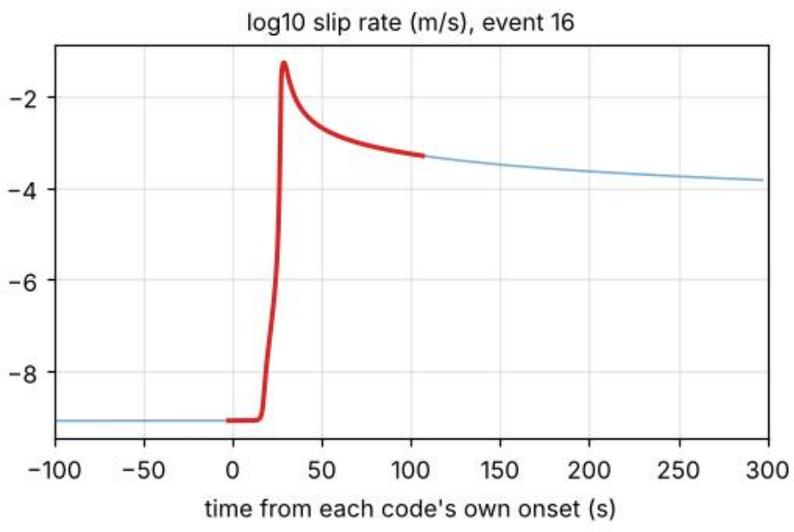


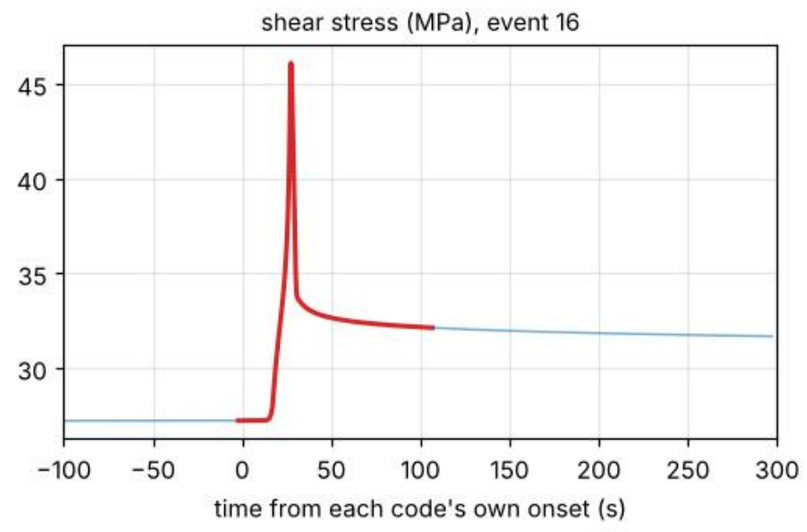


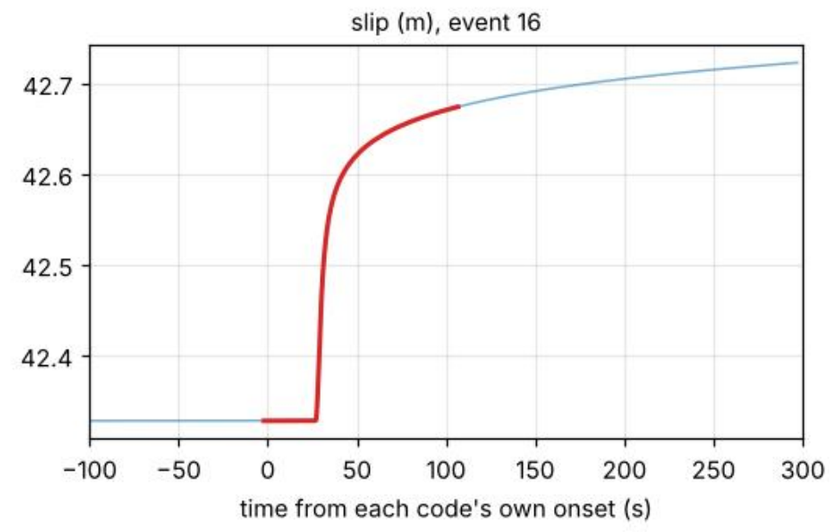


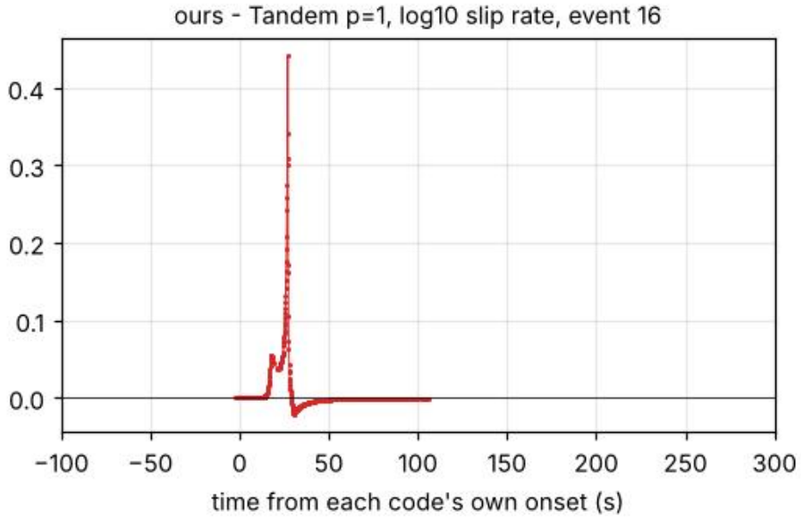


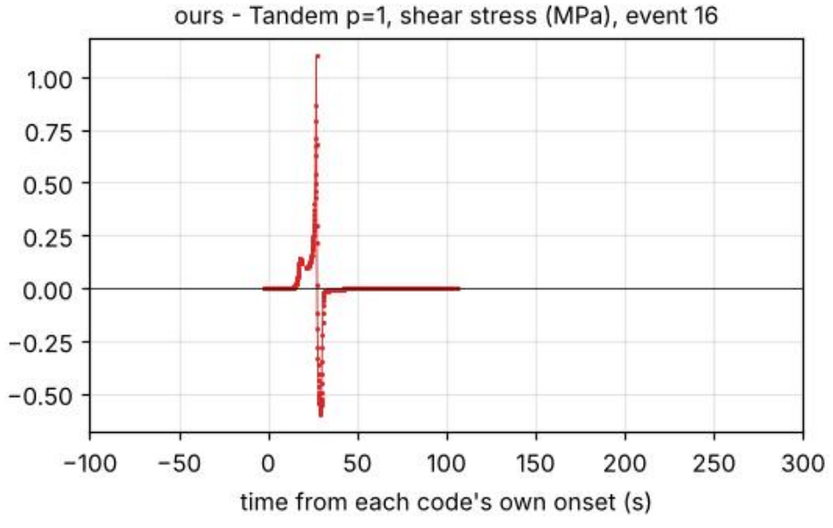


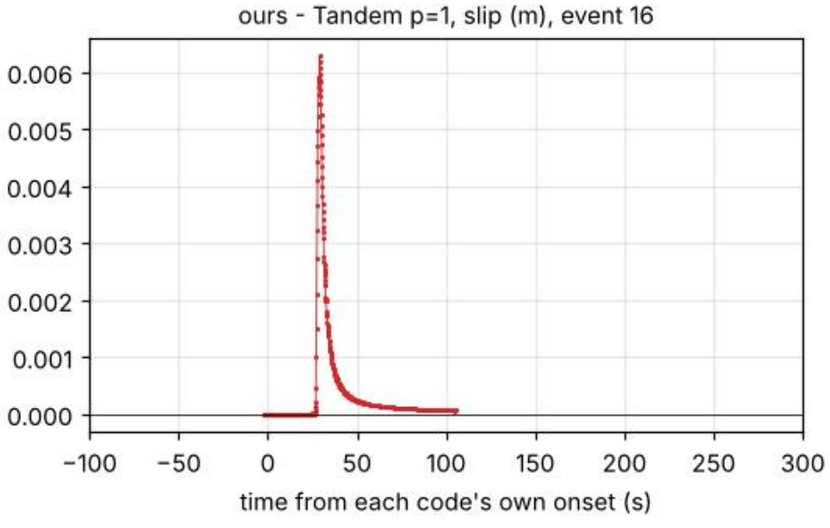






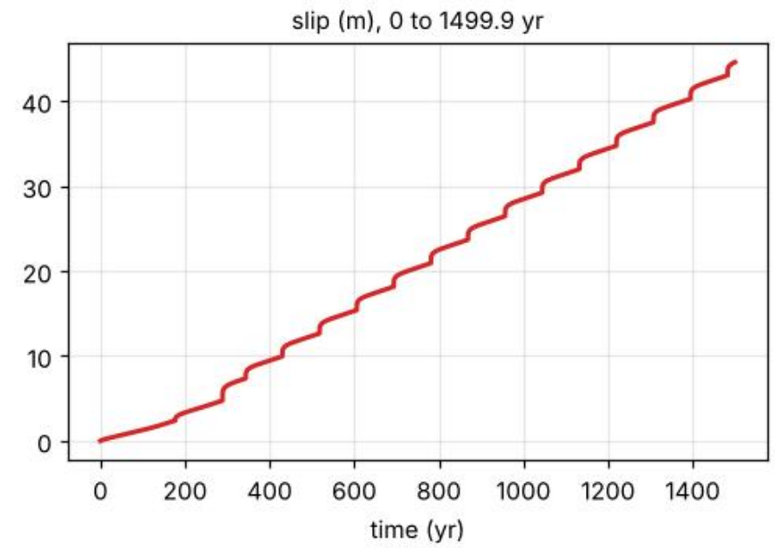


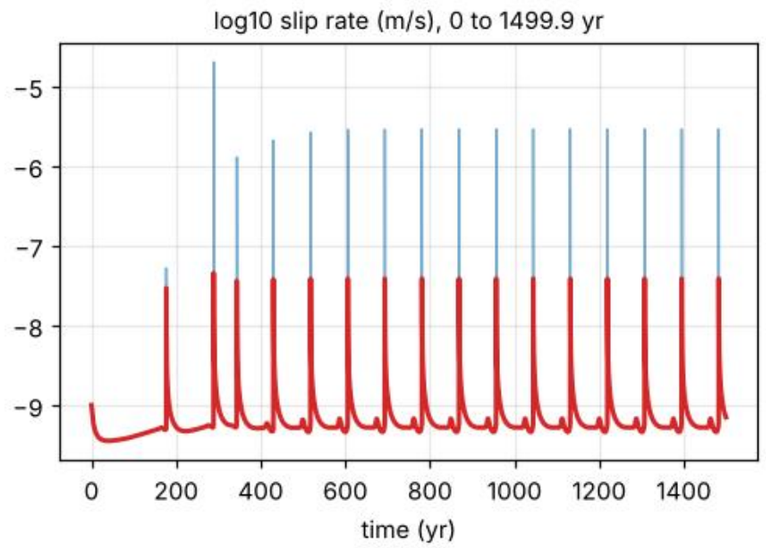


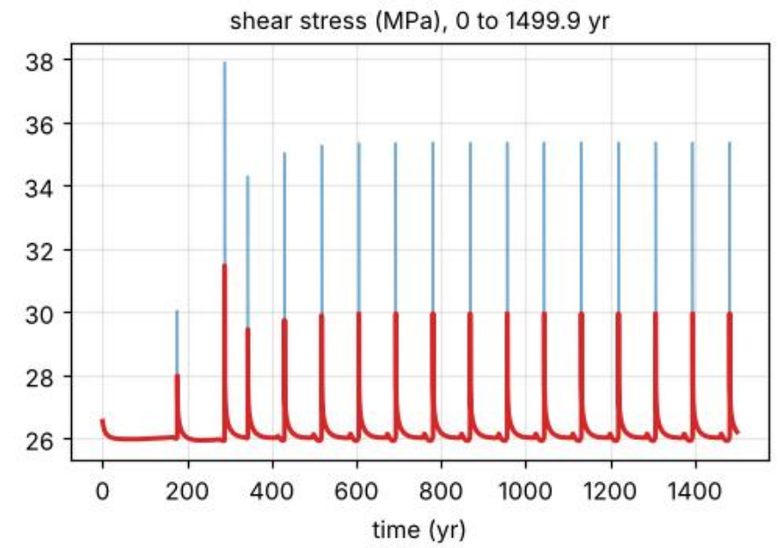


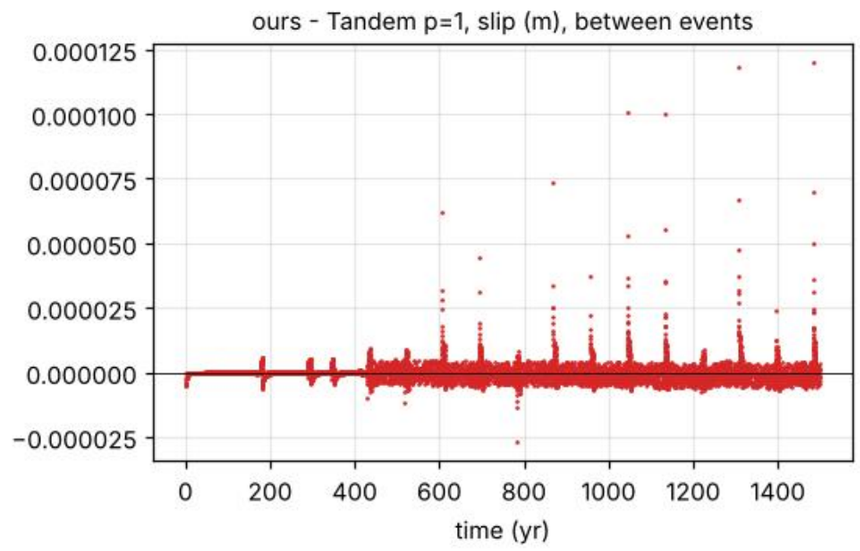


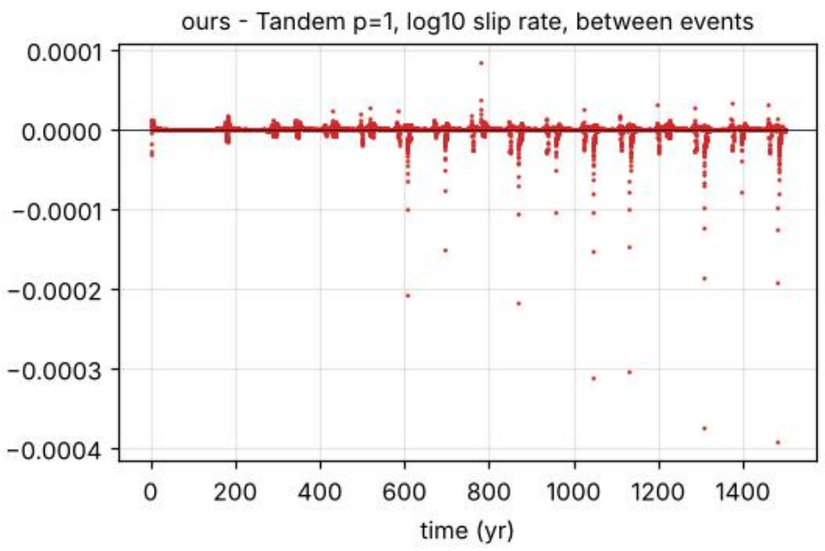
BP3-QD dip 60 thrust, fault station dp200 (20 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

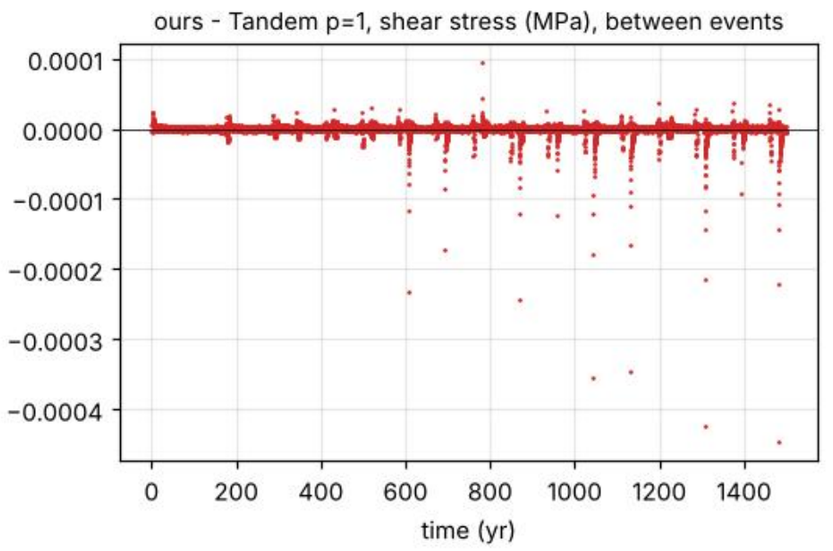


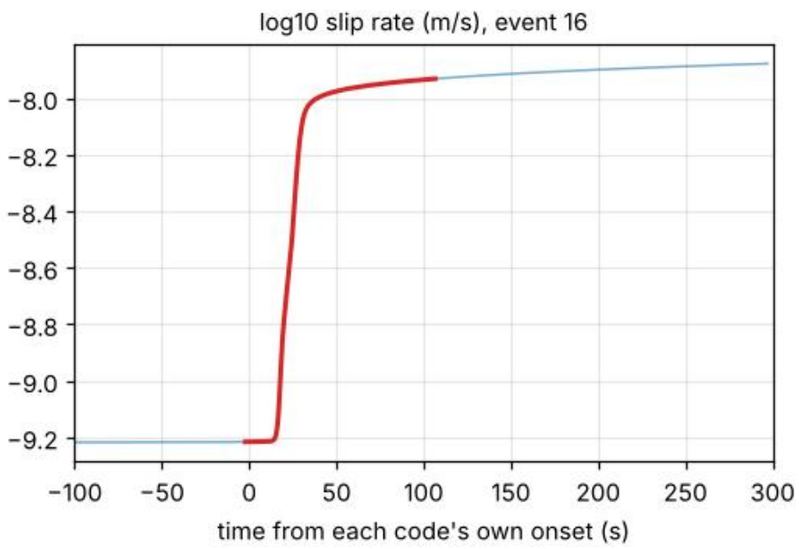


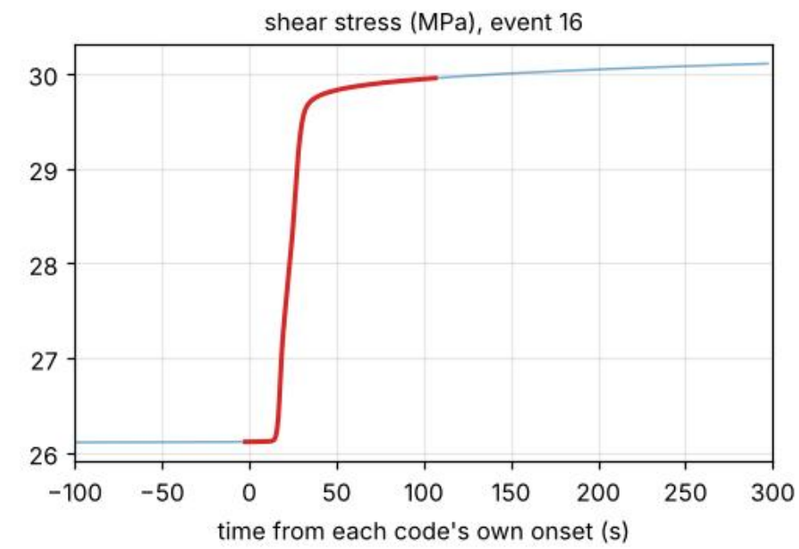


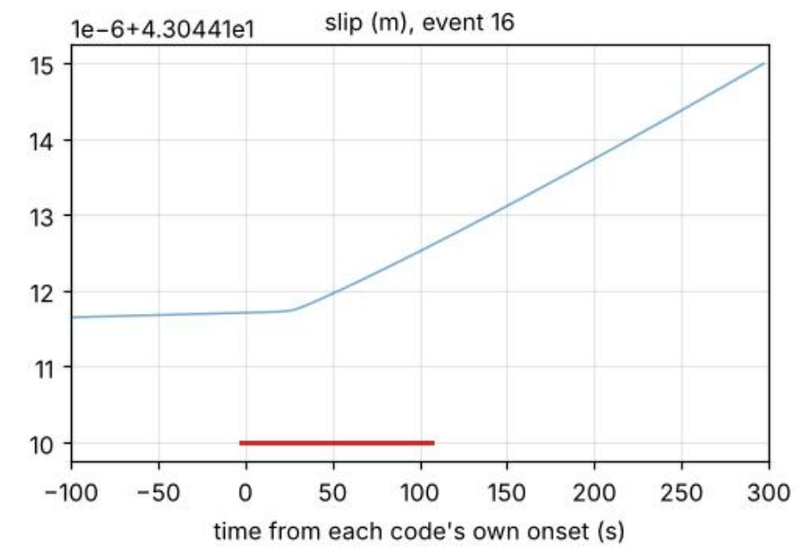


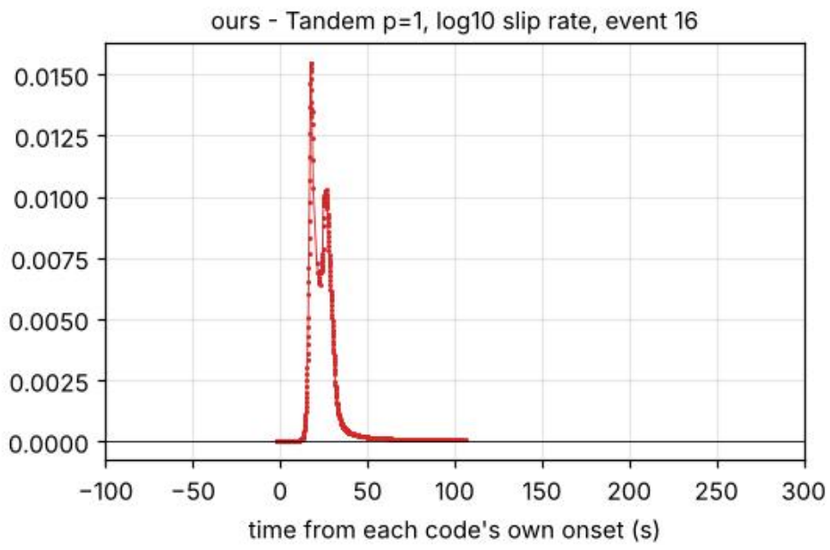


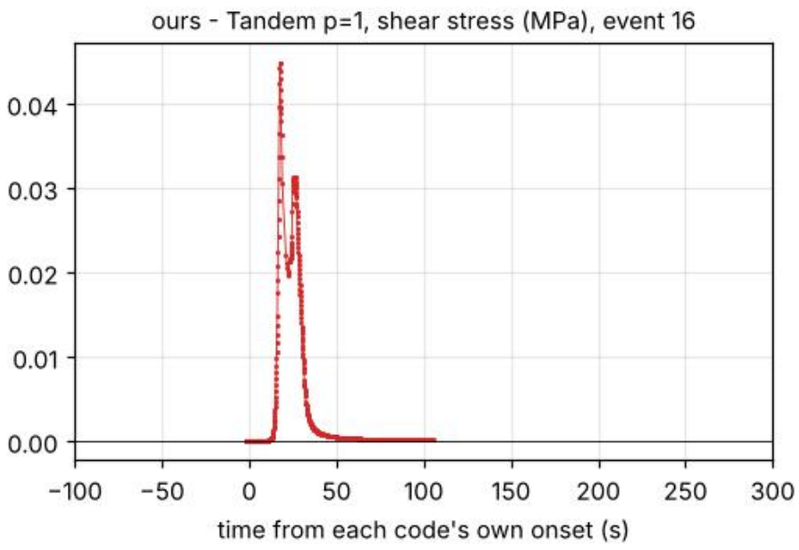


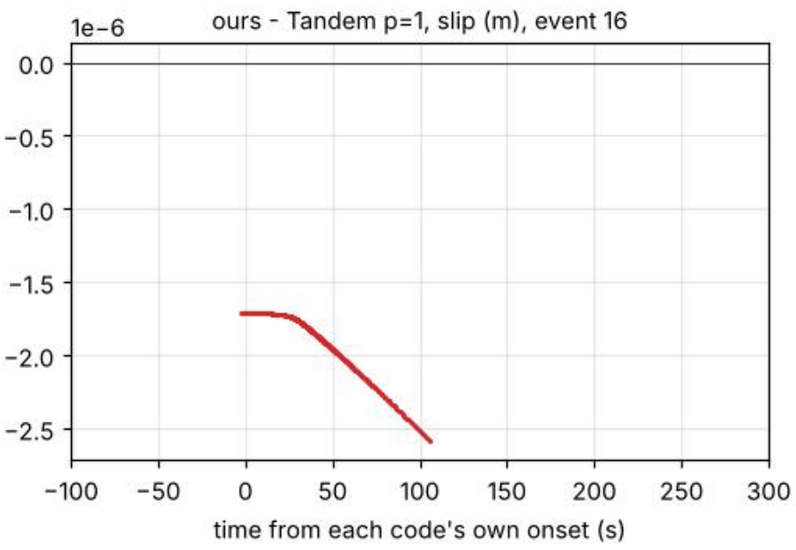






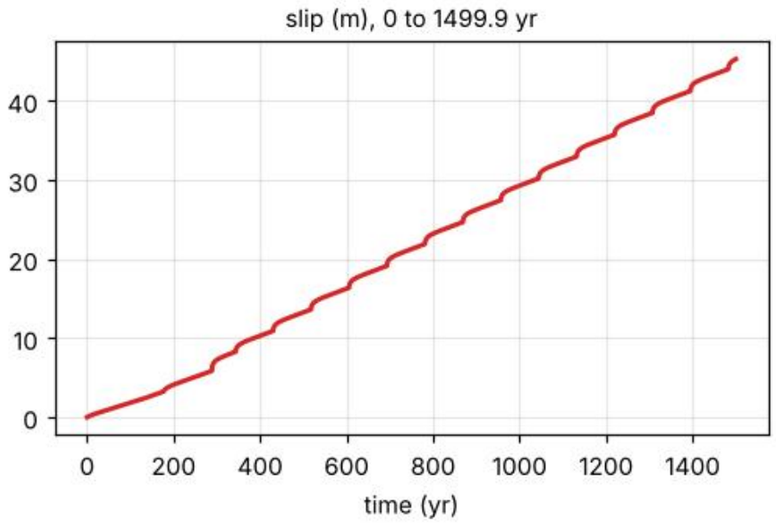


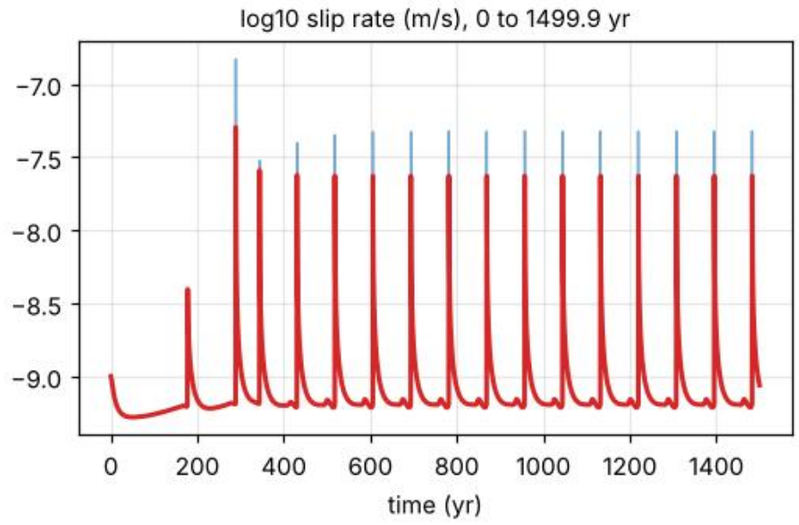


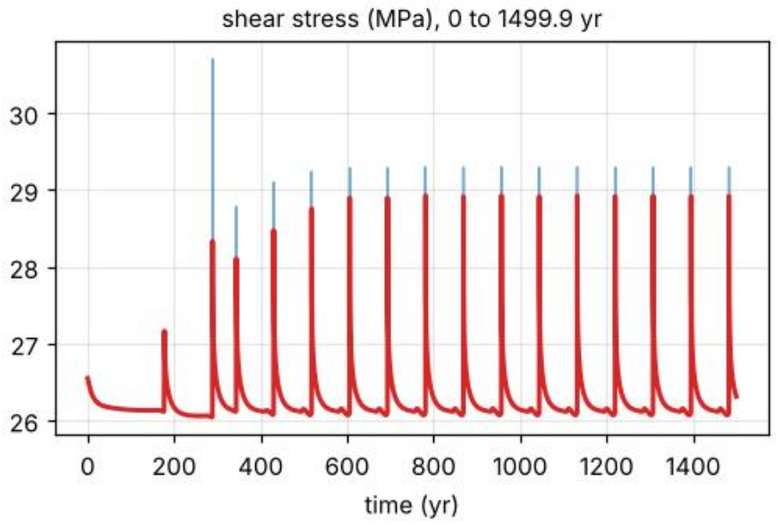


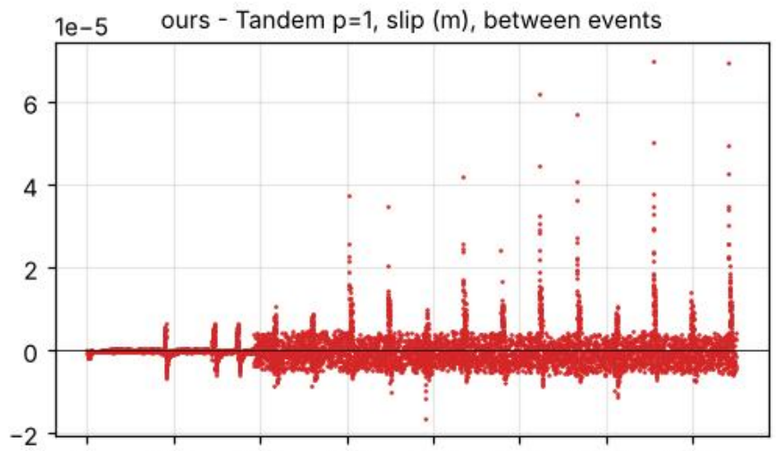
Tandem p=1 (Mac) seas-mfem p=1

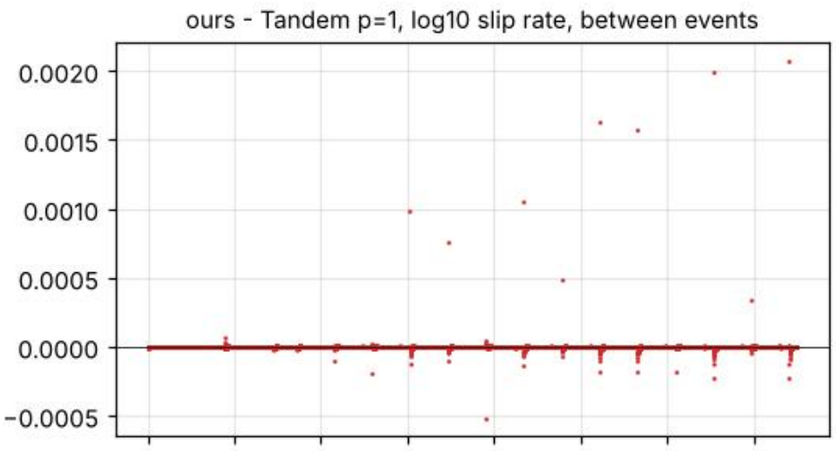
BP3-QD dip 60 thrust, fault station dp250 (25 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

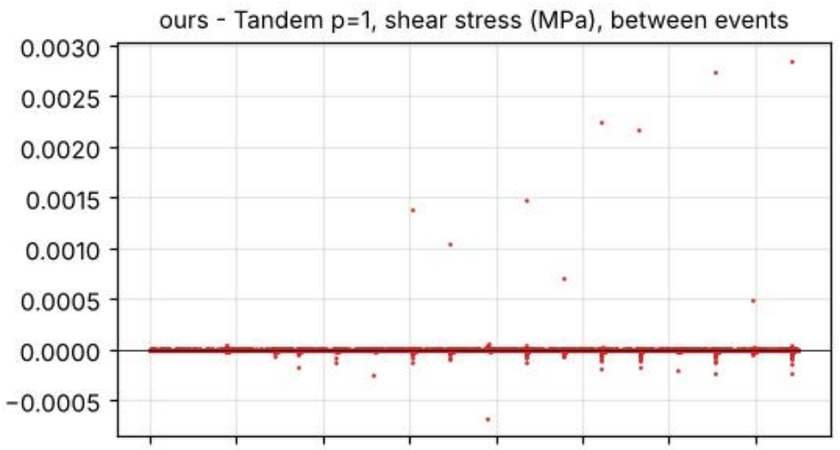


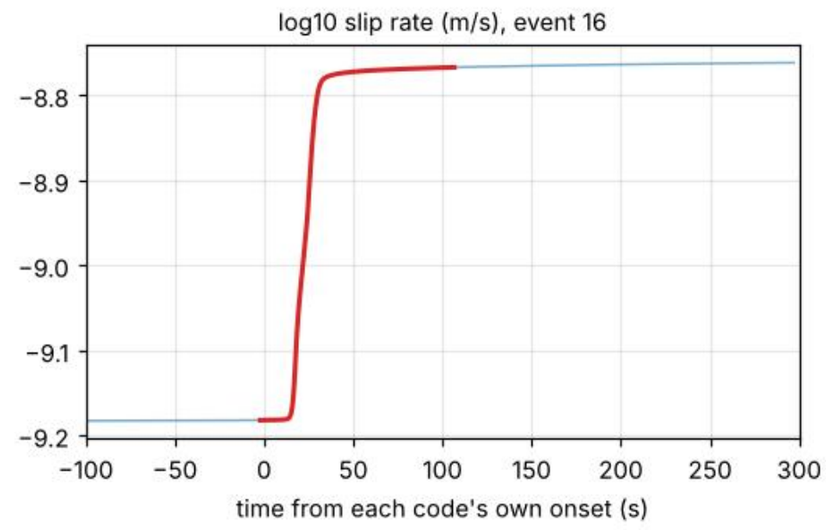


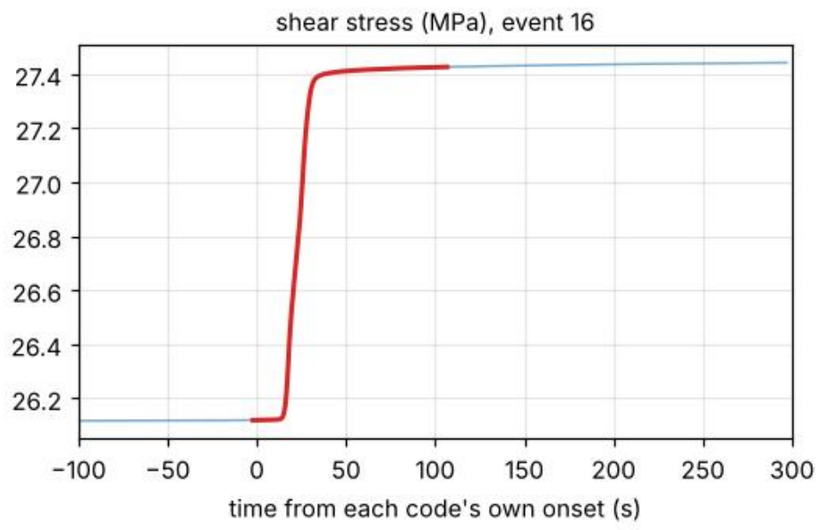


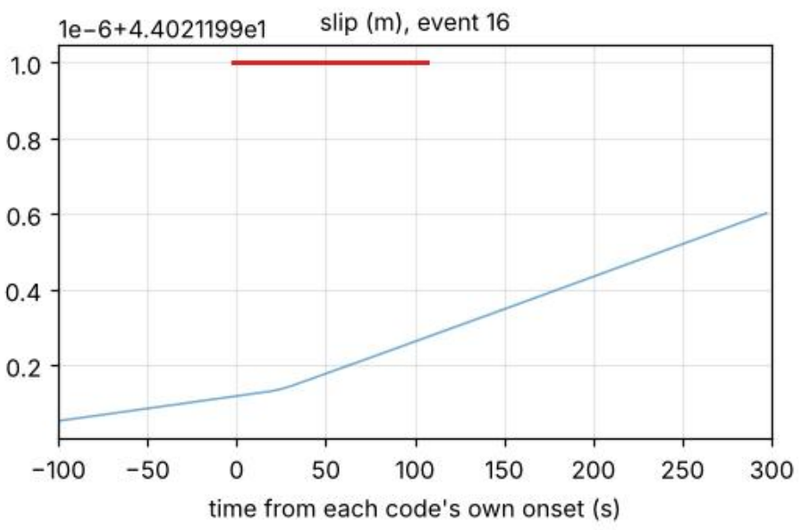


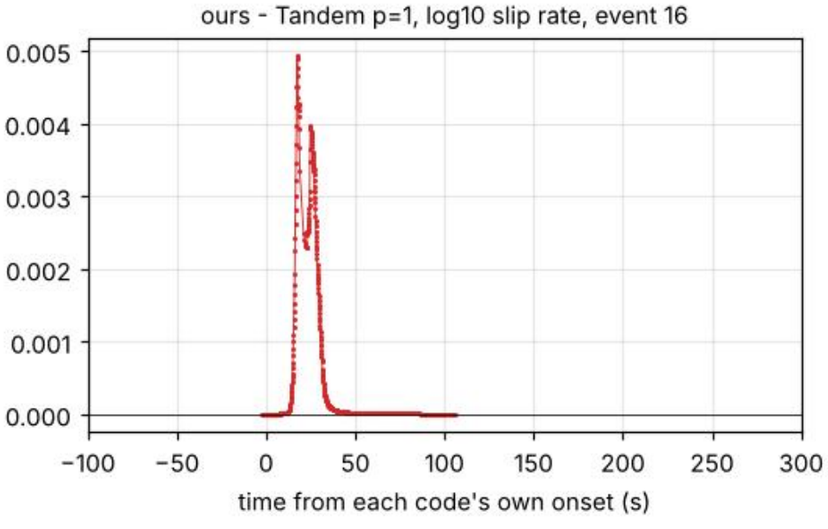


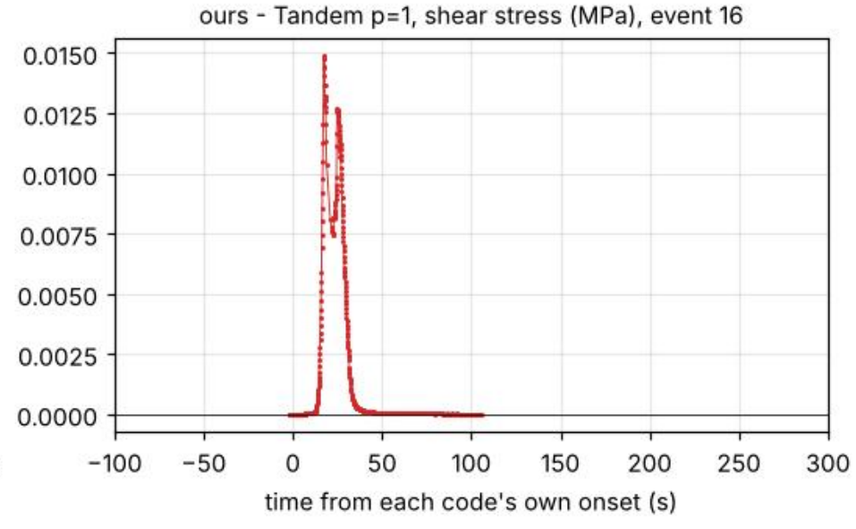


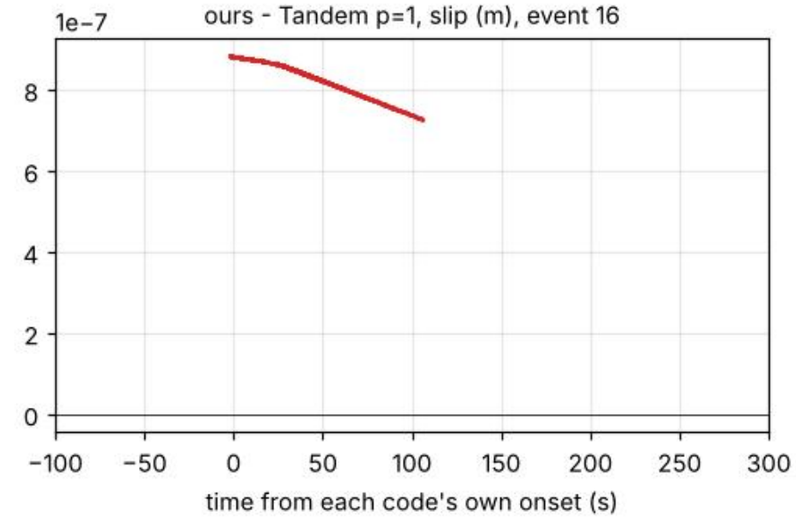






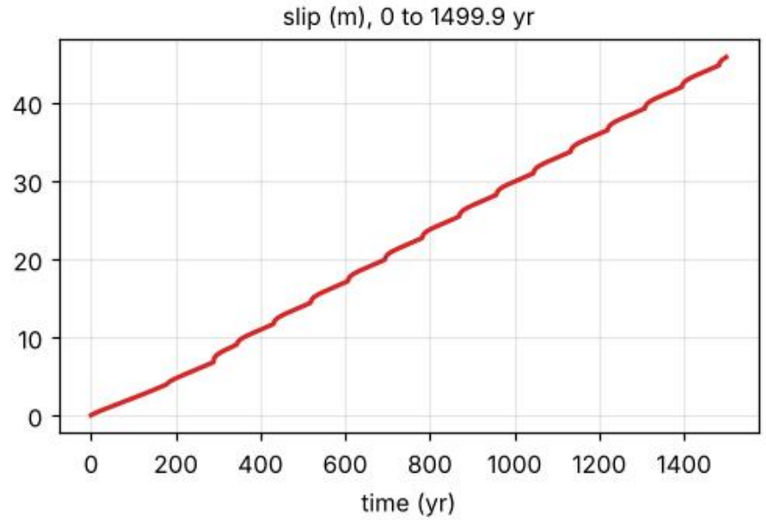


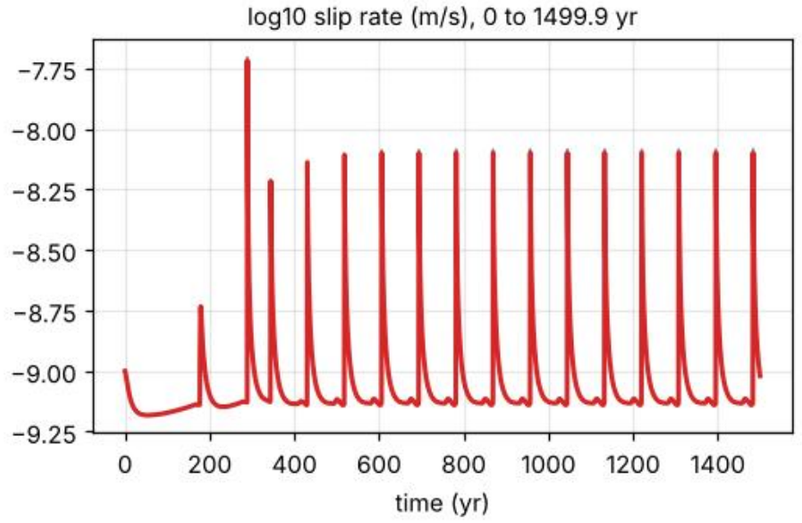


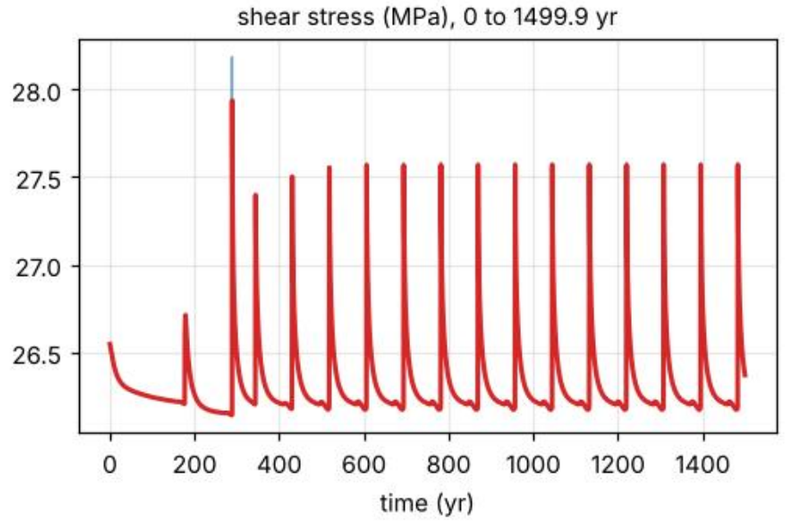


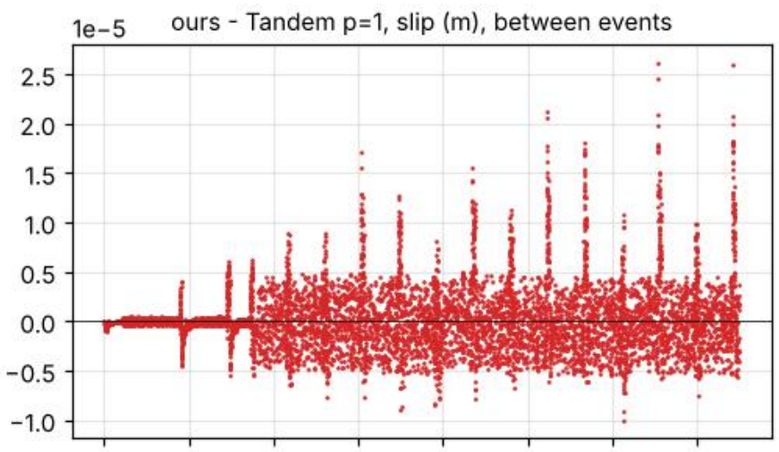


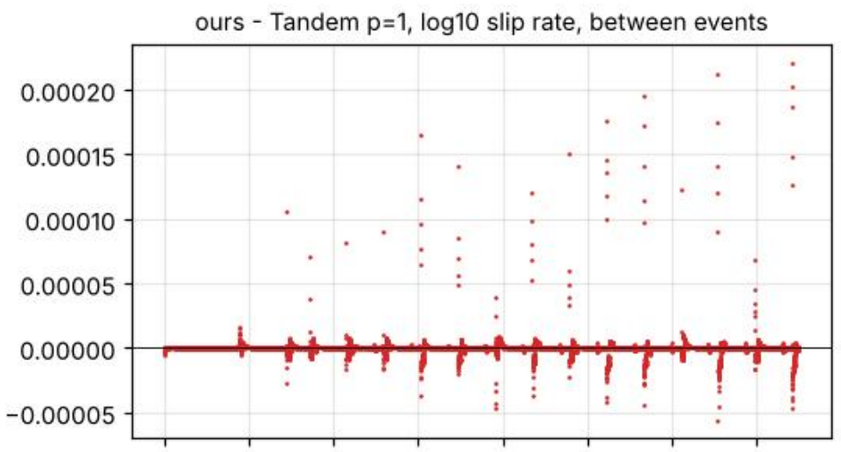
BP3-QD dip 60 thrust, fault station dp300 (30 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

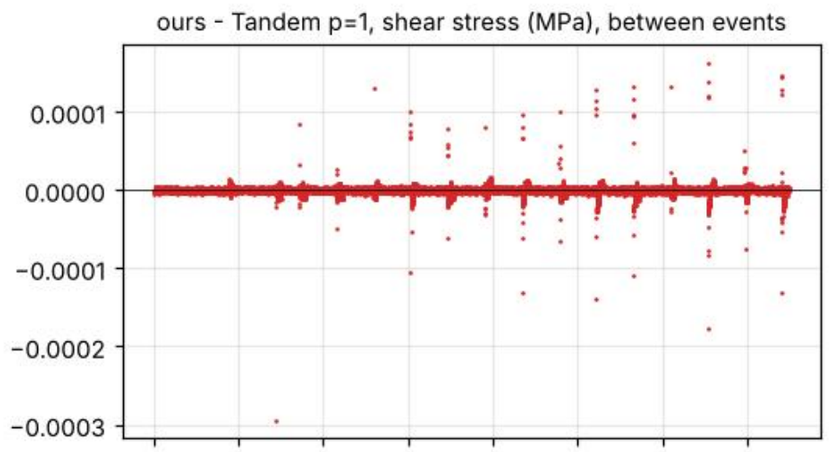


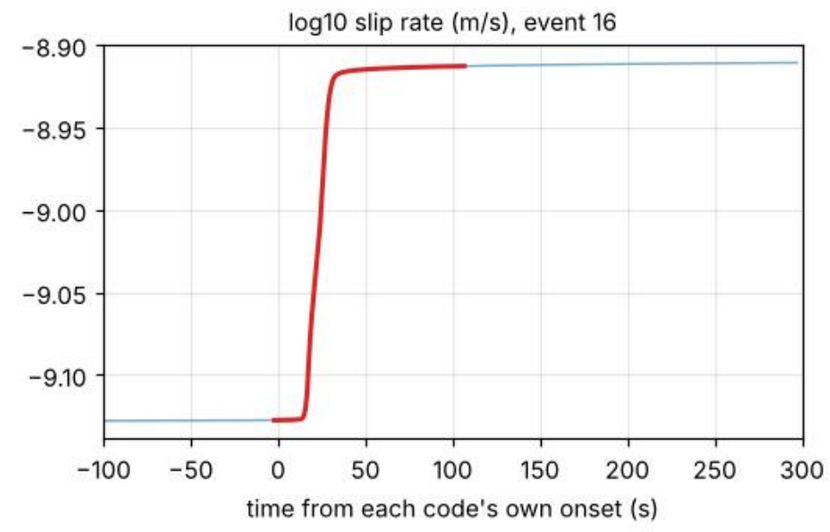


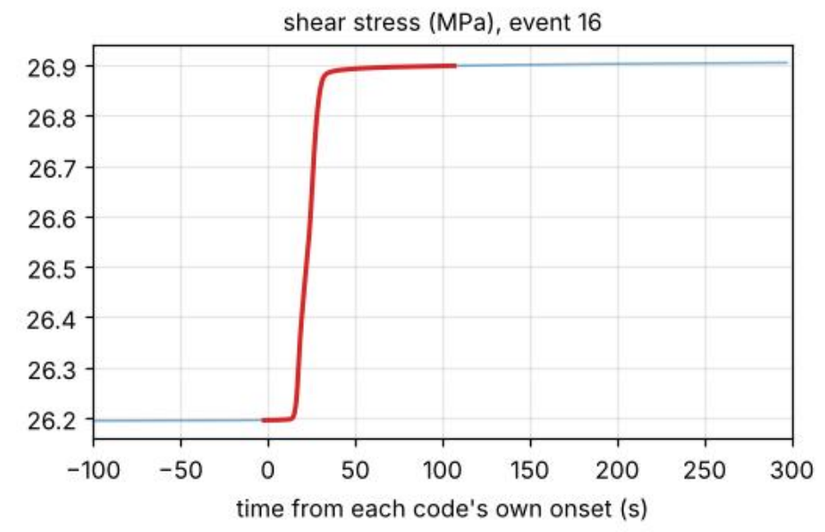


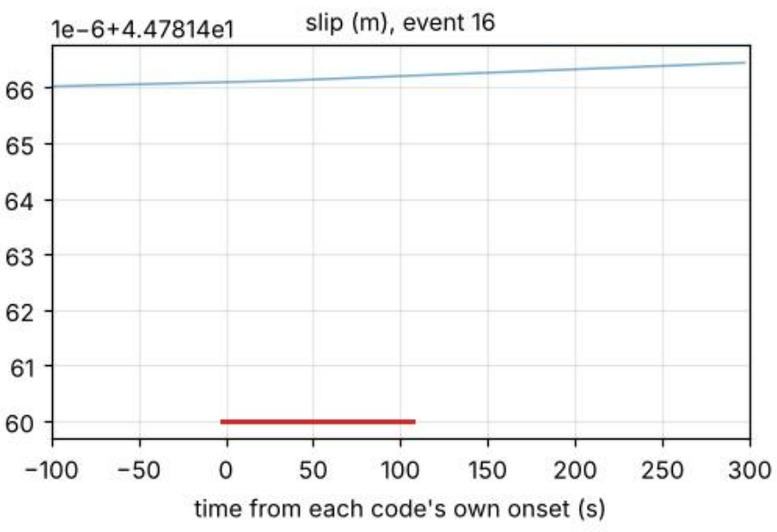


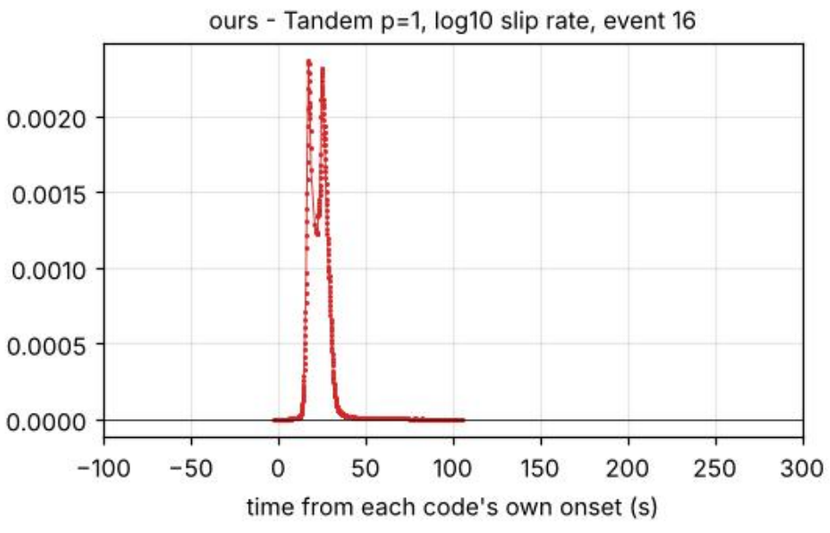


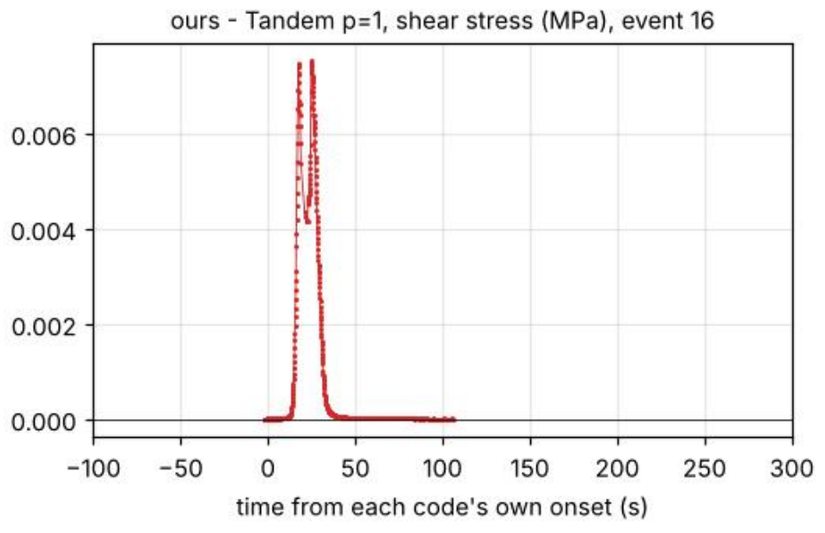


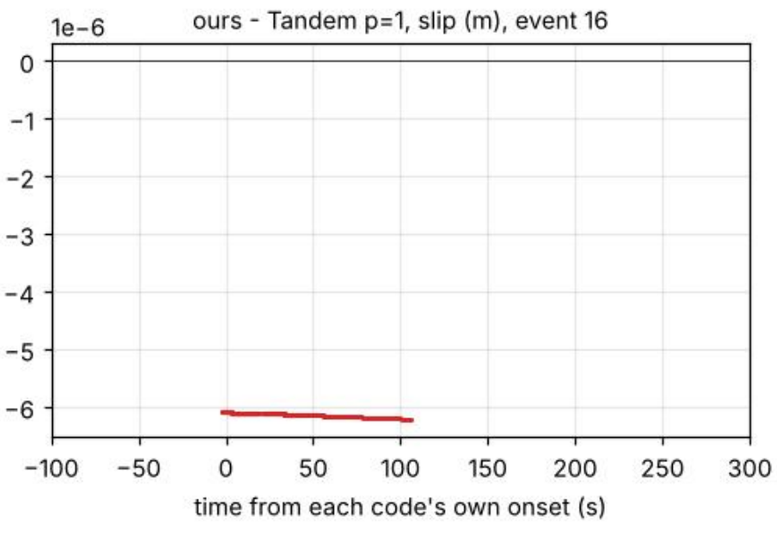






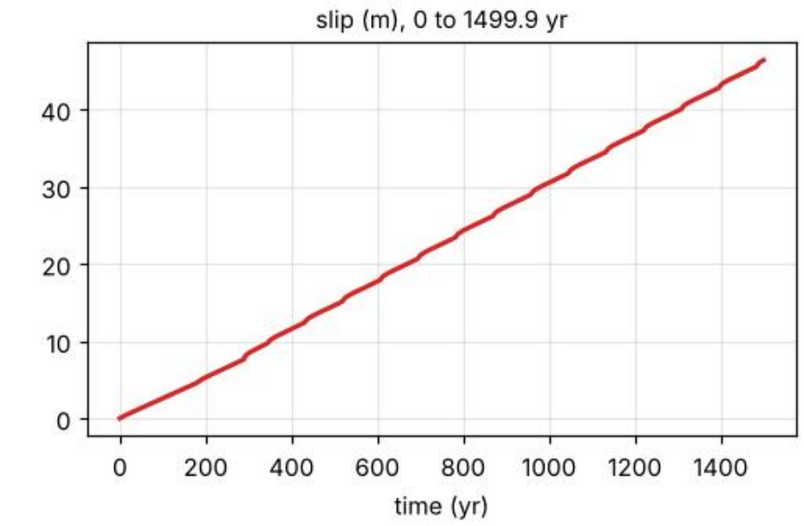


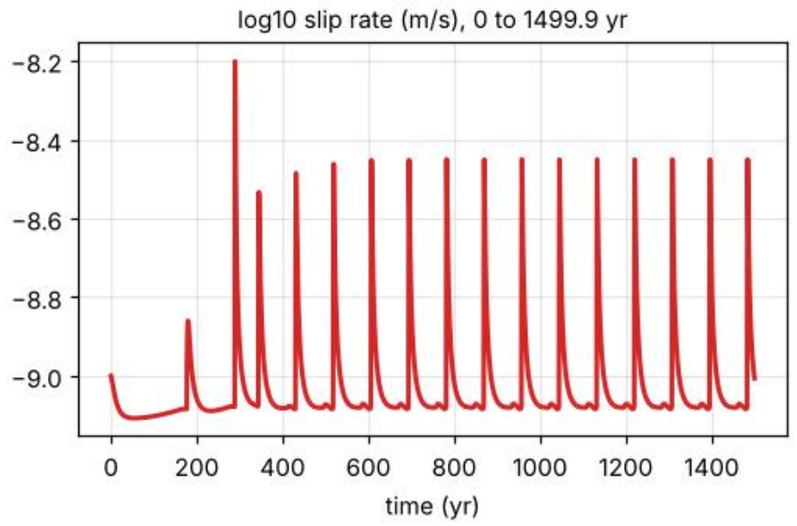


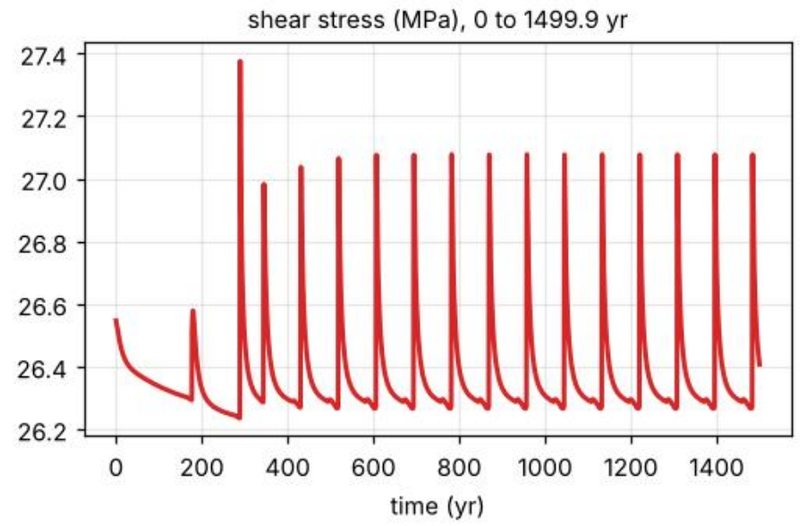


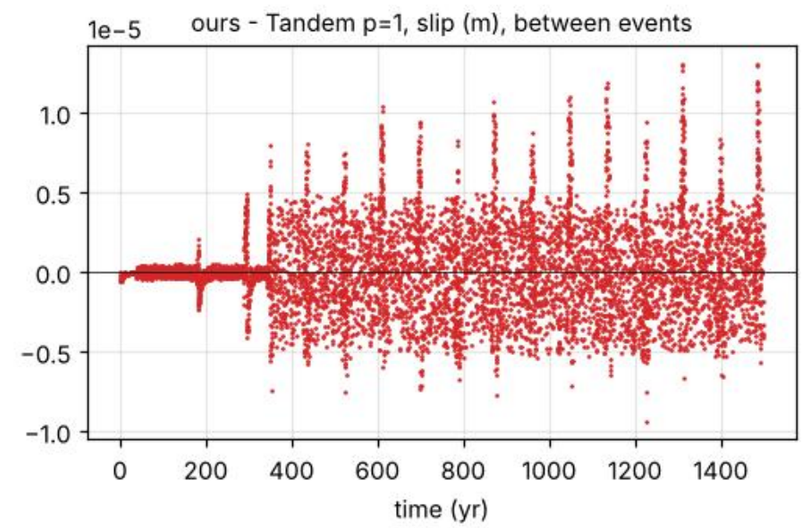
Tandem p=1 (Mac) seas-mfem p=1

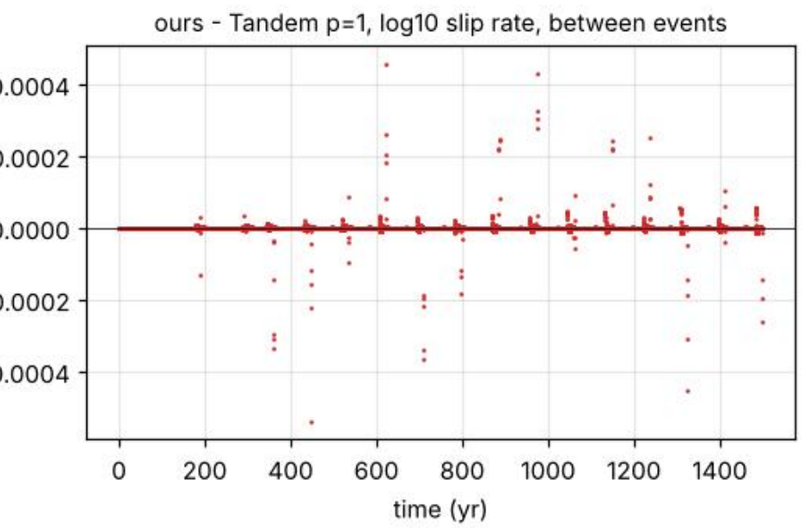
BP3-QD dip 60 thrust, fault station dp350 (35 km down-dip): seas-mfem p=1 against Tandem p=1
event 16 onsets (yr): Tandem p=1 (Mac) 1482.5815, seas-mfem p=1 1482.5814; between events = onset - 1 day to onset + 30 days left out

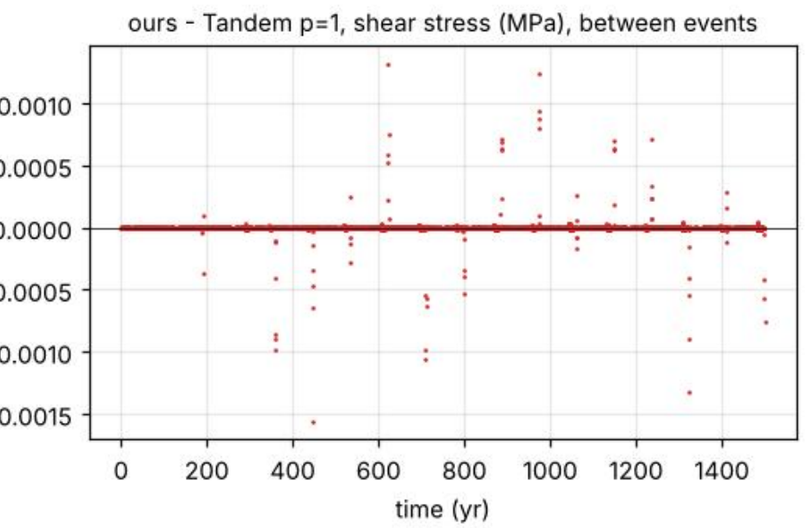


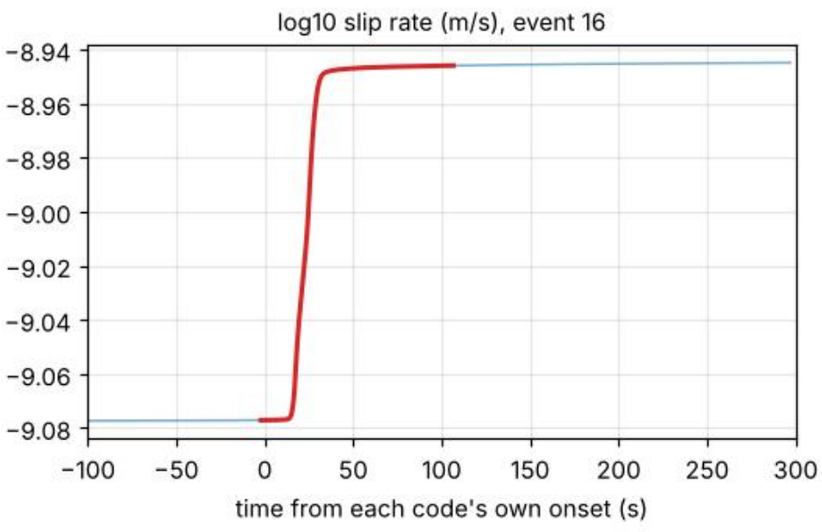


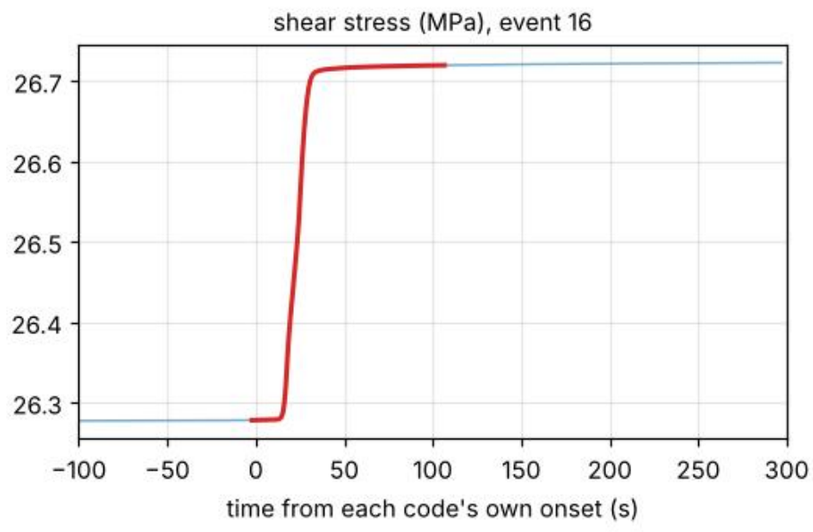


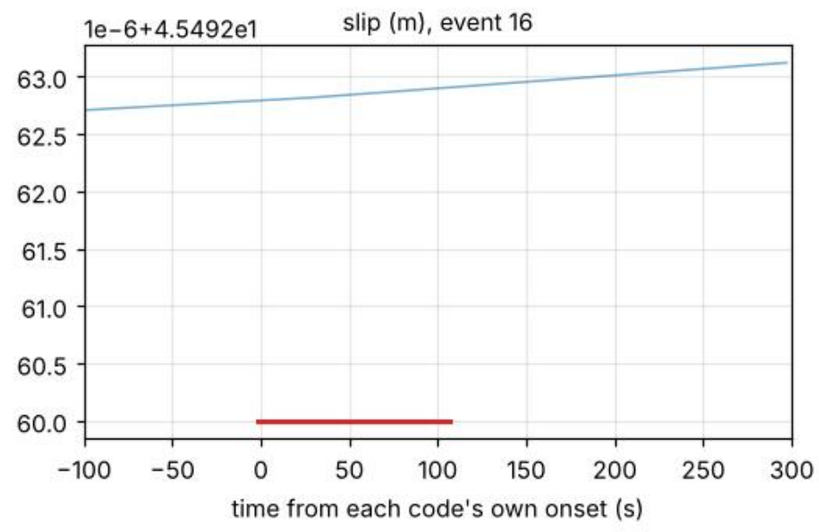


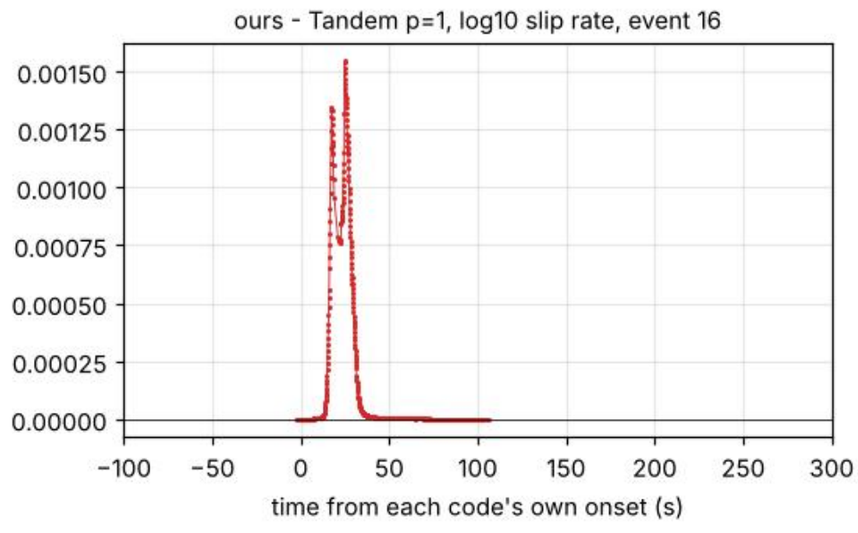


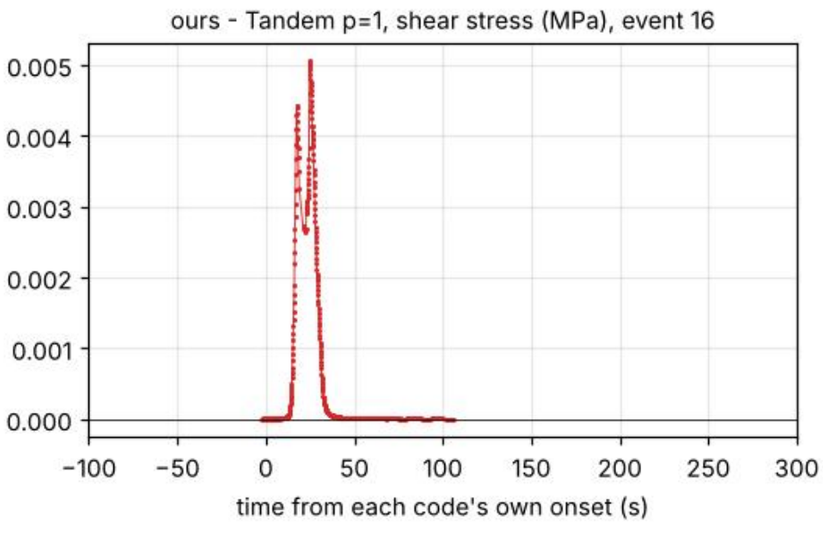


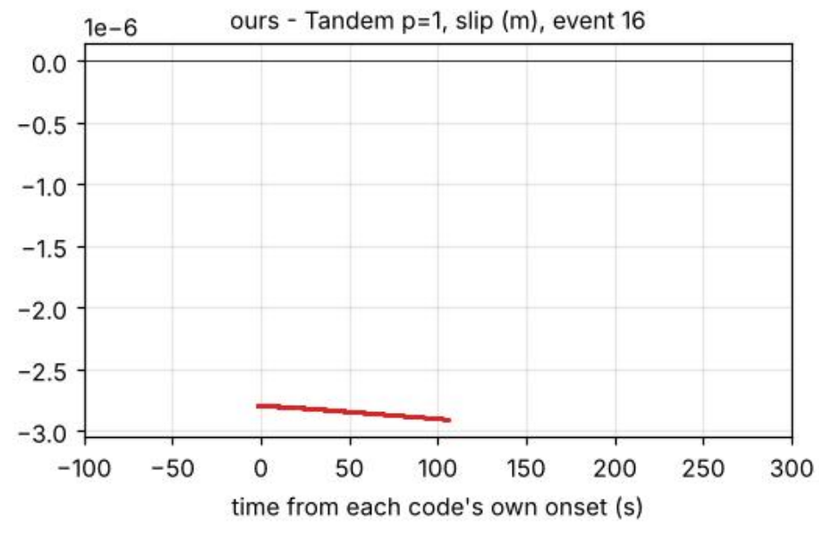






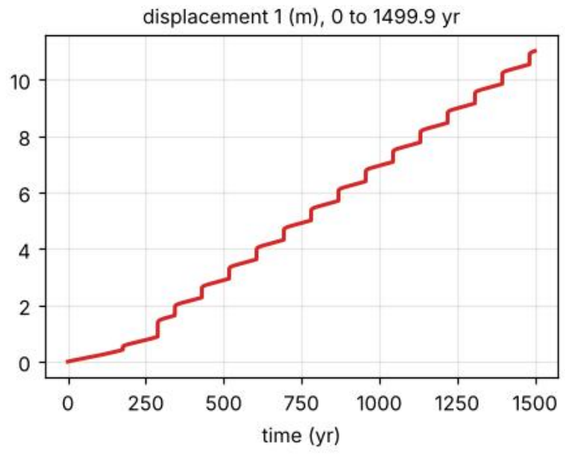


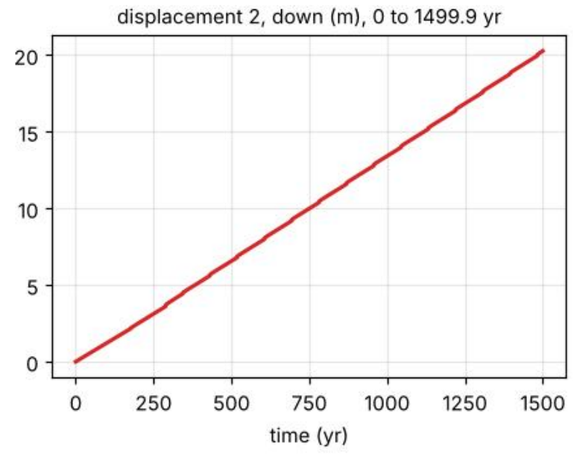


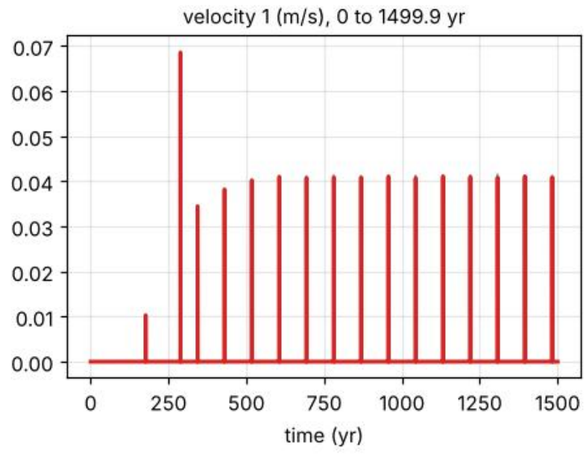


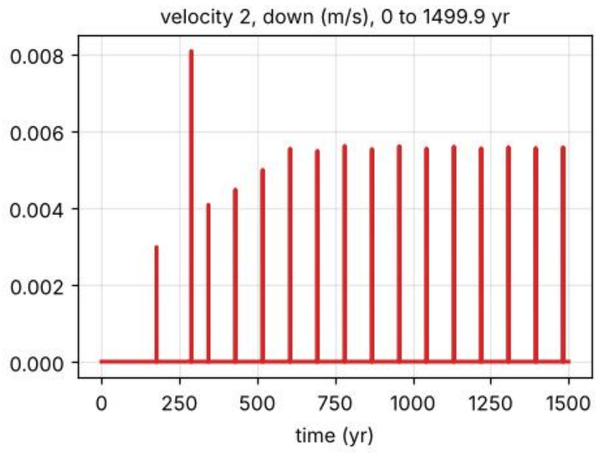


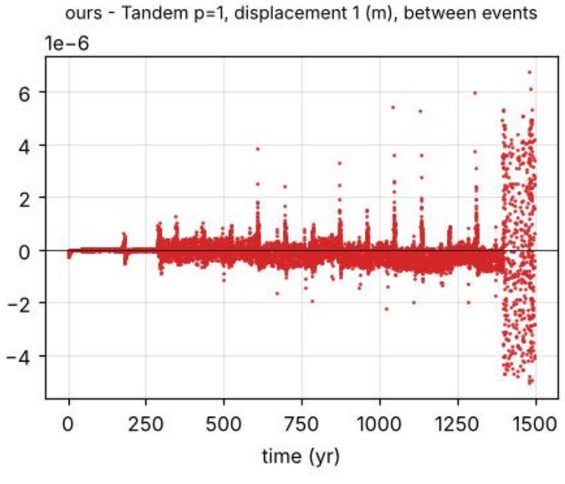
BP3-QD dip 60 thrust, surface station fn-32 (x1 = -32 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

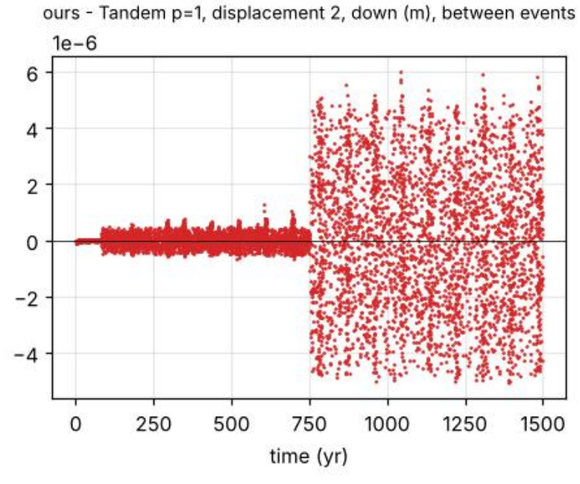


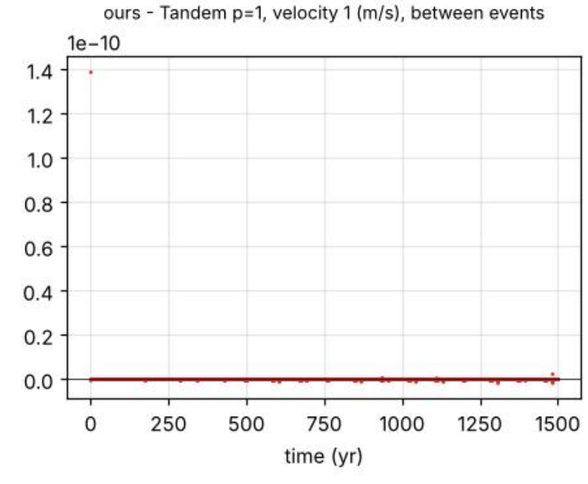


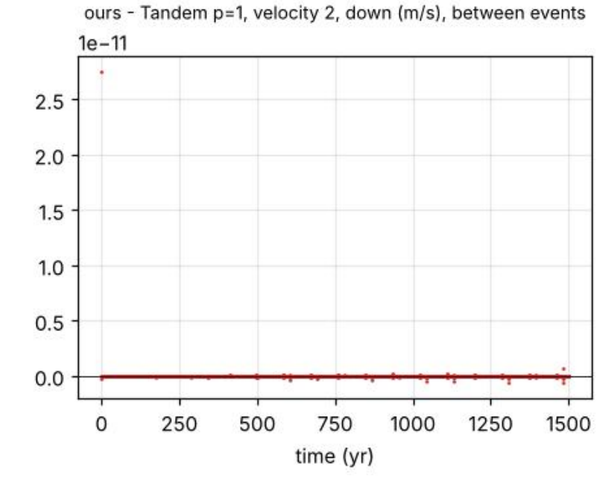


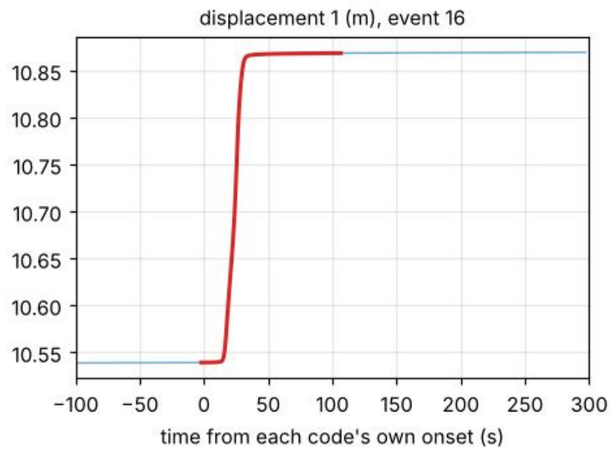


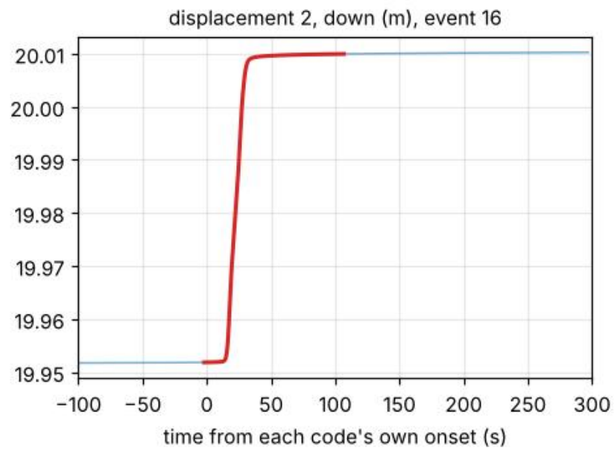


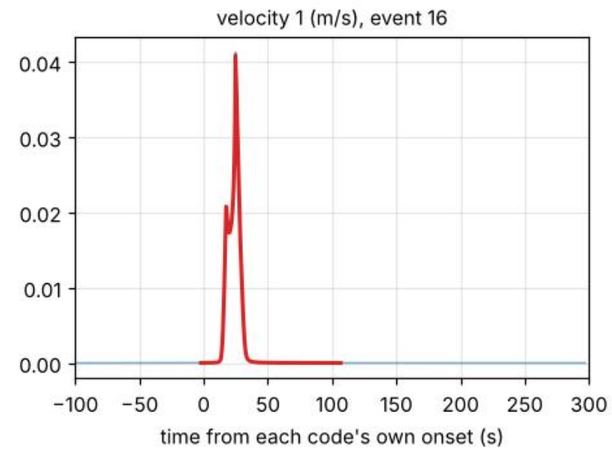


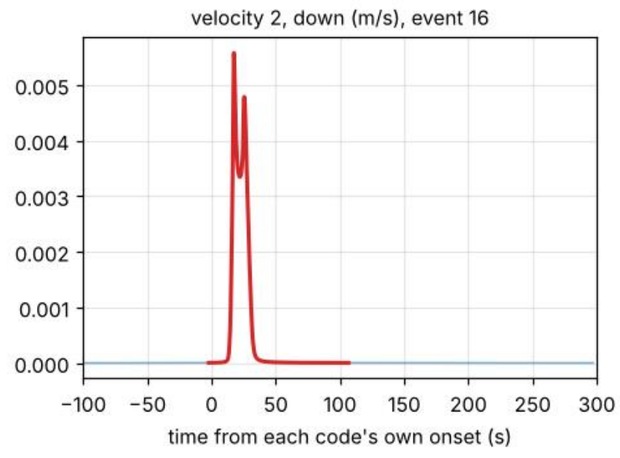


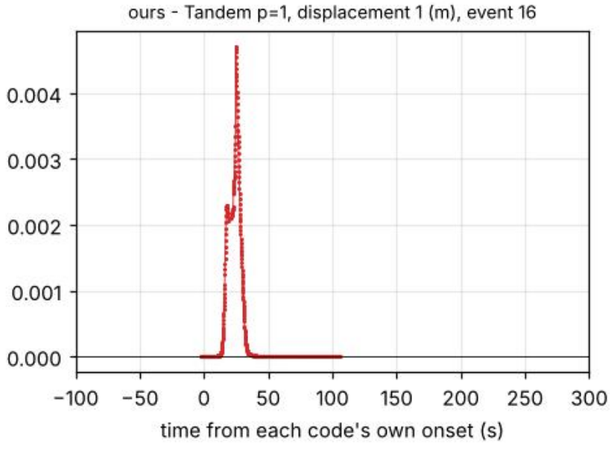


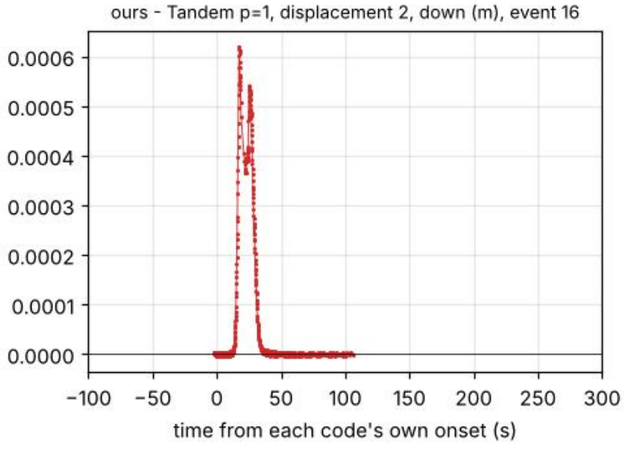


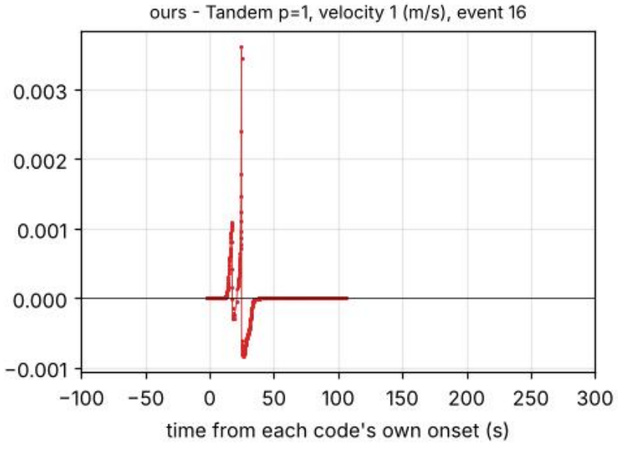


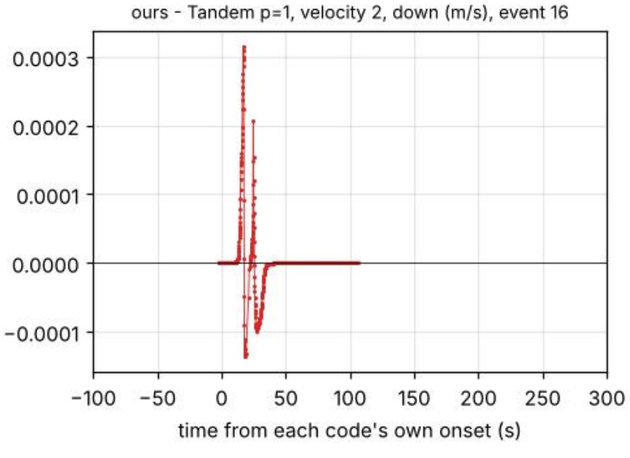






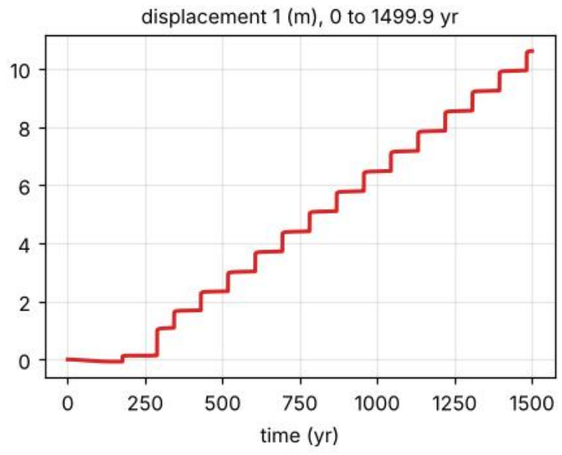


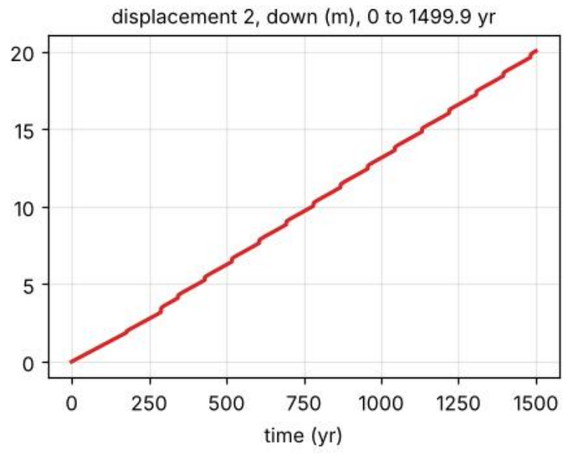


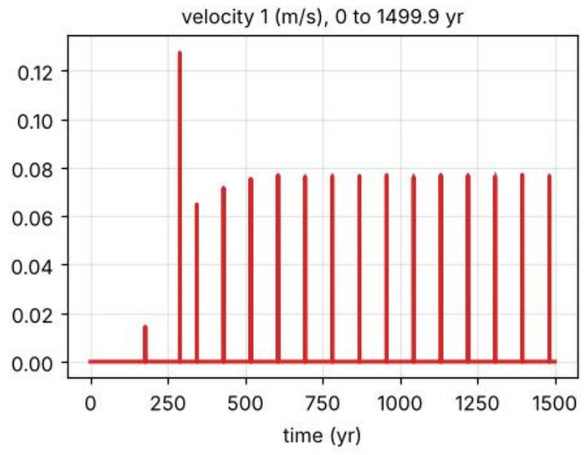


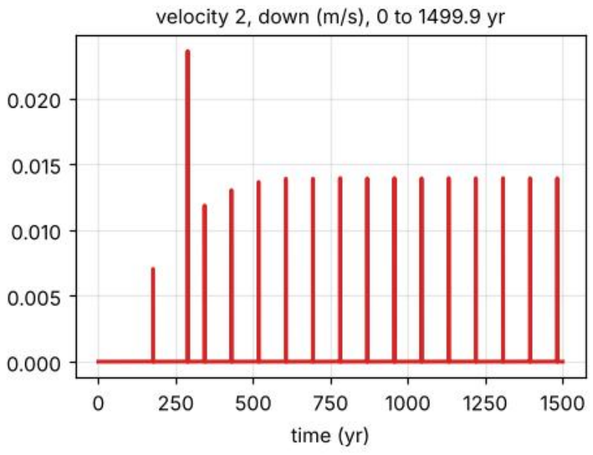
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

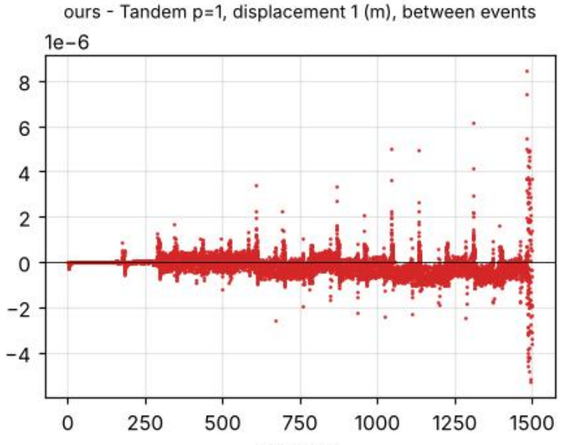
BP3-QD dip 60 thrust, surface station fn-16 (x1 = -16 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

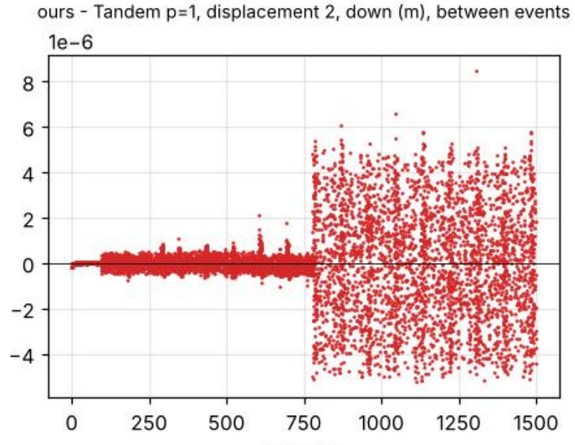


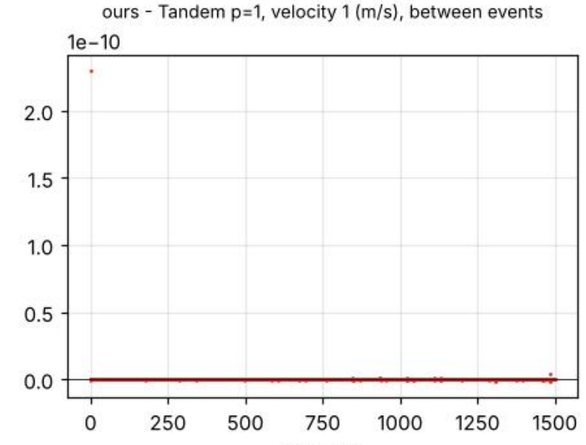


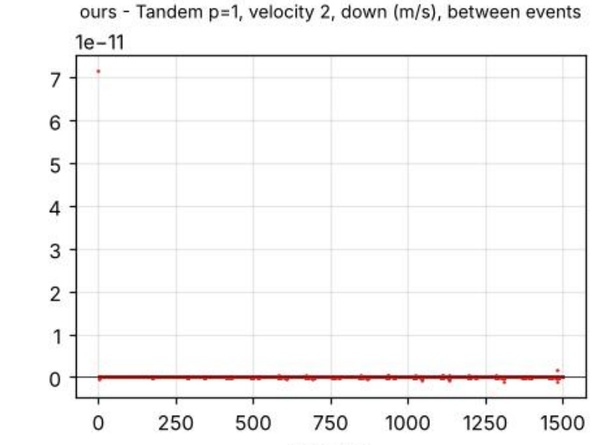


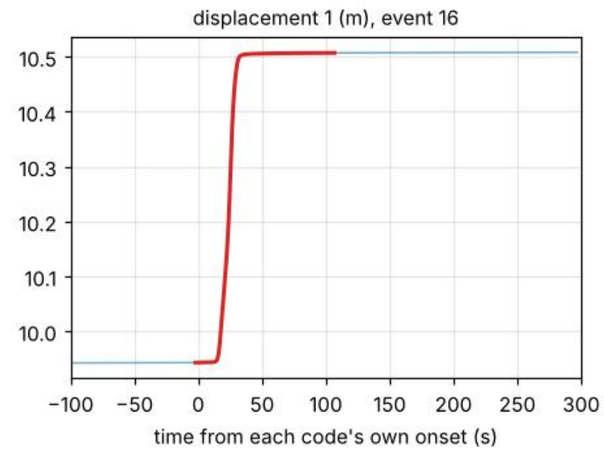


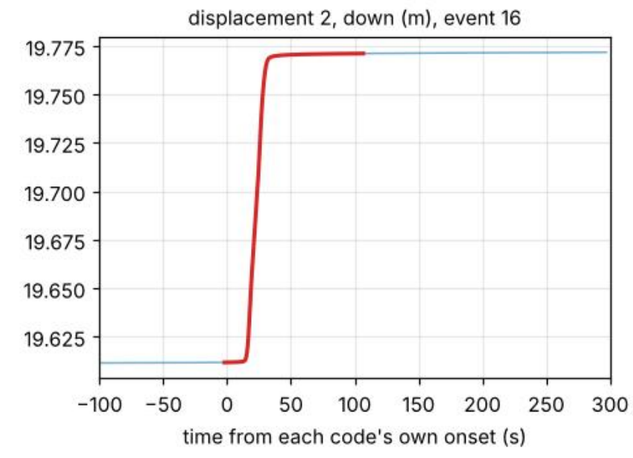


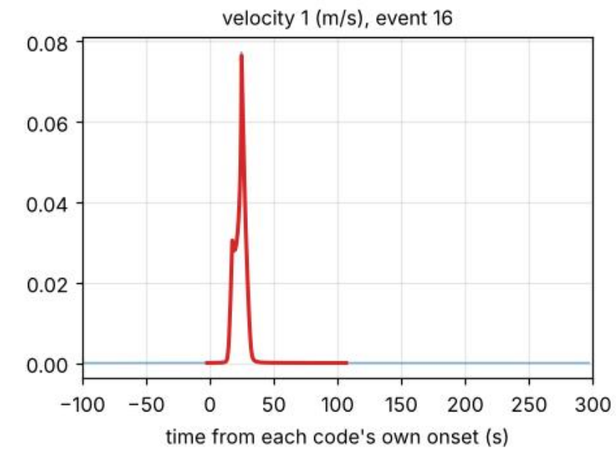


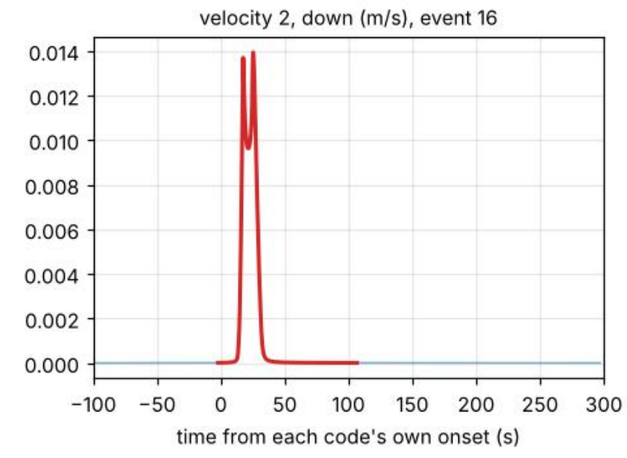


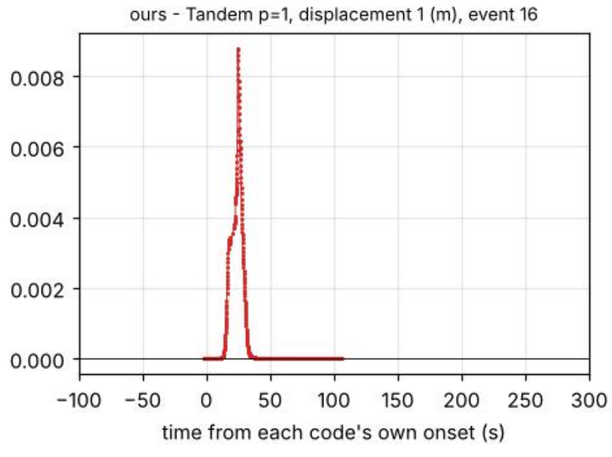


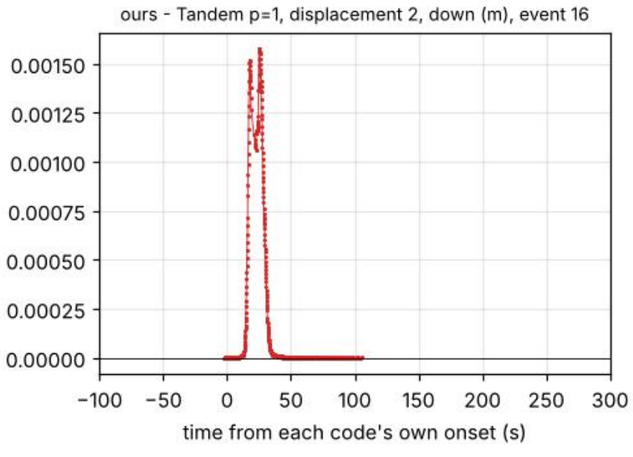


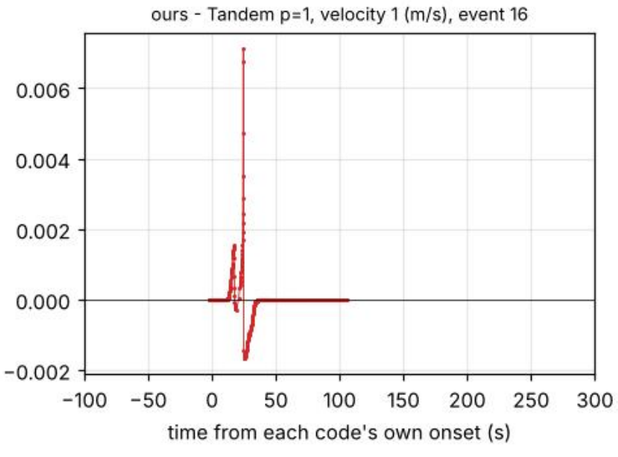


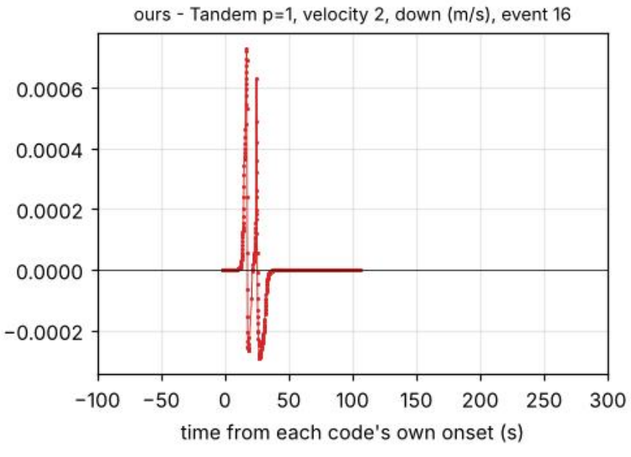






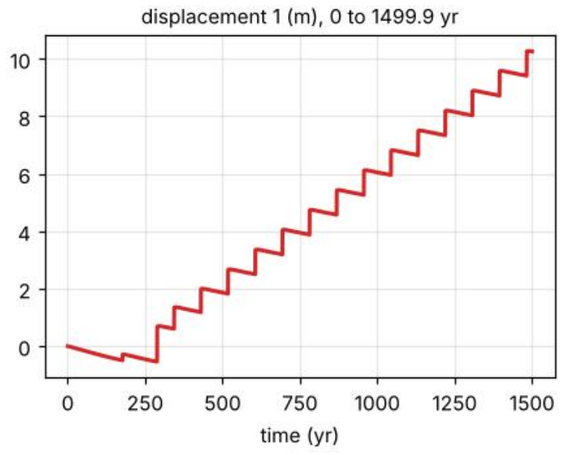


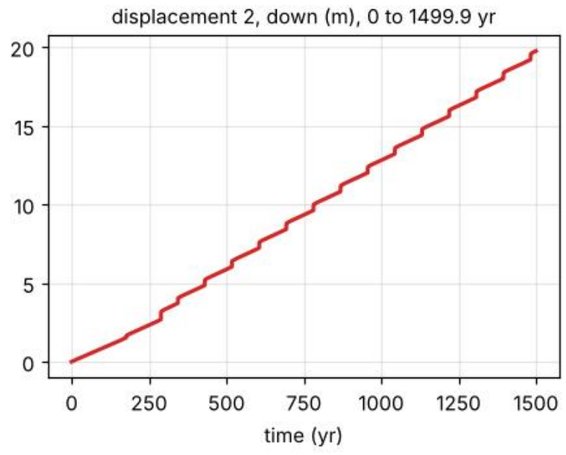


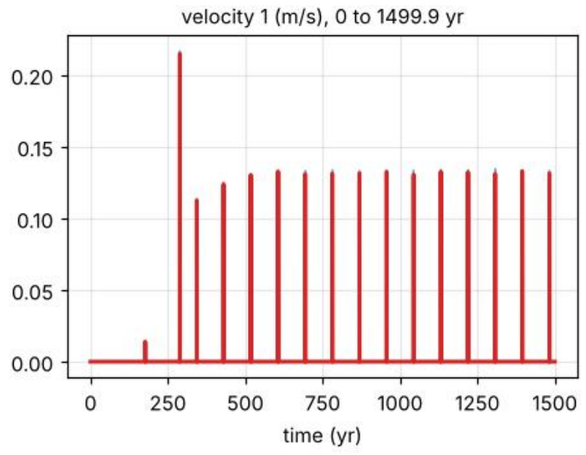


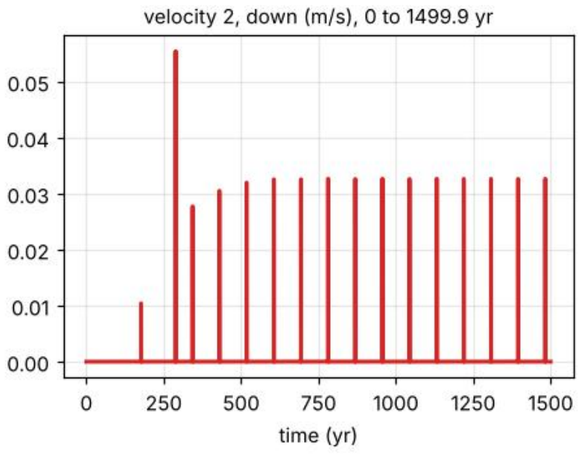
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

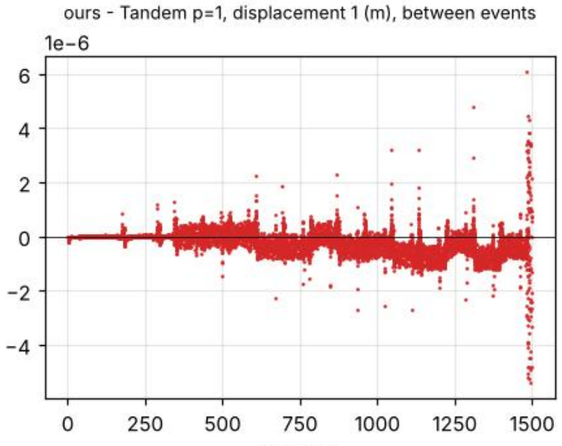
BP3-QD dip 60 thrust, surface station fn-08 (x1 = -8 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

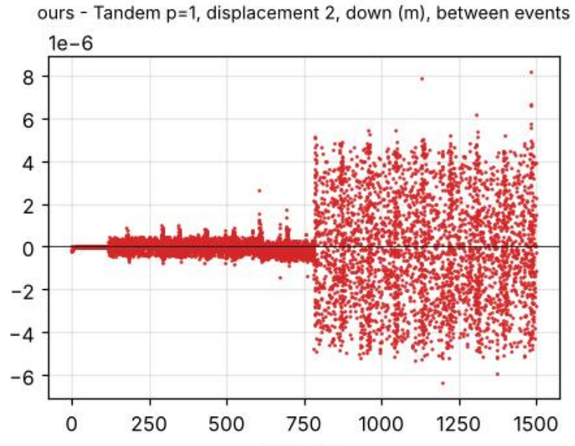


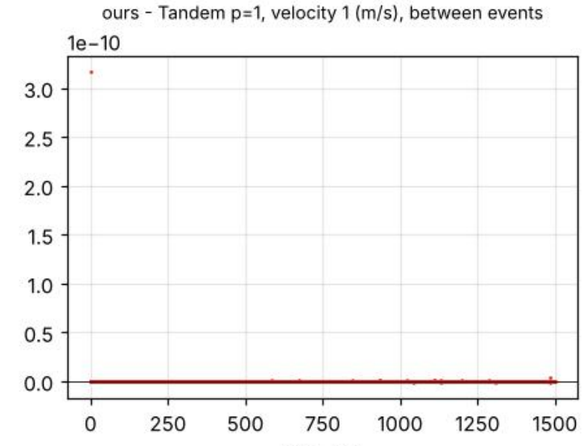


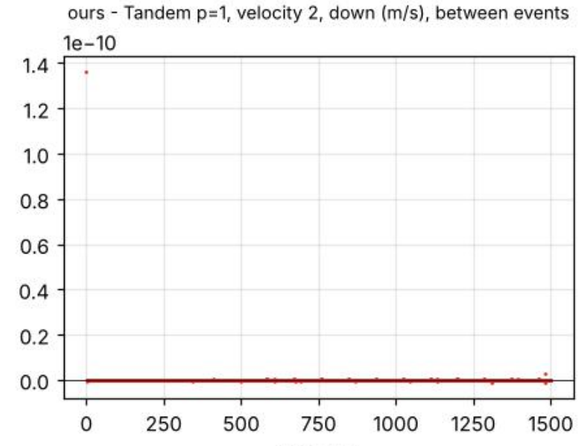


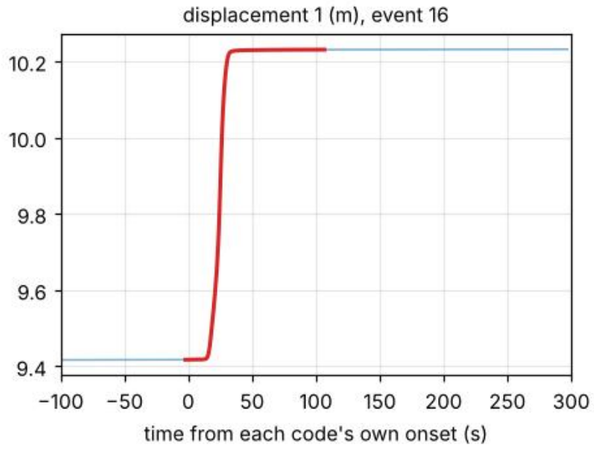


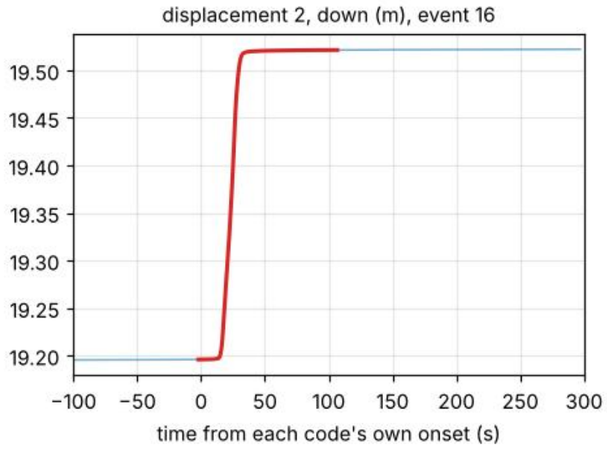


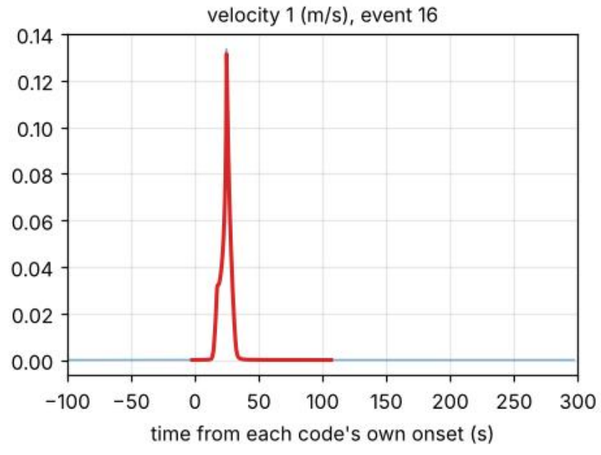


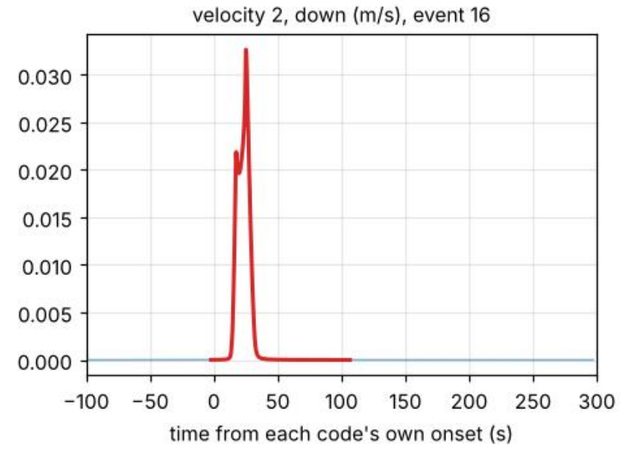


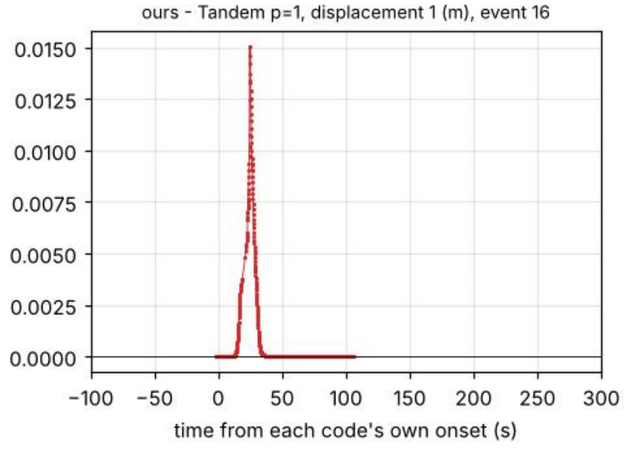


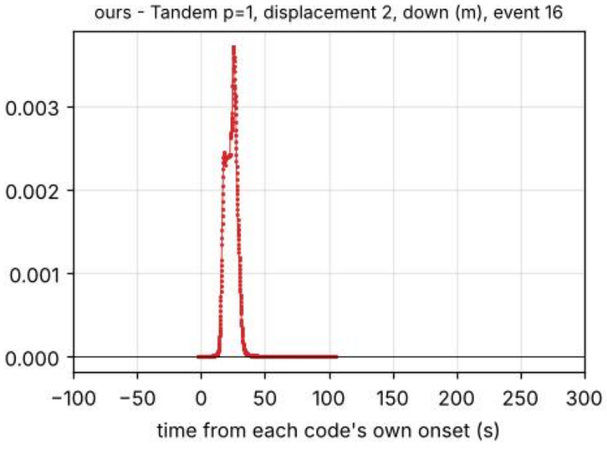


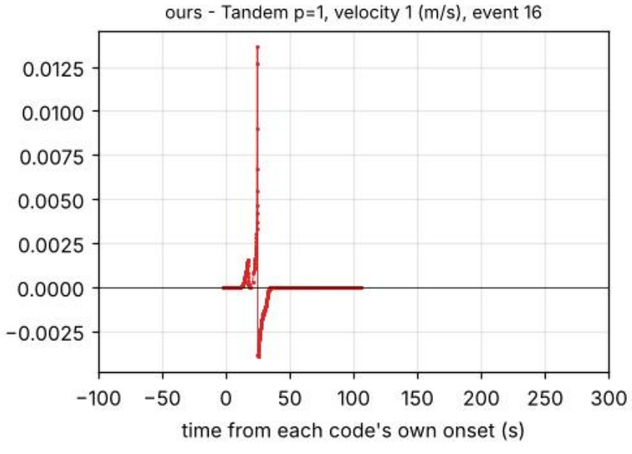


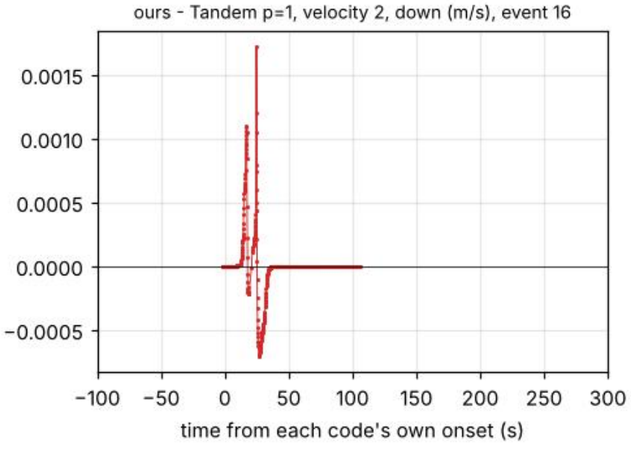




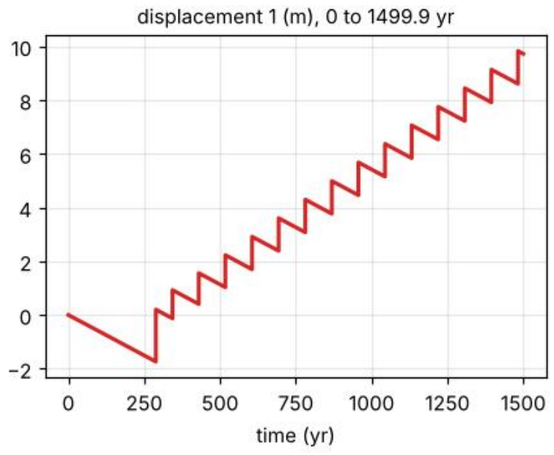


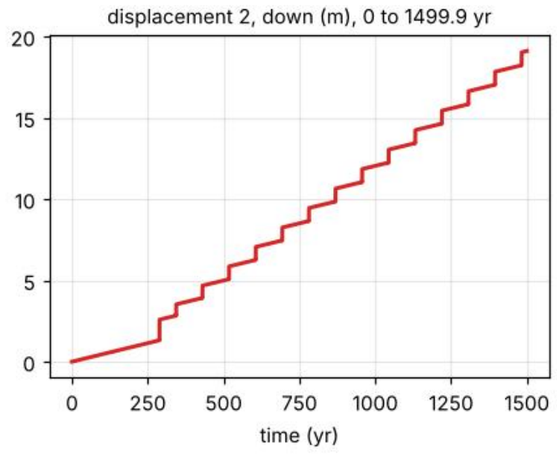


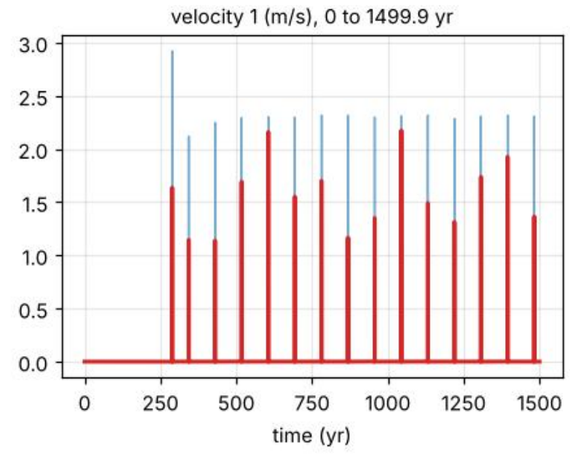


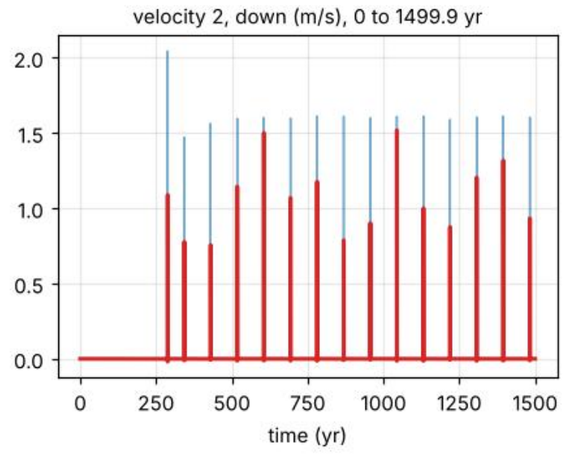


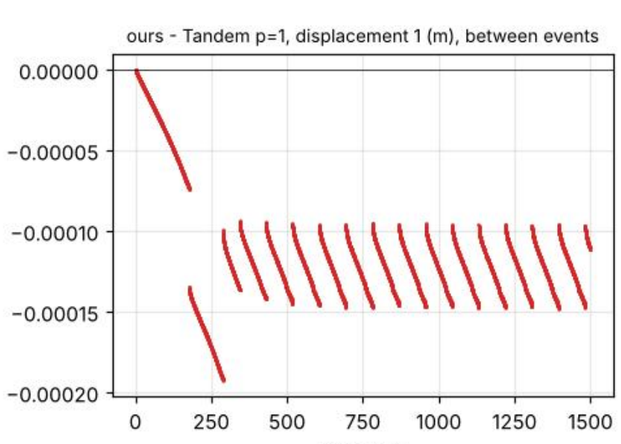
BP3-QD dip 60 thrust, surface station fn-00 (x1 = 0- km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

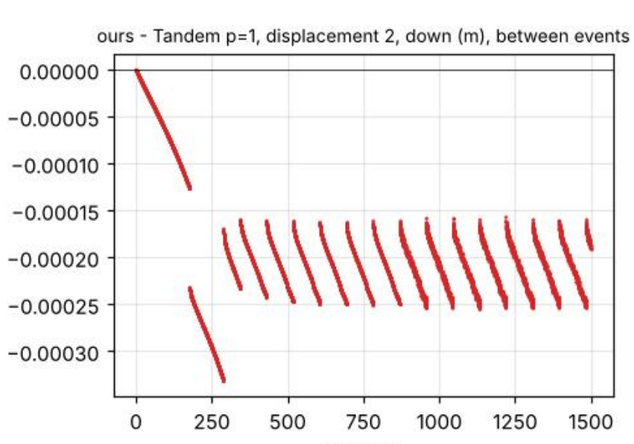


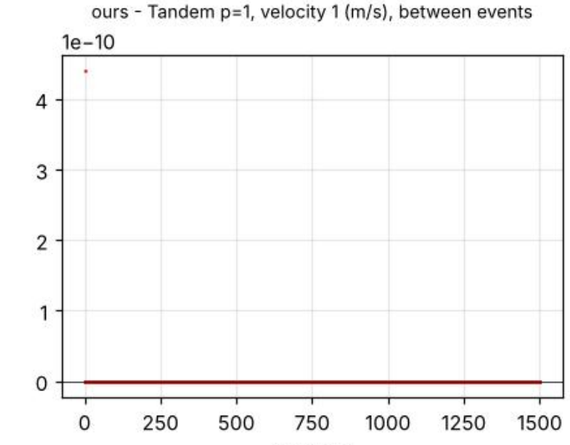


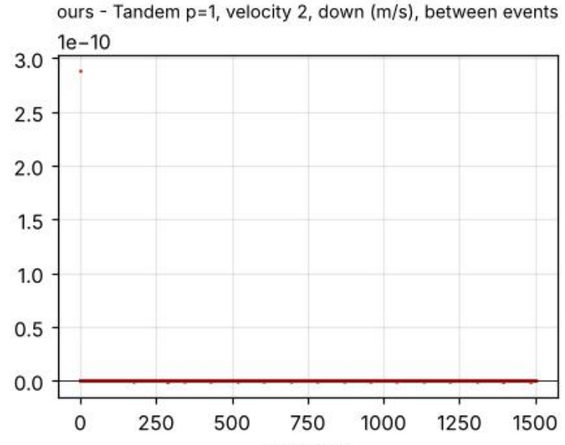


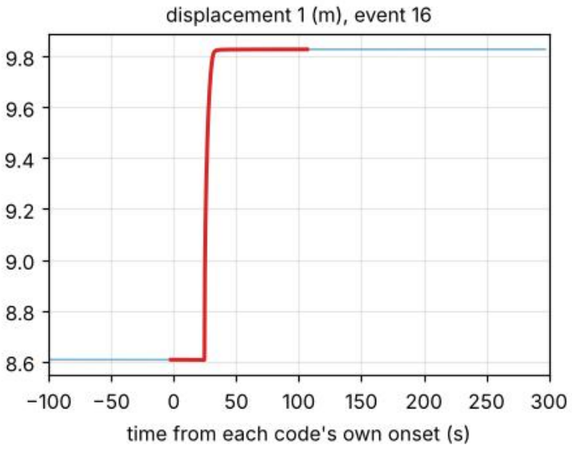


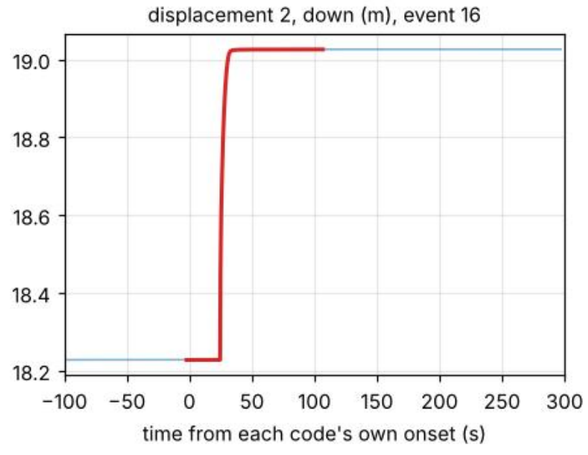


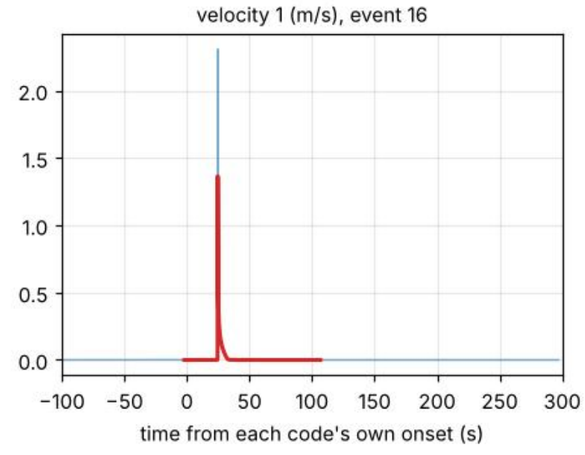


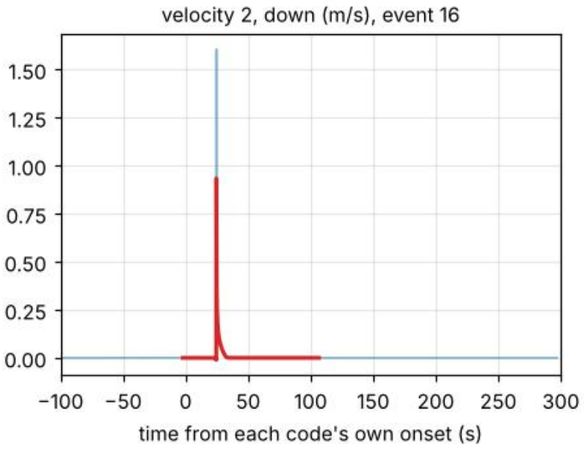


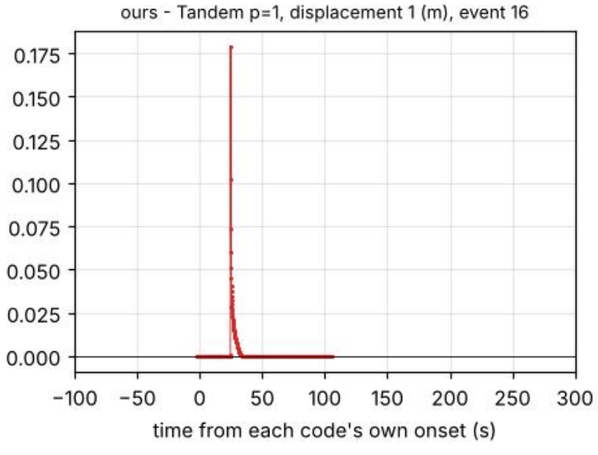


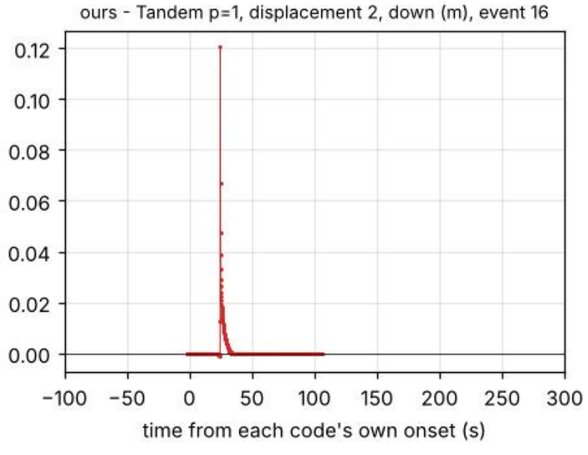


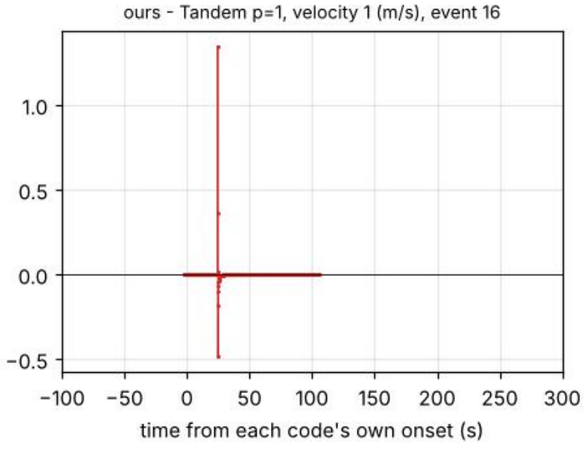


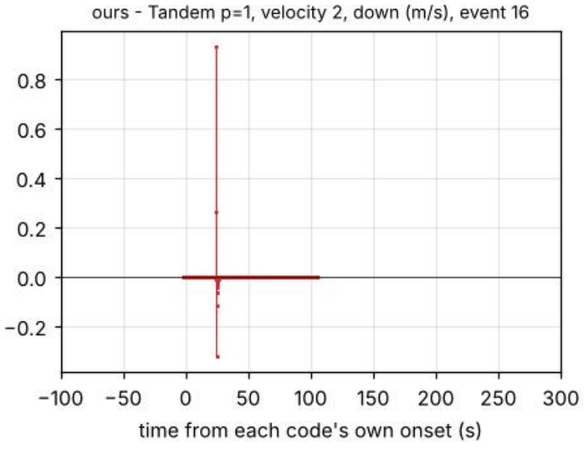






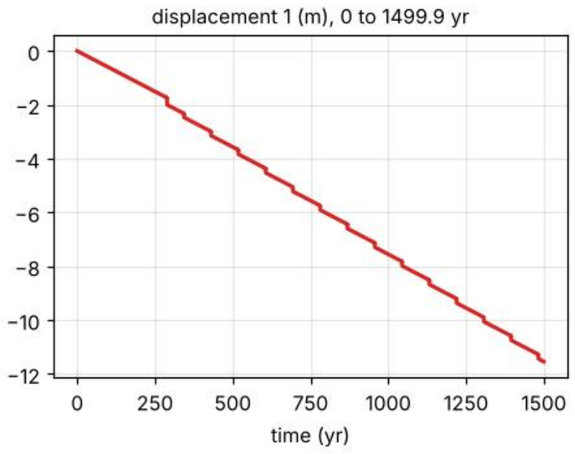


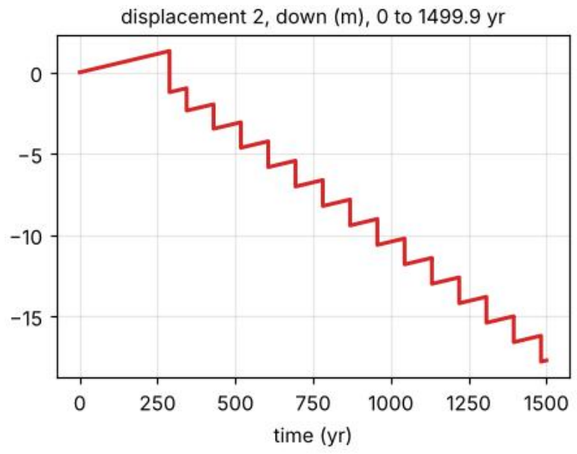


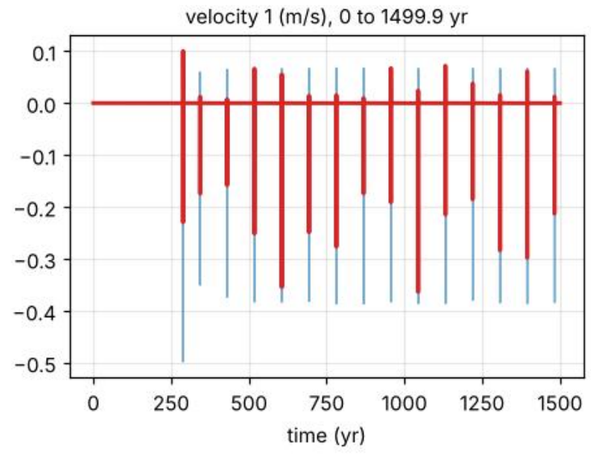


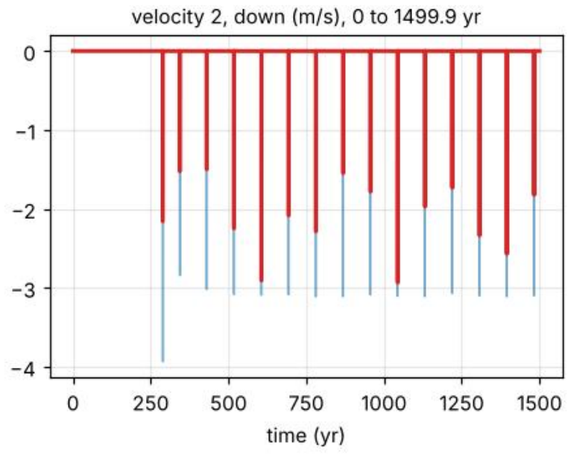
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

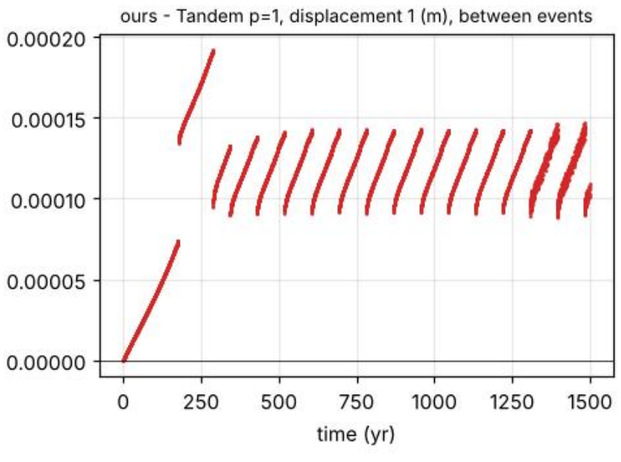
BP3-QD dip 60 thrust, surface station fn+00 (x1 = 0+ km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

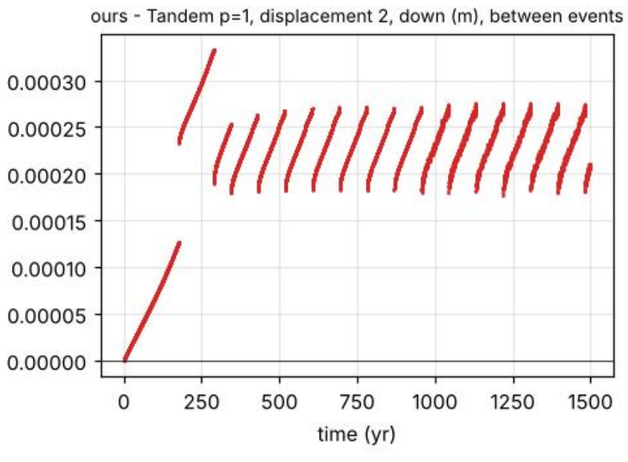


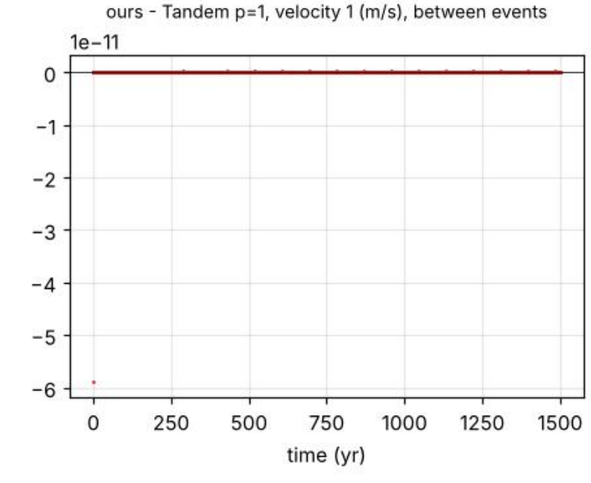


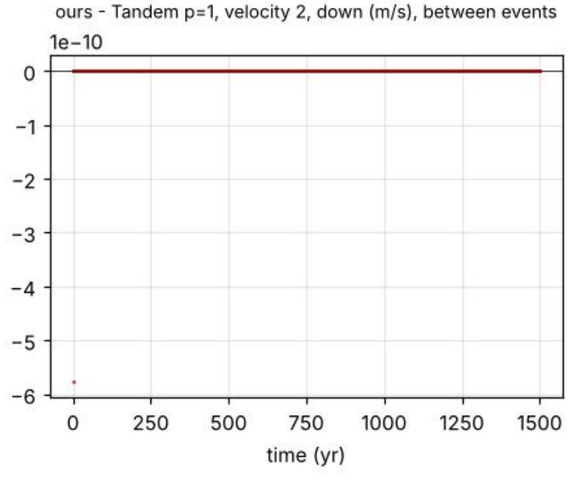


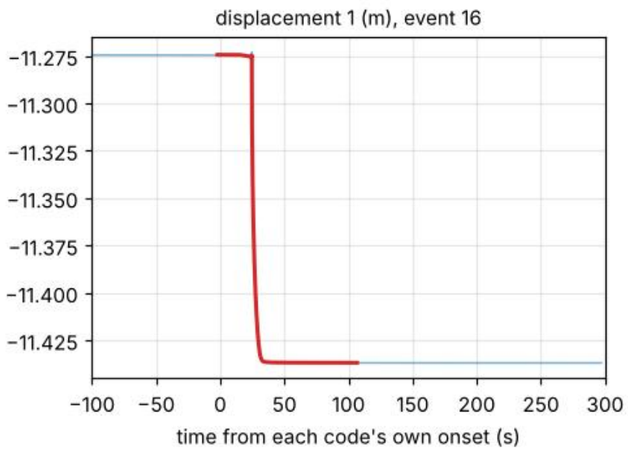


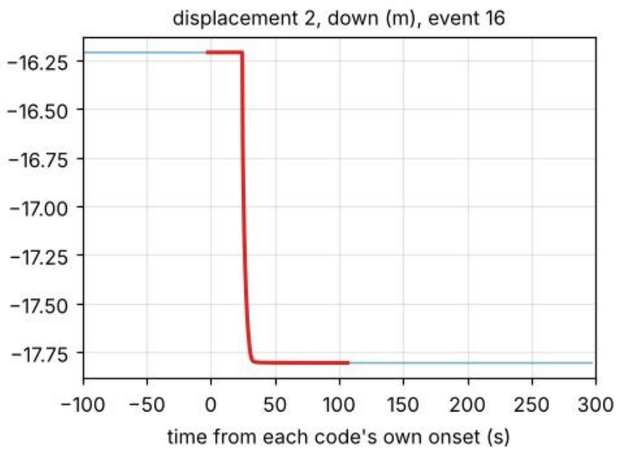


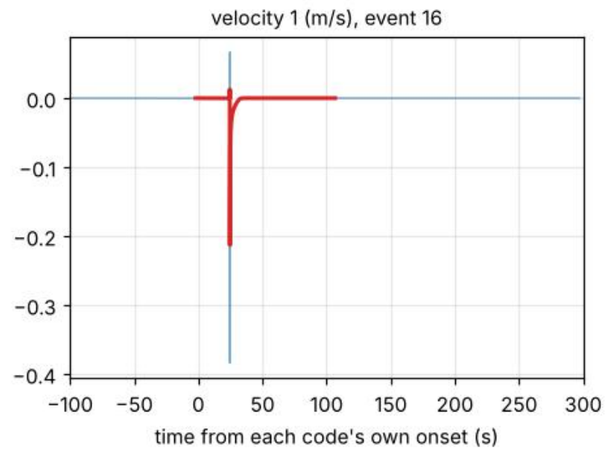


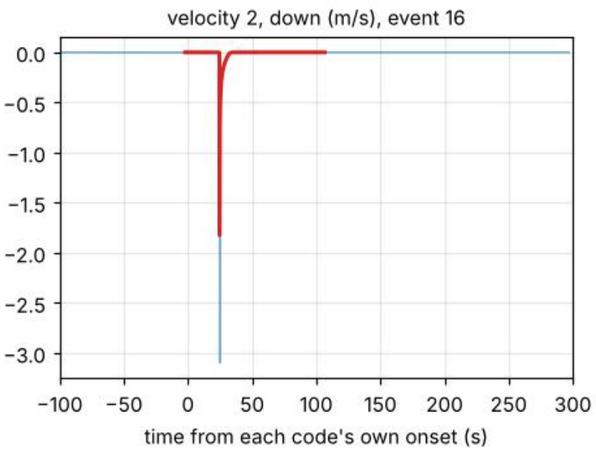


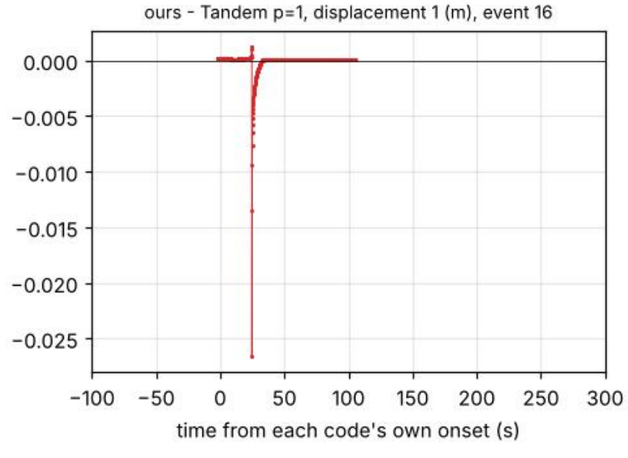


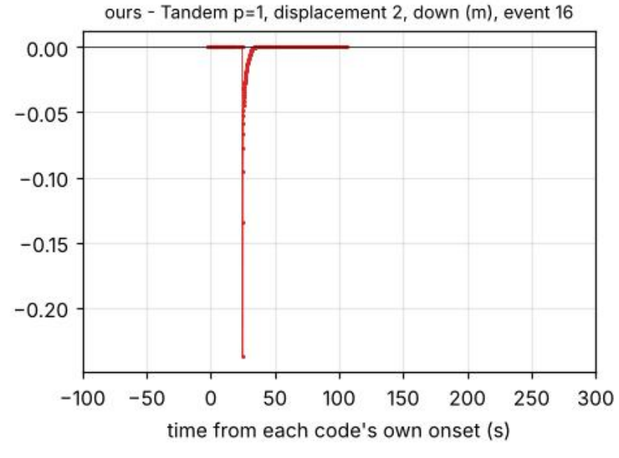


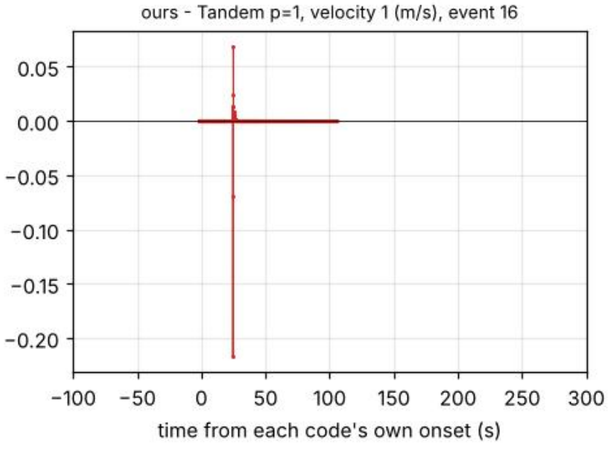


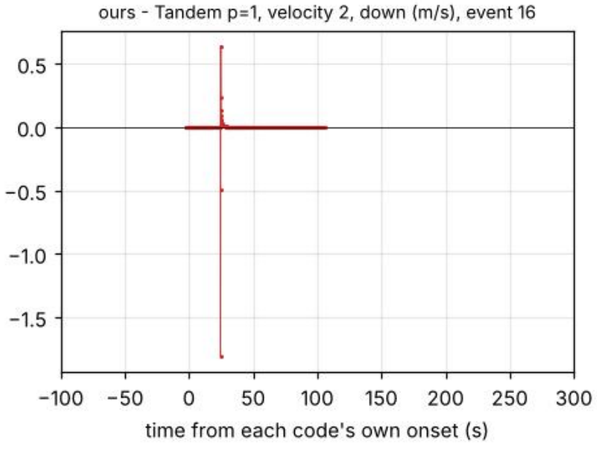






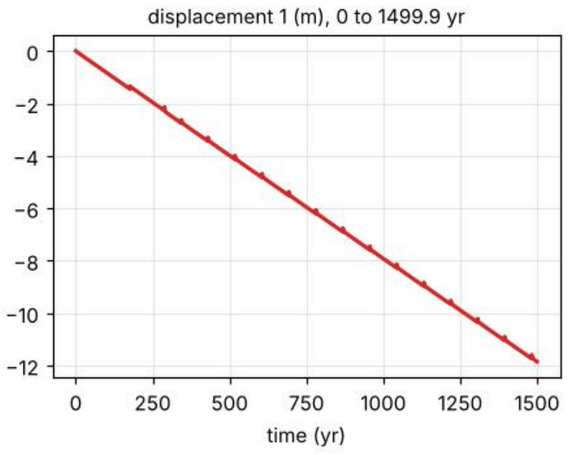


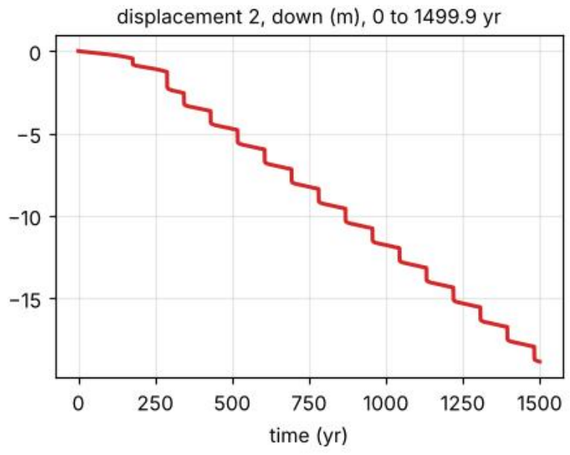


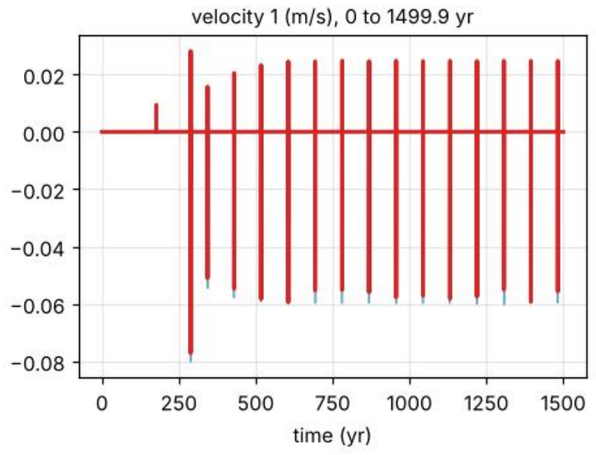


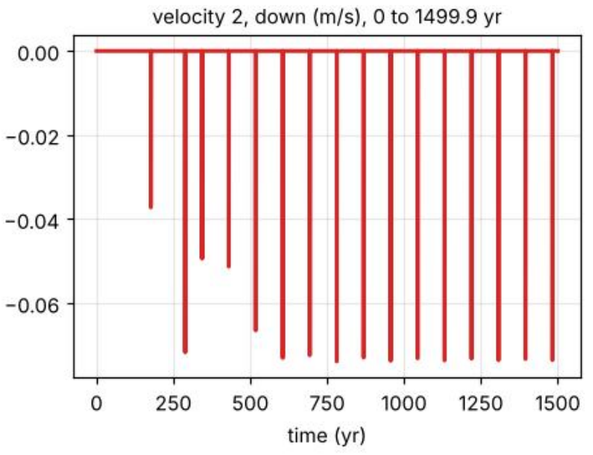
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

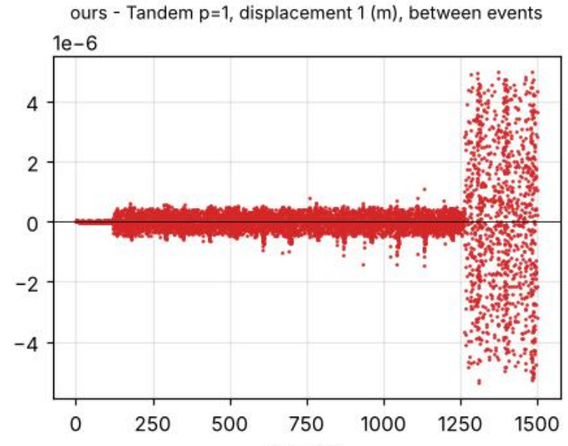
BP3-QD dip 60 thrust, surface station fn+08 (x1 = 8 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

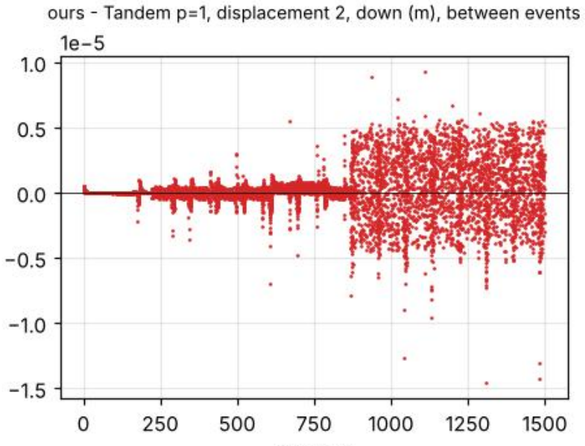


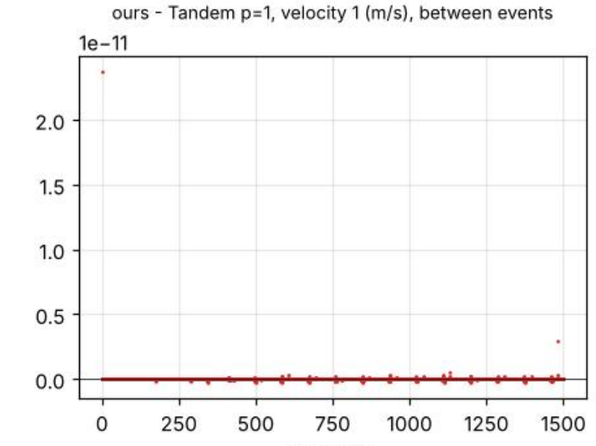


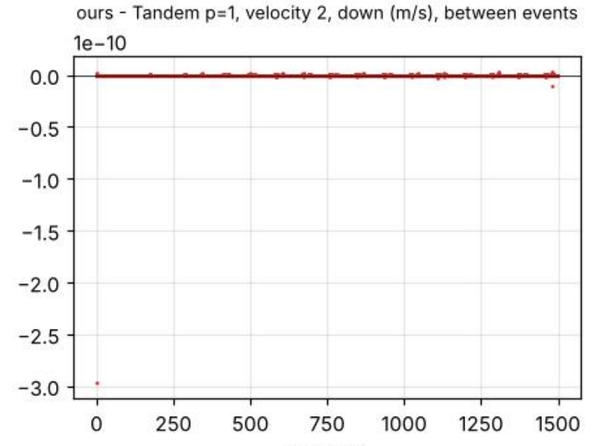


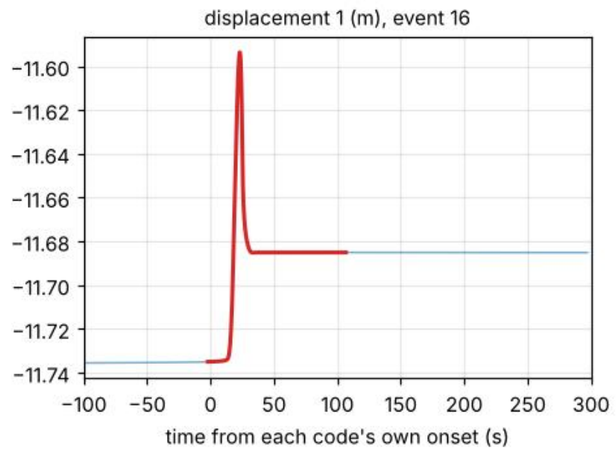


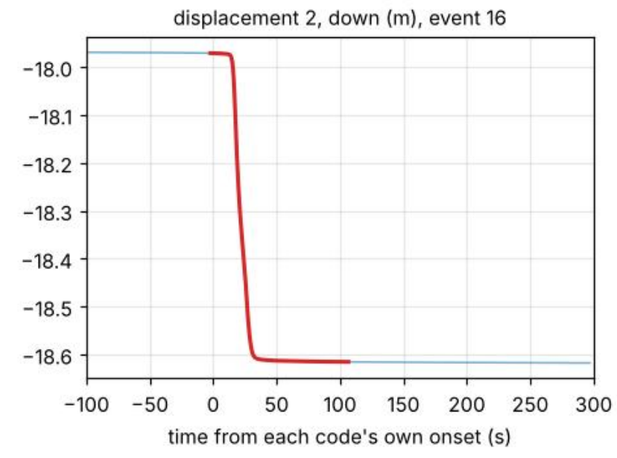


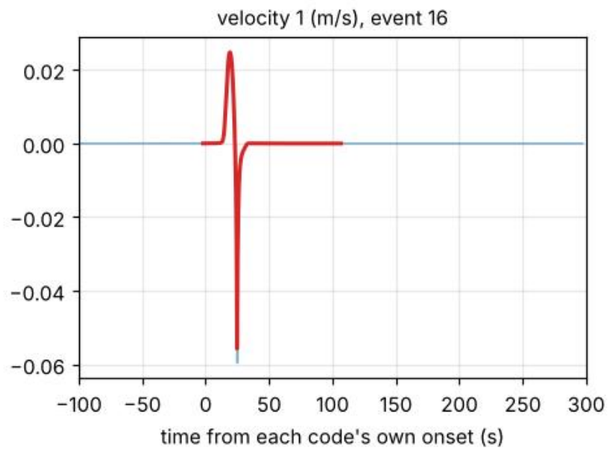


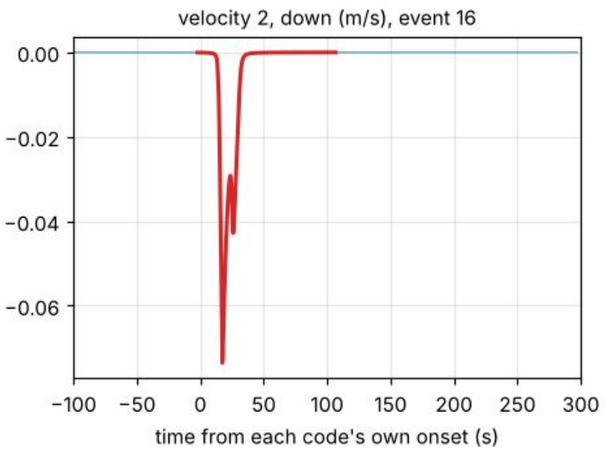


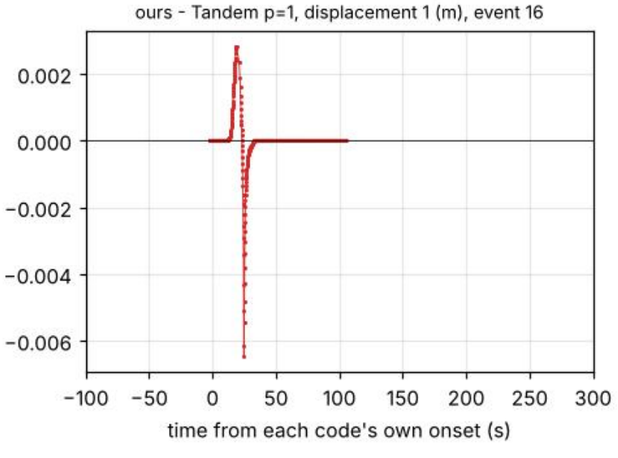


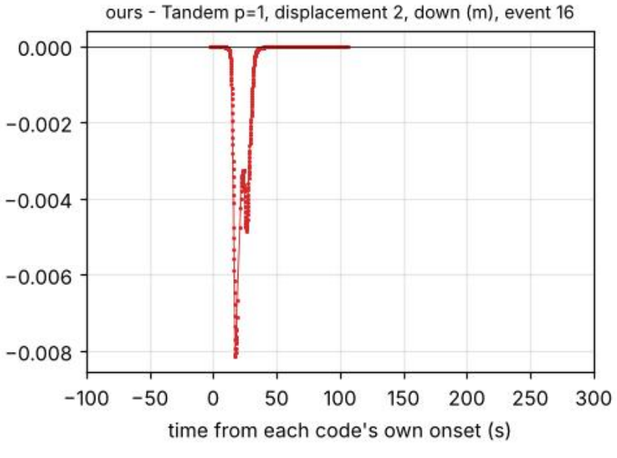


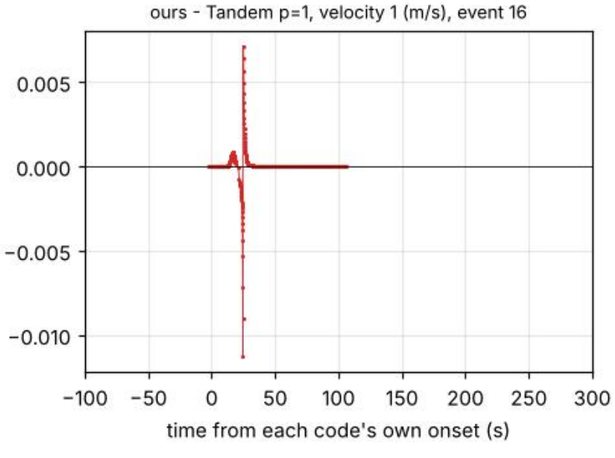


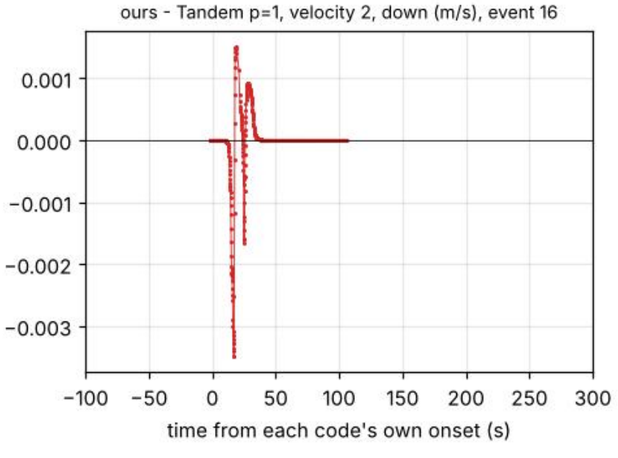






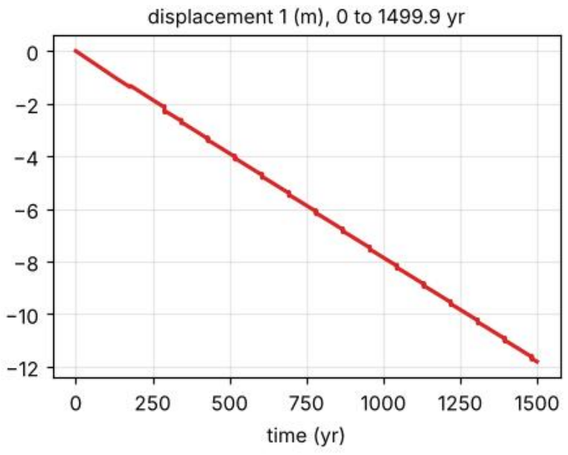


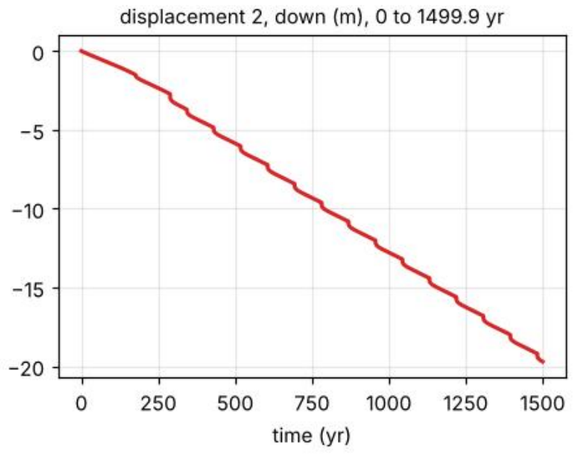


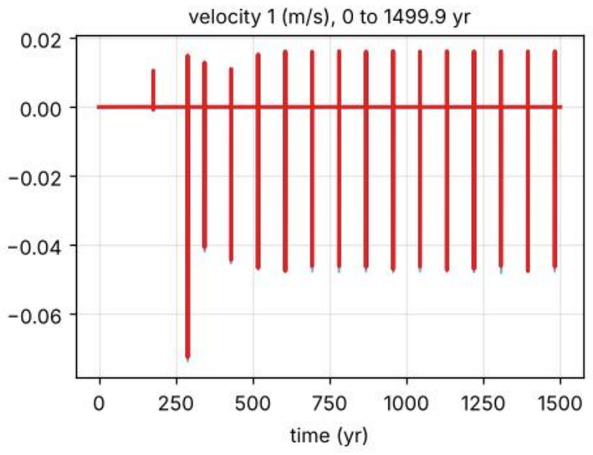


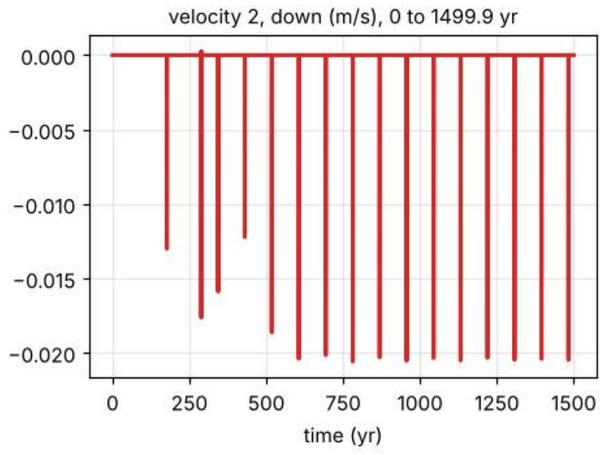
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

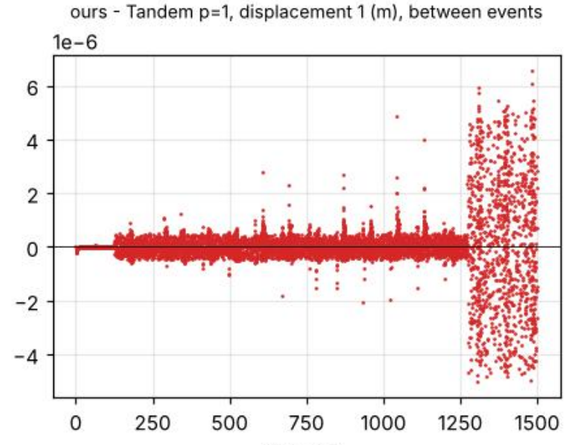
BP3-QD dip 60 thrust, surface station fn+16 (x1 = 16 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

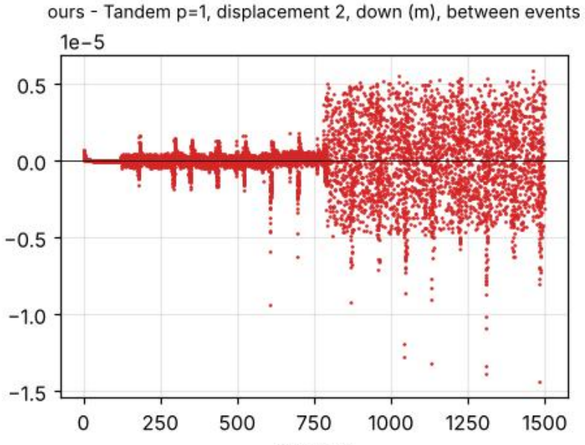


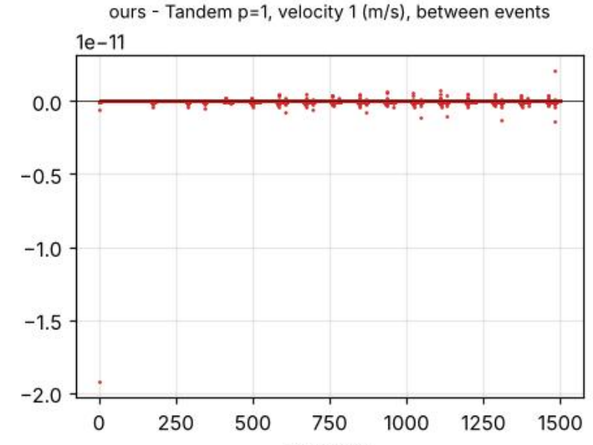


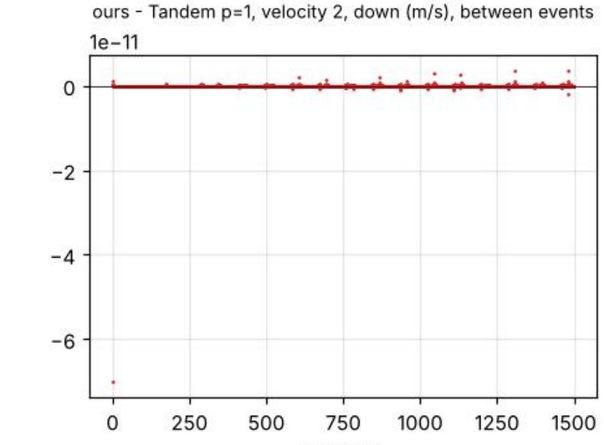


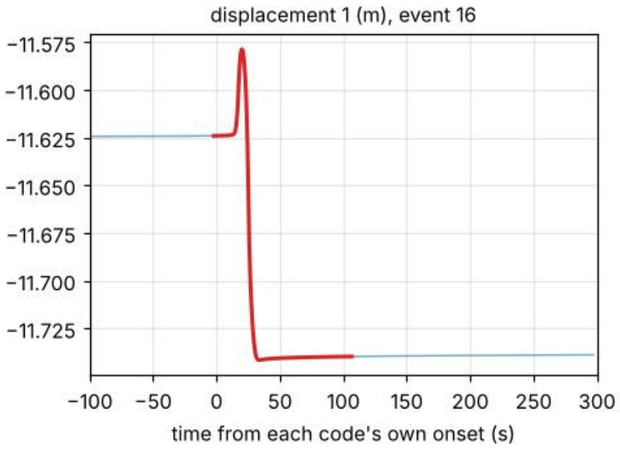


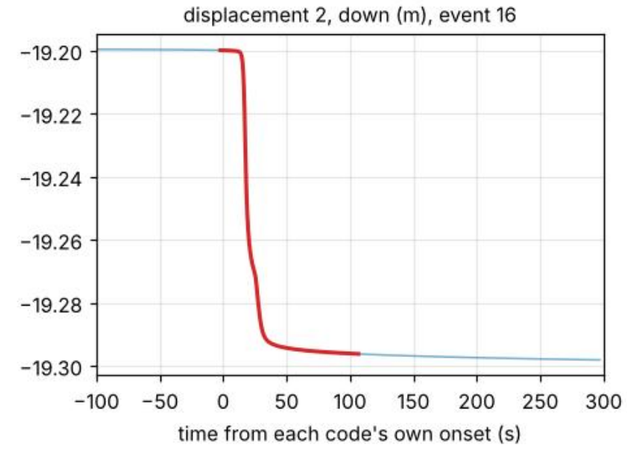


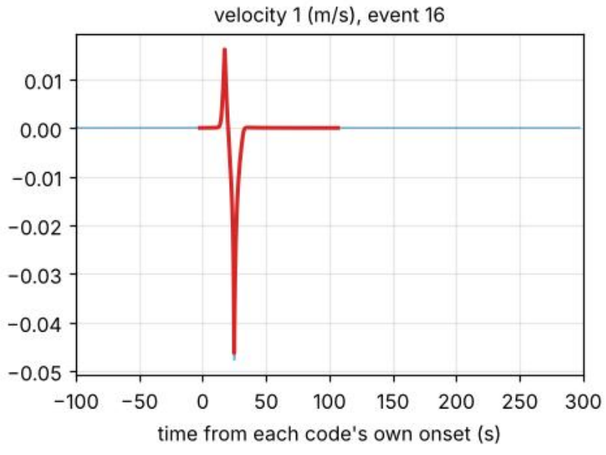


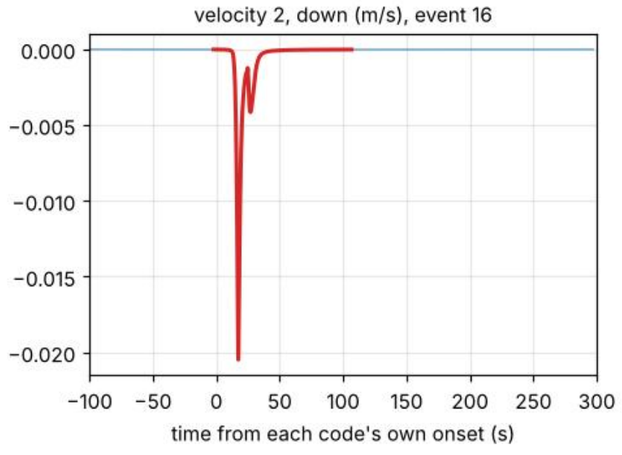


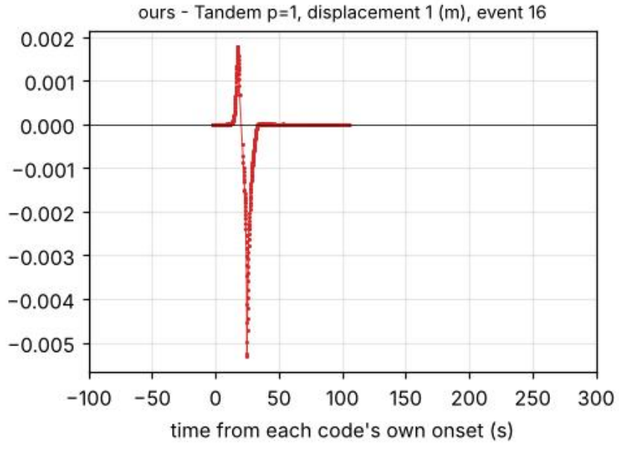


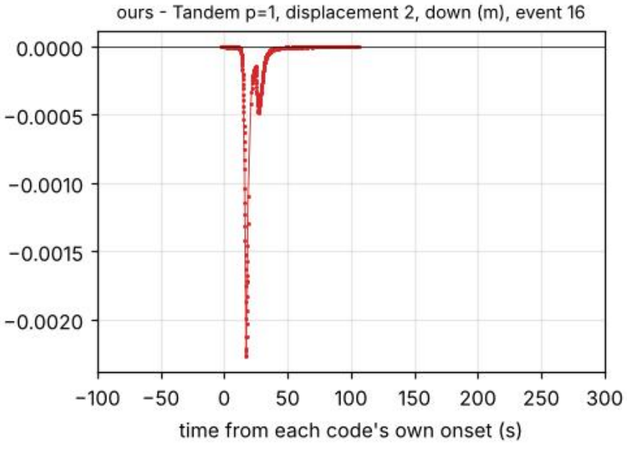


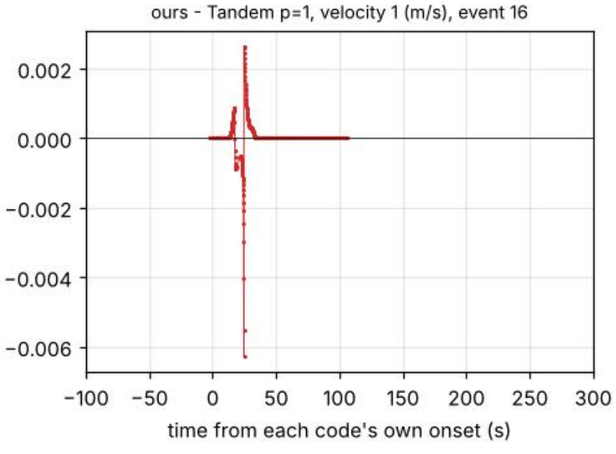


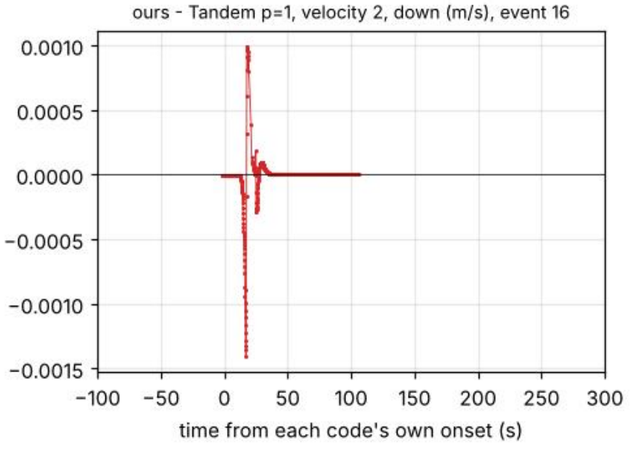






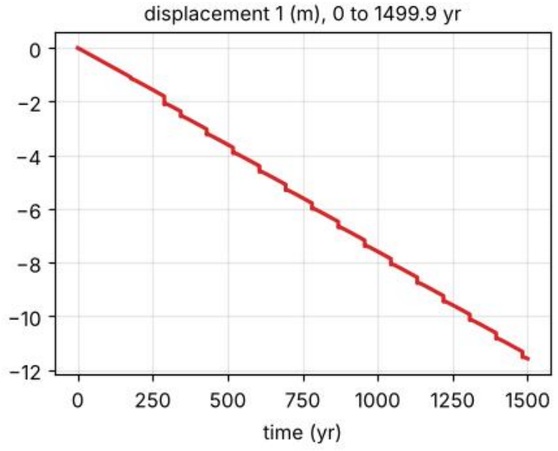


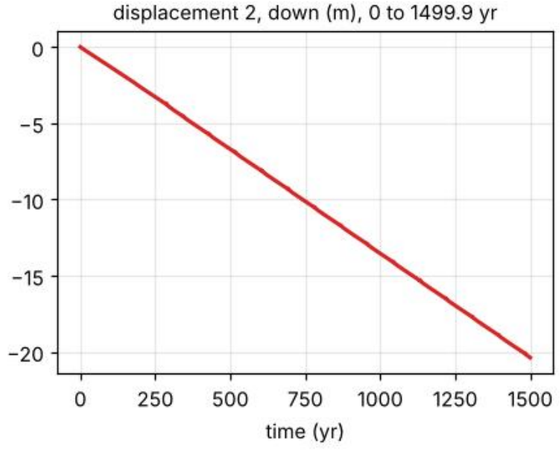


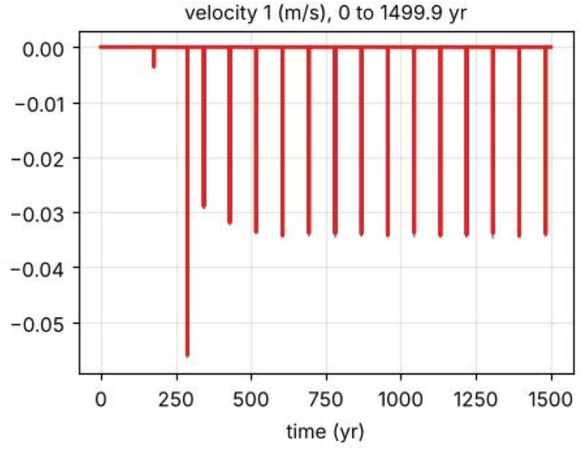


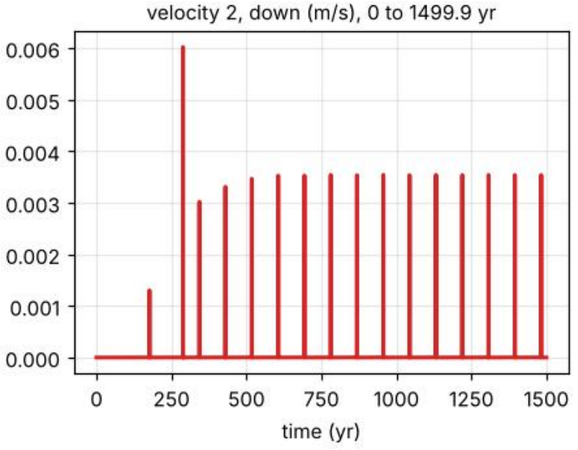
Tandem p=1 (Mac) (du/dt) seas-mfem p=1

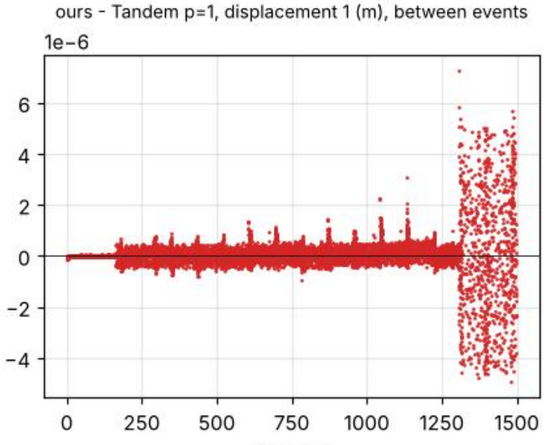
BP3-QD dip 60 thrust, surface station fn+32 (x1 = 32 km): seas-mfem p=1 against Tandem p=1 (Tandem velocity = du/dt; between events = onset - 1 day to onset + 30 days left out)

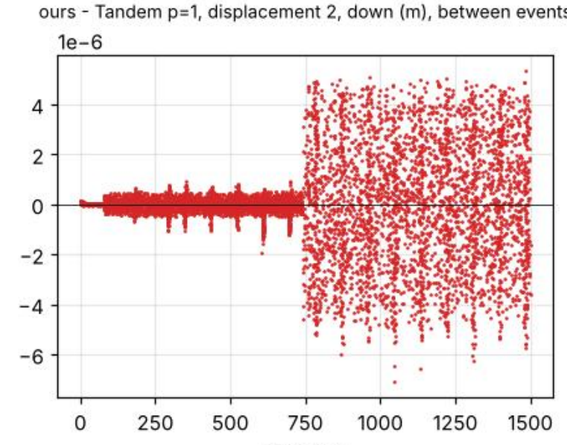


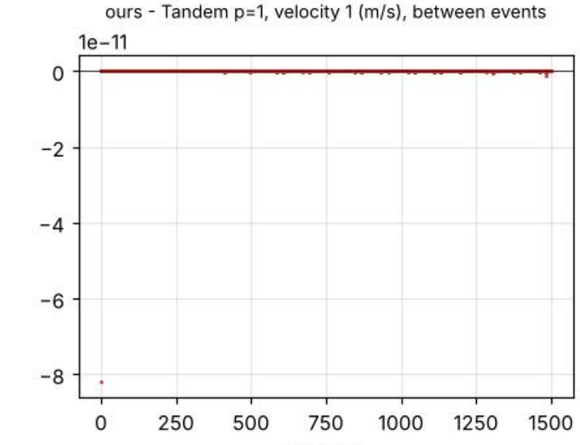


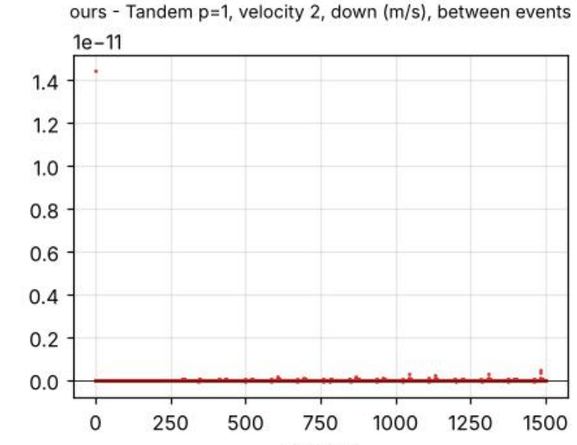


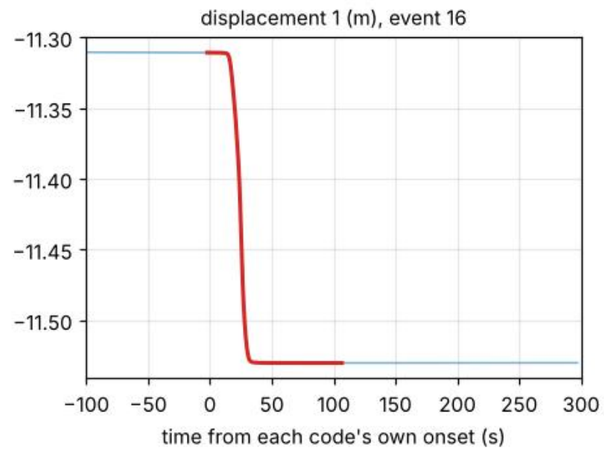


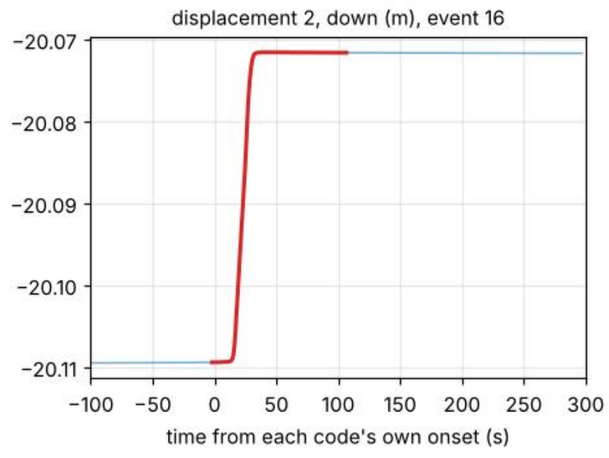


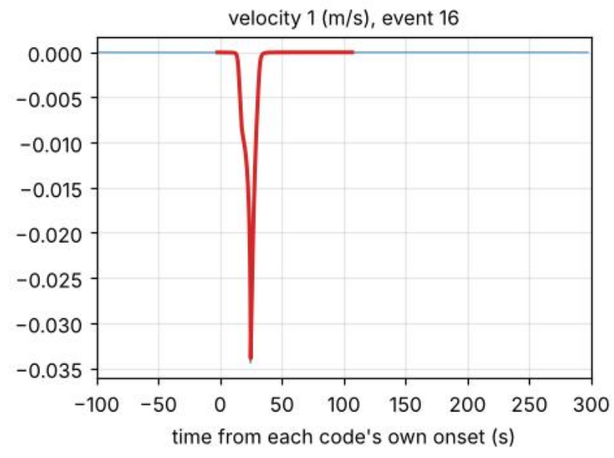


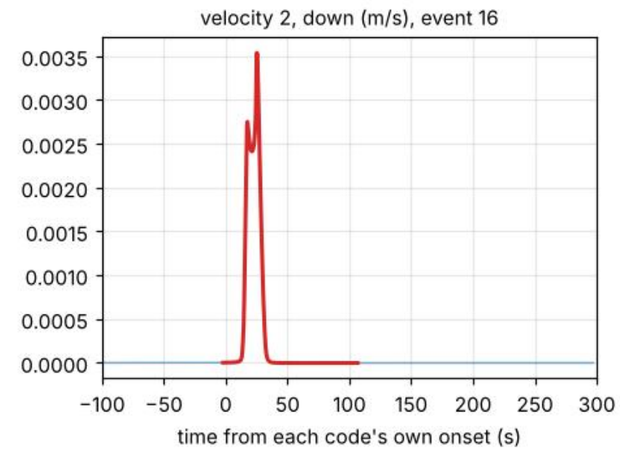


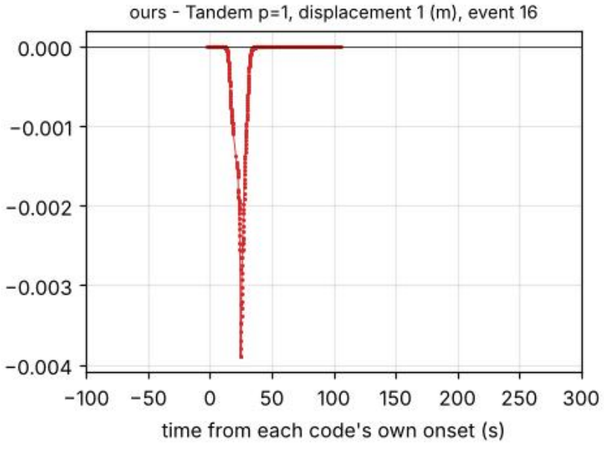


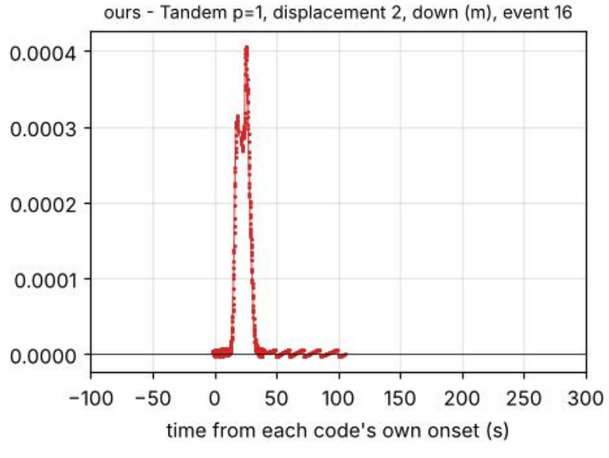


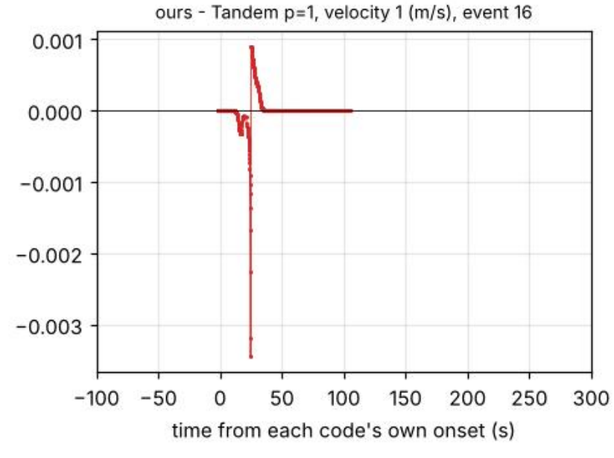


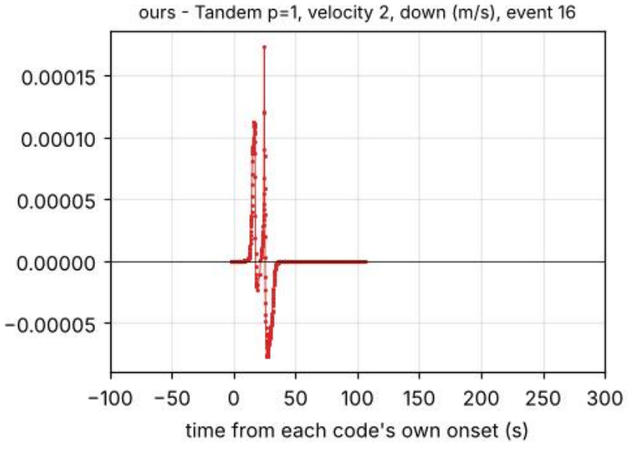












Tandem p=1 (Mac) (du/dt) seas-mfem p=1

Notes

1. Cut. Ours ends at 1500.0000 yr and Tandem p = 1 at 1499.9083 yr; both are cut at the earlier,
   1499.9083 yr. Each has 16 events, the last starting 17.3 yr before the cut.
   Ours = the run's SCEC export (cycle/scec_bp3_export.py).
   Onsets of both codes come from the station data with one definition (first time max V over the 12 fault stations
   > 1e-3 m/s after more than 1 yr below). Events are matched by index.

2. The post-event output gap. Ours writes a row every 0.1 s while max V > 1e-3 m/s, then, 10 s after it drops below,
   every dt_slow = 0.1 yr (median row spacing 1 day to 2 yr after event 16: 36.3 days). The afterslip in the first
   hours to days after each event is therefore not sampled: off-rupture slip-rate peaks that the references show
   15 to 30 minutes after the event (dp025, dp175, dp200) fall into that gap. Up to the end of our dense rows our
   slip rate matches Tandem p = 1. It is an output-schedule limit, not a solution difference.
   The interseismic differences below therefore exclude onset - 1 day to onset + 30 days of every event.

3. The near-trace surface pair fn-00 / fn+00. Ours evaluates both stations exactly on the fault trace (x1 = 0), as
   the limit from each side, so they differ only by the trace slip (dp000, ~1e-5 m). Tandem's sit at x1 = -1 m and
   +1 m, so they also carry the displacement gradient across 2 m of the hanging and foot walls; the pair
   difference is larger there. Neither is an error; the station definitions differ.

4. Peak magnitudes differ because of OUTPUT SAMPLING, adaptive (Tandem) against fixed (ours), not the solution.
   Tandem writes a row when the fault-wide max V has changed by 1 % since its last row (rtol 0.01), or after 0.3 yr;
   ours writes every 0.1 s while max V > 1e-3 m/s, else every 0.1 yr. So Tandem writes every few ms where max V
   changes fast and nothing for up to about 2 s on the mid-rupture plateau. From event 2 on the rupture breaks the
   surface and the peak is at dp000, a spike of about 0.05 s: Tandem catches 5.1 to 7.1 m/s, ours whichever 0.1 s row
   lands in it (2.6 to 5.2 m/s); Tandem on our rows (each code's own onset as zero) gives 2.5 to 5.5 m/s.
   At dp050, events 3 on, the front passes inside Tandem's mid-rupture gap: its file misses the arrival and the peak.
   Use the slip and onset columns, not peak V.

5. dp150 (15 km down-dip, where a starts to rise) differs from Tandem p = 1 before event 1, between about 30 and
   70 yr: shear stress up to -0.38 MPa and log10 V up to -0.32 (ours minus Tandem, 0 to 100 yr). After 100 yr, between
   events, the largest is -0.0055 MPa and -0.0033 in log10 V. dp125 and dp175 agree to 4e-04 MPa over 0 to 100 yr.
   Both codes write densely there, so this is not sampling. The probe sits on the mesh vertex at the kink of a
   (x_d = 15 km), where two fault faces meet and each keeps its own value; the codes likely read different faces
   (ours the up-dip one). Not confirmed.

6. Tandem p = 1 velocity at the surface stations is du/dt by finite differences (its output has no velocity).

7. Differences are ours minus reference, the locally denser series interpolated onto the sparser one's times.
   Wherever one series is locally denser, it is interpolated onto the other one's samples (decided sample by sample).